

Internet Appendices

for

A Comprehensive 2022 Look at the Empirical Performance of Equity Premium Prediction

by Amit Goyal, Ivo Welch, Athanasse Zafirov

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IA.I Expected Change of T-statistics With 10 More Years

This appendix investigates whether we would have expected so many estimators to deteriorate (as what we see in the data) when we add 10 years to original 50-year samples. The maintained hypothesis is that the models are stable and have equal forecasting power with a common parameter measuring the unknown forecasting power. This presumes that observed T-statistics are draws from this equal-strength distribution.¹ Researchers may have reported so many 50-year T-statistics (typically above 2.0 and below 5.2) not because the relationships are so strong or because they did anything wrong, but because other papers may not have been accepted to journals and cited as frequently.

The experiments here show that we can state comfortably that the observed deterioration of T-statistics is not inconsistent with a scenario in which the models were stable with all predictive power having been spurious. They also show that it is not possible to decisively assert zero predictive power with statistical confidence of 90% or more.²

A Empirical Observation

Our analysis in this appendix is based on Table 2, in which we replicated 29 papers. In these papers, authors worked (on average) with about 50 years of data, and we added about another 10 years. We call the (absolute value of the) T-statistic calculated over 50 years “ T_0 ,” that over 60 years “ T_{01} .” Recall:

- Authors reported T-statistics with a mean (over 29 papers) of about $T \approx 3$ and a median of about 2.85. No reported T-statistics were under 2.0. We do not use this information.
- Our empirical replications were very close to authors’ findings. These are our (29) “ T_0 ” statistics. We observed a mean of about $T_0 \approx 2.85$, with a median of 2.58. (One

¹Note that we cannot draw inference without a homogeneity assumption. If 9 estimators truly have an extremely negative beta and 20 estimators truly have an extremely positive beta, we would expect to see 9 estimators to have their T_{01} fall below 2.0 and 20 estimators not to fall below 2.0. We could say no more about estimators. Similar assumptions underlie every other statistical estimator.

²The confidence that there is no X-Y association seems to be about 75-90% — and even this is under the incorrect assumption of 29 uncorrelated estimators. In reality, many estimators are similar in nature, which makes the number of actual observations effectively smaller than 29 (papers) and thus any hypothesis more difficult to test. A better conclusion is thus not that we can rule out that the 29 estimators truly worked; but that we cannot rule out that the 29 estimators truly did not work. If we had to place a bet at even odds, it should be that they did not work.

outlier, CGMS, was responsible for much of this deterioration. However, a handful of other papers also had T_0 statistics close to 2.0.)

- After we added forward data, we obtained a mean T-statistic of 2.25 (median of 2.10). These are our (29) “ T_{01} ” statistics.

In our analysis, we will determine the probabilities of observing specific T_{01} statistics, given the actual empirical values of the T_0 statistics.

B Experiment Intuition

Suppose that all model estimators are stable with a (normalized) true beta of β . We call a specific true beta to be one scenario. The variables are normalized so that each true β maps monotonically into a particular T-statistic for a given number of statistics. For example, if $\beta=0$, then we expect the T-statistic given 600 (monthly) draws to be zero. If $\beta=1$, we expect it to be higher, on average some T_0 statistic in the first 600 draws. We name scenarios to the reader not by their true β s but only by their expected T_0 average statistics.

Given some unknown true beta scenario, the variations in our T-statistics now arise only because of sampling variation. Intuitively, if the true beta is 0, we would expect to see a 50-year T_0 statistic of, say, 4.0 with a 50-year history only extremely rarely. If the true beta is high, observing an estimate T_0 of 4.0 with a 50-year history would be more common.

However, this T_0 is not our statistic of primary interest. Instead, conditional on having observed a T_0 statistic of 4.0 (over the first 50 years), we can ask how often we would now observe a T_{01} statistic (that adds 120 monthly draws) that is less than T_0 ; or one that is below 2.0.

Intuitively, under a scenario in which the true beta is zero, we should frequently observe a drift towards a T-statistic of zero when we add more months. Under a positive beta scenario, we should often observe a drift towards higher T-statistics, stronger when the true underlying beta is stronger.

C Statistics of Interest

In our paper, we refer to three statistics:

- S1: The fraction of T_{01} statistics that are below T_0 . In the empirical data, 20 out of 29 T-statistics deteriorated. (9 improved.)
- S2: The fraction of T_{01} statistics that are below 2.0.³ In the empirical data, 16 out of 29 full-sample T_{01} deteriorated. (13 improved.)
- S3: The delta statistic ($T_0 - T_{01}$). In the empirical data, this was 0.60.

D Data Generation Process

For a given true-beta scenario, in each simulation, we draw 720 months of data from two independent normal distributions to represent X and Y. We calculate the T_0 statistic in the first 600 months and the T_{01} statistic over the entire 720 months.

We did this under various true-beta scenarios about the true association between X and Y. For example, the (null) hypothesis is that there is no true association between X and Y (common parameter $\beta=0$). In this case, the expected T_0 is zero. However, for large realizations of T_0 (say, 3.0), the expected S1 is greater than 50%, the expected S3 statistic is positive. One (alternative) hypothesis could be a positive association strong enough that the expected T_0 statistic over 50 years would be 2.5. For the same large realtion of T_0 (say 3.0), the expected S1, S2, and S3 statistics would now be smaller than they were under the null hypothesis.

³The CGMS T_0 is only 1.6, and thus is more likely to be below 2.0. There are another 5 T_0 statistics very close to 2.0. The S2 statistic is interesting because 2.0 roughly represents statistical significance in published papers.

E Simulated and Empirical Performance

In a first step, we ran 100 million simulations (of 720 months each) under a scenario and recorded T_0 and T_{01} statistics. Many simulations are necessary to draw cases in which T_0 is large (say, above 4).

Figure 1 shows the output of these simulations: statistics based on the mappings from T_0 to T_{01} , given specific scenarios.

The blue lines show the distribution of the S1 statistic (frequency of cases in which the T_{01} statistic declined relative to the T_0 statistic). The black lines show the distribution of the S2 statistic (T_0 statistics below 2.0). For true higher associations, the T statistics are less likely to deteriorate, inducing lower S1 and S2 statistics. For example:

- **Black Curve:** The zero-association scenario panel shows that *if the T_0 statistic is about 2.5*, then we predict the T_{01} statistic to be below T_0 (i.e., 2.5) about 75% of the time. If the relationship is truly strong, we would expect this to occur only about 30% of the time.
- **Blue Curve:** The zero-association scenario panel shows that *if the T_0 statistic is about 2.5*, then we predict the T_{01} statistic to be below 2.0 about 25% of the time. If the relationship is truly strong, we would expect this to occur only about 3% of the time.

In a second step, we use these probabilities to assess the empirically observed S statistics of $S1=20$, $S2 = 16$, and $S3=0.60$, given the empirically observed distribution of T_0 .

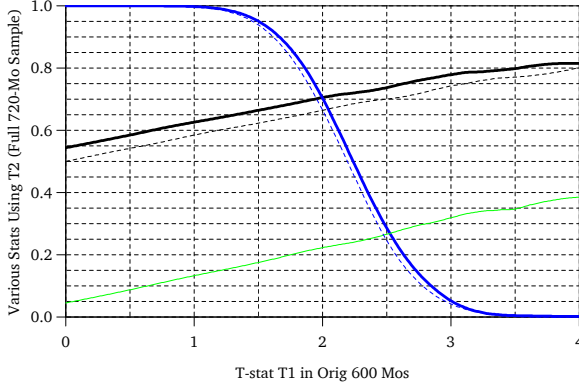
Although it is tempting to use the reported mean T_0 statistic of 2.85 and look up the extended mean T_{01} statistic in the graphs, this would be incorrect. This is because the variance matters, too.⁴ Therefore, we have to look up the probability for each paper given its T_0 separately. We draw a realization of T_{01} given the empirical T_0 for each of the 29 papers and add the sum of $T_{01} > 2.0$ over the 29 papers. We repeat this 100,000 times to obtain a distribution of how often we would expect to see S statistics given this T_{01} configuration.

The drawing itself is a generalization of the binomial distribution, in which each of 29 throws has its own separate probability. For example, under the null hypothesis (no

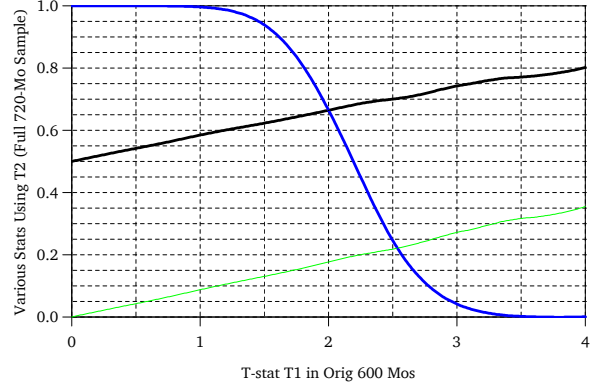
⁴29 papers all with T-statistics of $T_0 = 2.85$ would have a lower probability of a subsequent T-statistic below 2.0 than, say, 5 papers with a T_0 -statistic of 0.0 and 24 papers with a T_0 -statistic of 100.

Figure 1: Simulation-Based Probabilities

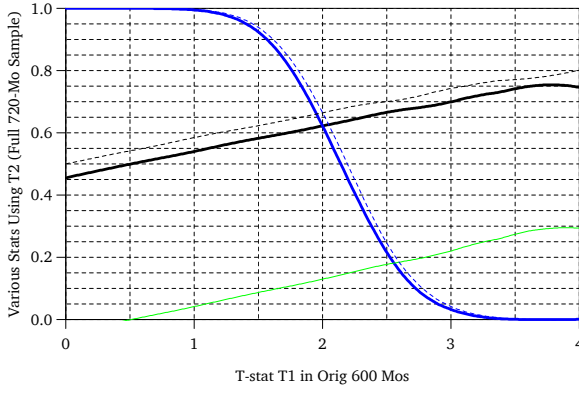
True Association: “Negative” ($E T_0 \approx -0.2$)



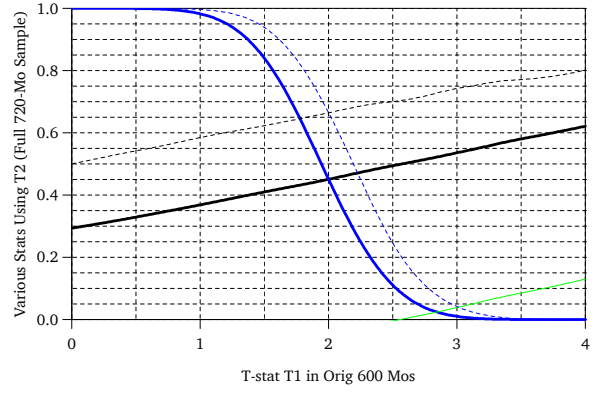
True Association: “Zero” ($E T_0 \approx 0$)



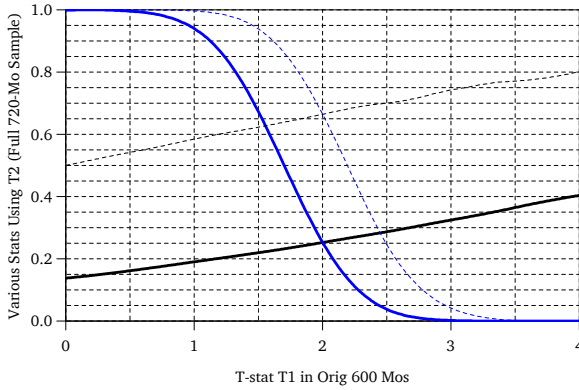
True Association: “Tiny” ($E T_0 \approx 0.2$)



True Association: “Modest” ($E T_0 \approx 1.22$)



True Association: “Strong” ($E T_0 \approx 2.8$)



These are single draw probabilities. The black line is the S1 statistic, the blue line is the S2 statistic, the green line is the S3 statistic. The thinner dashed equivalent lines for S1 and S2 are the null hypothesis (for visual comparison). **Interpretation:** Over the domain of scenarios considered here, S1, S2, and S3 decrease monotonically in the true beta. That is, given a true higher association of X and Y, the statistics are less likely to deteriorate. However, the exact probability change also depends conditionally on the T_0 statistics.

association between predictors and market premia), because there is one T-statistic of $T_0 \approx 1.48$ (skew), there is always a very high probability that its T_{01} statistic will be below 2.0.

Thus, on a set of 29 independent draws (conditional on the empirically observed distribution of the 29 T_0 statistics), CGMS alone means that there are very few outcomes in which the S2 statistic is not greater than 0 (i.e., at least one T_{01} below 2.0).⁵ Equivalently, because there is one T_0 -statistic of 5.13 (gip), its T_{01} -statistic always has a very high probability of being above 2.0. Thus, on a set of 29 draws (conditional on the observed distribution of T_0 statistic), there are almost no outcomes in which the S1 statistic is 29. The most common outcome is where 21 out of 29 estimators would show a T_{01} -statistic above 2.0 (and, of course, 8 a T_{01} -statistic below 2.0). If there is truly a positive association, conditional then more of the T_{01} will be above 2.0. In the extreme, if the true underlying associations were very strong, almost all T_{01} statistics would drift above 2.0, and the s1 statistic would condense around 29.

⁵With a good number of T_0 statistics below 2.3, under the null hypothesis, it is virtually impossible to have cases in which s1 is 0.

	S1: # $T_{01} > T_0$						S2: # $T_{01} > 2.0$				
$E(T_0)$:	-0.25	0.0	0.25	1.23	2.45	$E(T_{01})$:	-0.24	0.0	0.2	1.5	2.8
0	0.0	0.0	0.0	0.0	0.0	0	0.0	0.0	0.0	0.1	2.9
1	0.0	0.0	0.0	0.0	0.0	1	0.0	0.0	0.0	1.0	14.4
2	0.0	0.0	0.0	0.0	0.2	2	0.0	0.1	0.2	5.0	27.3
3	0.0	0.0	0.0	0.0	0.6	3	0.2	0.6	1.3	13.1	27.7
4	0.0	0.0	0.0	0.0	1.8	4	1.1	2.5	4.8	21.5	17.6
5	0.0	0.0	0.0	0.0	4.2	5	3.8	6.9	11.0	23.9	7.4
6	0.0	0.0	0.0	0.1	7.6	6	8.9	13.5	18.1	18.6	2.1
7	0.0	0.0	0.0	0.1	11.7	7	15.6	20.0	21.8	10.5	0.4
8	0.0	0.0	0.0	0.4	15.0	8	20.4	21.2	19.2	4.3	0.1
9	0.0	0.0	0.0	1.0	16.1	9	20.3	17.3	13.3	1.4	0.0
10	0.0	0.0	0.0	2.4	14.6	10	15.3	10.8	6.8	0.3	0.0
11	0.0	0.0	0.0	4.4	11.5	11	8.7	5.0	2.6	0.1	0.0
12	0.0	0.0	0.2	7.5	7.8	12	3.9	1.7	0.8	0.0	0.0
13	0.0	0.1	0.5	10.8	4.6	13	1.3	0.5	0.2	0.0	0.0
14	0.1	0.3	1.3	13.3	2.5	14	0.4	0.1	0.0	0.0	0.0
15	0.3	0.9	2.9	14.9	1.1	15	0.1	0.0	0.0	0.0	0.0
16	0.9	2.1	5.3	14.4	0.4	16*	0.0	0.0	0.0	0.0	0.0
17	2.2	4.0	8.7	11.8	0.2	17	0.0	0.0	0.0	0.0	0.0
18	4.2	7.3	12.2	8.5	0.1	18	0.0	0.0	0.0	0.0	0.0
19	7.6	11.1	14.8	5.4	0.0	19	0.0	0.0	0.0	0.0	0.0
20*	11.6	14.8	15.8	2.9	0.0	20	0.0	0.0	0.0	0.0	0.0
21	15.4	16.3	14.4	1.3	0.0	21	0.0	0.0	0.0	0.0	0.0
22	17.1	15.8	11.0	0.5	0.0	22	0.0	0.0	0.0	0.0	0.0
23	16.0	12.7	6.9	0.2	0.0	23	0.0	0.0	0.0	0.0	0.0
24	12.2	8.0	3.7	0.0	0.0	24	0.0	0.0	0.0	0.0	0.0
25	7.5	4.2	1.6	0.0	0.0	25	0.0	0.0	0.0	0.0	0.0
26	3.4	1.7	0.5	0.0	0.0	26	0.0	0.0	0.0	0.0	0.0
27	1.2	0.5	0.1	0.0	0.0	27	0.0	0.0	0.0	0.0	0.0
28	0.2	0.1	0.0	0.0	0.0	28	0.0	0.0	0.0	0.0	0.0
29	0.0	0.0	0.0	0.0	0.0	29	0.0	0.0	0.0	0.0	0.0

*: Empirically observed.

Recall that the empirically observed S1 was 20, the empirically observed S2 was 16. The S1 statistic suggests that the true associations between X and Y were extremely weak (with the true beta suggesting an average T_{01} statistic closer to 0.1 than the reported empirical average of 2.8). Most of what was reported could have been due to random realizations from thousands of variables having been explored and only the best ones having been published. The S2 statistic is even more decisive, suggesting that the true

X-Y association may have been negative (the opposite of what was reported in papers) rather than zero.

On both measures, the subsequent performance of T_{01} given T_0 can be judged as very poor and not consistent with strong X-Y associations. However, we cannot reject a modest association with high statistical confidence. For example, on the S1 statistic, the probability of observing $S1 = 20$ was 2.9% under “modest” and 14.7% under the “null.” Thus, the likelihood ratio suggests a posterior of 84% that the true association is the null and 16% that it is modest.

One incorrect assumption in the above analysis is that the 29 predictors were independent. We know that a number (e.g., VIX and SVIX) were highly correlated. When SVIX predicted well, so would VIX. When SVIX fell below 2.0, so would VIX.

To check the intuition that this should not greatly influence our conclusion (that the null hypothesis fits better), consider a version in which only three estimators are repeated: one with a T_0 statistic of 2.06 (the fifth) is repeated 10 times; one with a T_0 statistic of 2.58 (the 15th) is repeated 9 times; and one with 3.60 (the 25th) is repeated 10 times. Thus, when any of them is above or below 2.0, so will 8 or 9 others. This is an extreme case of correlated estimators.

In this case, the empirical observation still suggests a zero or tiny association between X and Y.

	S1: # $T_{01} > T_0$						S2: # $T_{01} > 2.0$				
$E(T_0)$:	-0.25	0.0	0.25	1.23	2.45	$E(T_{01})$:	-0.24	0.0	0.2	1.5	2.8
0.0	1.4	2.1	3.1	11.1	32.7	0.0	26.4	30.9	35.5	55.0	76.6
9.0	4.2	5.2	6.2	11.0	13.5	9.0	8.1	7.5	7.3	4.9	1.9
10.0	9.4	12.0	14.5	25.3	31.1	10.0	50.4	49.5	47.8	36.8	20.9
19.0	27.7	28.4	29.2	25.5	12.9	19.0	14.9	12.0	9.5	3.2	0.6
20.0	14.6	15.4	15.5	13.7	6.9	20.0	0.1	0.0	0.0	0.0	0.0
29.0	42.6	36.9	31.5	13.5	2.9	29.0	0.0	0.0	0.0	0.0	0.0

F S3 Statistic

As for the $S3$ statistic, under the null hypothesis, we would observe an $T_0 - T_{01}$ improvement of about -0.13 (for the smallest $T_0 \approx 1.48$) to -0.35 (for the largest $T_0 \approx 5$). Under the modest association hypothesis, the improvement would be between -0.09 and $+0.13$. Under the strong association hypothesis, we would expect to see an improvement of about 0.1 to 0.3 . The typical standard deviation is about 0.4 . This suggests (again) that even the null hypothesis can barely fit a T_0 to T_{01} deterioration as pronounced as what we observed in the empirical data. And, again, the mild association cannot be rejected (with its implied T -statistic of about 1.5) even under the assumption of independent estimator draws.

G Potential Alternative Underlying Models

Note that similar procedures could be used either to estimate not only the best beta parameter, but also a second parameter related to a common decline of predictive ability. However, we felt that this would push the analysis too far — our paper is already twice the length of a typical paper. We thus leave such explorations to future papers.

IA.II Extended Variable Description

For a complete definition, the reader can also refer to the original paper.

Paper 1	Consumption Fluctuations and Expected Returns
Author(s)	Atanasov, Müller, and Priestley (2019)
Variable	pce : Aggregate consumption relative to its trend
Main Data Sources	FRED
Construction	From FRED, we obtain “Real personal consumption expenditures per capita: Services (A797RX0Q048SBEA)” and add it to “Real personal consumption expenditures per capita: Goods: Nondurable goods (A796RX0Q048SBEA),” and then take log of the sum to create the primary measure of consumption. On this resulting series $\{y_t\}$ we run the linear projection method of Hamilton (2018) to separate the trend from the cyclical component to obtain the predictor v_{t+24}, with 24 lookahead periods (parameter h) and the method’s usual 4 quarterly lags: $y_{t+24} = \beta_0 + \beta_1 y_t + \beta_2 y_{t-1} + \beta_3 y_{t-2} + \beta_4 y_{t-3} + v_{t+24}$ The OOS version of the variable uses data only up to the period for running the filter.

Paper 2	Improving the predictability of real economic activity and asset returns with forward variances inferred from option portfolios
Author(s)	Bakshi, Panayotov, and Skoulakis (2011)
Variable	impvar : Implied stock market variance
Main Data Sources	OptionMetrics, CRSP, Kenneth French's website
Construction	Taking the put and call options data on the S&P 500, we first drop all in-the-money options. We drop options with negative bids and negative bid-ask spreads. We calculate days to expiry and place each option into one of n buckets where $n = 1, 2, 3, 4$, which have an approximate corresponding number of days till expiry equal to τ_n, where $\tau_1 = 19, \tau_2 = 49, \tau_3 = 79, \tau_4 = 109$. Options are then allocated to one of these buckets based on how close their days to expiry are to each τ_n. For each month, we calculate the following for each bucket: $\omega(K) = -\frac{\frac{8}{\sqrt{14}} \cos\left(\arctan(1/\sqrt{7}) + \frac{\sqrt{7}}{2} \ln\left(\frac{K}{S_t}\right)\right)}{\sqrt{S_t K^{3/2}}}$ Where K is each option's respective strike price and S_t is the market index level observed at the end of month t. For calls and puts respectively, we take each option's bid-ask midpoint $C_t^{(n)}[K]$ and $P_t^{(n)}[K]$ for each month t and days till expiry τ_n. We then calculate the price of the claim: $H_t^{(t,n)} = e^{-r^* \tau_n} + \int_{K>S_t} \omega[K] C_t^{(n)}[K] dK + \int_{K<S_t} \omega[K] P_t^{(n)}[K] dK$ Where r is the spot one day risk-free rate obtained from Kenneth French's website observed at the end of the month. Each integral in the equation above is calculated at time t as follows: (for call options) after sorting the options by strike price K, for all $K_i < K_j$ we take the average value of $\omega(K_i) C_t^{(n)}[K_i]$ and $\omega(K_j) C_t^{(n)}[K_j]$ and multiply it by the difference of K_j and K_i to get $\omega[K] C_t^{(n)}[K] dK$. These are then all summed to get $\int_{K>S_t} \omega[K] C_t^{(n)}[K] dK$ (the procedure is equivalent for put options). We use forward variance between τ_1 and τ_2 as $f_t^{(2)} \equiv \ln H_t^{(t,1)} - \ln H_t^{(t,2)}$ calculated at time t as our measure.

Paper 3	The VIX, the Variance Premium, and Stock Market Volatility
Author(s)	Bekaert and Hoerova (2014)
Variable	vp : variance premium
Main Data Sources	Marie Hoerova
Construction	The monthly data series was directly acquired, without any modification, from Marie Hoerova.

Paper 4	Government investment and the stock market
Author(s)	Belo and Yu (2013)
Variable	govik : Public sector investment rate
Main Data Sources	FRED
Construction	The public sector investment rate (GIK_t) is equal to public sector investment (GI_t) divided by the stock of capital (K_t). From FRED, we get “Gross government investment (A782RC1Q027SBEA; Table 3.9.5 line 3 in NIPA accounts)” from which we subtract “Federal Government: National Defense Gross Investment (DGI; Table 3.9.5 line 19 in NIPA accounts)”. The annualized series are adjusted by their respective chain-type price index: “Price index, Government gross investment (Table 3.9.4 line 19 in NIPA accounts)” and “Price index, Federal defense investment (B788RG3Q086SBEA; Table 3.9.4 line 13 in NIPA accounts)”. From this procedure we obtain GI_t. Since the stock of capital K_t is not available at the quarterly level, we follow the authors’ steps and derive it from the process $GIK_t = \frac{I_t}{I_{t-1}} \frac{GIK_{t-1}}{(1 - \delta + GIK_{t-1})}$ where quarterly depreciation parameter $\delta = 2.6\%$. To derive K_t, the value of GIK_{t-1} is set to its “steady state” value GIK^* at 1947Q1. GIK^* is the fixed point derived from the above equation, which is equal to δ plus the average quarterly growth rate of GI_t. K_t is then estimated by depreciating the previous period’s capital stock by δ and adding current period public sector investment GI_t.

Paper 5	Expected Stock Returns and Variance Risk Premia
Author(s)	Bollerslev, Tauchen, and Zhou (2009)
Variable	vrp : Variance risk premium
Data Sources	Hao Zhou's website
Construction	The monthly data series was directly acquired, without any modification, from Hao Zhou's website.

Paper 6	Micro(structure) before macro? The predictive power of aggregate illiquidity for stock returns and economic activity
Author(s)	Chen, Eaton, and Paye (2018)
Variable	lzrt : log zero return days
Main Data Sources	CRSP
Construction	We use the <i>Zeros measure (ZEROS)</i> which is the fraction of zero return days out of the total trading days during the monthly period. For each stock-month, we calculate the fraction of zero days in that month. Then, we take average across NYSE stocks and finally take the log to give us the monthly series. Break-adjustment is introduced at two points in the log zeroes series. In the first case, a dummy variable D_1 is constructed which is equal to 1 for observations dated after June 1997, and 0 otherwise. From July 1997 until December 2000, for each observation we regress the series on D_1 in windows expanding from 1948 to the observation's date. The log zeroes series is then adjusted by subtracting the resulting slope coefficient on D_1 from each observation. We construct another dummy variable D_2 which is equal to 1 for observations dated after December 2000, and 0 otherwise. This variable is then similarly added to the expanding window regressions run above for post-1947 samples running past December 2000. The log zeroes observation is then adjusted by subtracting the resulting regression coefficients on both D_1 and D_2. This gives us our final break-adjusted variable.

Paper 7	Skewness in Expected Macro Fundamentals and the Predictability of Equity Returns: Evidence and Theory
Author(s)	Colacito, Ghysels, Meng, and Siwasarit (2016)
Variable	skew : Skewness of GDP growth forecasts
Main Data Sources	Federal Reserve Bank of Philadelphia, Blue Chip Economic Indicators
Construction	Individual GDP growth forecasts are collected from two different surveys: the Livingston Survey published by the Federal the Reserve Bank of Philadelphia, and a proprietary private survey provided by Blue Chip Economic Indicators. The former of which is used as the basis for GDP forecast skewness observed before December 1998, and the latter for observations then on. Following the authors' steps for the Livingston survey, the bi-annual real GDP growth forecasts are calculated as the log-difference of expected nominal GDP and CPI growth. GDP growth forecast is obtained by taking the log-ratio of the survey's "GDP_X6M" and "GDP_XBP" series. And inflation is obtained by taking the log-ratio of the survey's "CPI_6M" and "CPI_BP" series. The annual growth rate is then derived by multiplying the resulting log-difference by 4/3, and multiplied again by 100 to express it in percentage units. For the Blue Chip survey, expected real GDP growth rates are computed in June and December of each year by taking the log of 1 plus the current, one- and two-quarters ahead GDP growth rates' forecasts and dividing by 300. The result is then multiplied by 100 to express it in percentage units. For both series, we calculate $E[growth]$ (cross-sectional median of forecasts for each period), $V[growth]$ (difference between the 75th and 25th percentile of the cross-section of forecasts) and $S[growth]$ (sum of the 75th and 25th percentile of the cross-section of forecasts minus twice the median). $S[growth]$ is then multiplied by the square root of $V[growth]$ and the cubic root of the result is calculated to obtain skewness.

Paper 8	Credit conditions and stock return predictability
Author(s)	Chava, Gallmeyer, and Park (2015)
Variable	crdst : credit standards
Main Data Sources	FRED
Construction	This is the unmodified series “Net Percentage of Domestic Banks Tightening Standards for Commercial and Industrial Loans to Large and Middle-Market Firms (DRTSCILM)” obtained from FRED.

Paper 9	Time-Varying Risk Premiums and the Output Gap
Author(s)	Cooper and Priestley (2009)
Variable	ogap : Production-output gap
Main Data Sources	FRED
Construction	We obtain “Industrial Production: Total Index (INDPRO)” from FRED. The gap is estimated from regressing the following linear model: $y_t = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot t^2 + \nu_t$ where y_t is the log of industrial production, t is a time trend, and ν_t is the error term, which is the output gap. The OOS version of the variable uses data only up to the period for running the regression.

Paper 10	Striking oil: Another puzzle?
Author(s)	Driesprong, Jacobsen, and Maat (2008)
Variable	wtexas : Changes in oil price
Main Data Sources	Global Financial Data
Construction	Using the month-end close price on West Texas Intermediate (WTI), we calculate month-over-month percentage price change.

Paper 11	Accruals, cash flows, and aggregate stock returns
Author(s)	Hirshleifer, Hou, and Teoh (2008)
Variable 1	<i>accrul</i> : aggregate accruals and innovations in accruals
Variable 2	<i>cfacc</i> : aggregate cash flows & innovations in cash flows
Main Data Sources	Compustat and CRSP
Construction	Accruals are calculated from Compustat data as the change in non-cash current assets less the change in current liabilities excluding the change in short-term debt and the change in taxes payable, minus depreciation and amortization expense: $Accruals = (\Delta CA - \Delta Cash) - (\Delta CL - \Delta ST \text{ Debt} - \Delta TP) - Dep$ Cash flow is calculated simply as income before extraordinary items minus accruals. Following the authors, these measures are retained only for firms with December fiscal year ends, and firms with any missing data points are dropped. Each of these two measures are scaled by lagged total assets, and then value-weighted by market capitalization (from CRSP using previous year data). Both of these are summed up by year to obtain aggregate accruals and cashflows for each year.

Paper 12	Investor Sentiment Aligned: A Powerful Predictor of Stock Returns
Author(s)	Huang, Jiang, Tu, and Zhou (2014)
Variable	sntm : distilled investor sentiment index
Main Data Sources	Jeffrey Wurgler's
Construction	We obtain the five investor sentiment proxies available on Jeffrey Wurgler's webpage: close-end fund discount rate (CEFD), number of IPOs (NIPO), first-day returns of IPOs (RIPO), dividend premium (PDND), and equity share in new issues (EQTI). We take 12-month moving average of NIPO and 12-month weighted average of RIPO (using NIPO as weights). PDND and RIPO are lagged by additional 12 months. We take negative of PDND and CEFD. We then regress these variables on contemporaneous growth rate of industrial production, employment, durable consumption, non-durable consumption, service consumption, and a recession indicator. The residuals are standardized to have a zero mean and standard deviation of one and serve as the basis for the PLS. We estimate monthly aligned sentiment indexes using the Partial Least Squares method. The variable is calculated separately for predicting simple and log returns and separately for predicting returns at monthly, quarterly, and annual frequency. The OOS version of the variable uses data only up to the period for estimation of basis variables as well as PLS.

Paper 13	New Orders and Asset Prices
Author(s)	Jones and Tuzel (2012)
Variable	ndrbl : ratio of new orders to shipments of durable goods
Main Data Sources	FRED
Construction	From FRED we obtain “Manufacturers’ New Orders: Durable Goods (DGORDER)” and divide it by “Manufacturers’ Value of Shipments: Durable Goods (AMDMVS)”, taking the log of the result. Both of these series begin in 1992 and are extended until 1958 using the SIC based Manufacturers’ New Orders of Durable Goods and Value of Manufacturers’ Shipments for Durable Goods obtained from the US Census Bureau. We do not make the authors’ adjustment to FRED data where they subtract shipments of semiconductors from shipments as semiconductors are now included in both orders and shipments.

Paper 14	Average Skewness Matters
Author(s)	Jondeau, Zhang, and Zhu (2018)
Variable	skvw : Average stock skewness
Main Data Sources	CRSP, Kenneth French’s website
Construction	For each stock-month, using D_t trading days in that month, we calculate skewness for that stock as $Sk_{i,t} = \sum_{d=1}^{D_t} \tilde{r}_{i,d}^3 / \left(\sum_{d=1}^{D_t} \tilde{r}_{i,d}^2 \right)^{1.5}$ where $\tilde{r}_{i,d} = r_{i,d} - \bar{r}_{i,t}$ and $\bar{r}_{i,t}$ is the average calculated using all D_t days except last two following the authors. $Sk_{i,t}$ is then value-weighted using the market capitalization at the end of month t to give the monthly time-series $skvw_t$. We keep only common stocks traded on the NYSE, AMEX and NASDAQ, stocks that have prices above \$1 at the end of month t, which have at least 10 trading days of data in month t, and whose Amihud (2002) illiquidity is below the 99.9th percentile.

Paper 15	Tail Risk and Asset Prices
Author(s)	Kelly and Jiang (2014)
Variable	tail : time-varying tail risk of cross-sectional returns
Main Data Sources	CRSP, Kenneth French's website
Construction	Each month, using daily individual stock return data from CRSP and factor data from Kenneth French's website, we run a three factor (excess market return, SMB, HML) model and estimate the residuals $\epsilon_{i,d}$ for firm i at day d. We keep only firms with at least 10 return observations per month for each month t. Let the fifth percentile of these daily residuals be $(p_t^{0.05})$. Then our monthly measure (λ_t) is: $\lambda_t = \frac{1}{K_t} \sum_{k=1}^{K_t} \ln \frac{\epsilon_{i,d} I(\epsilon_{i,d} \leq p_t^{0.05})}{p_t^{0.05}},$ where $I(\cdot)$ is an indicator function and K_t is the total number of exceedences where $\epsilon_{i,d}$ is below the fifth percentile within month t.

Paper 16	Market Expectations in the Cross-Section of Present Values
Author(s)	Kelly and Pruitt (2013)
Variable	fbm : factor from cross-section of book-to-market ratios
Main Data Sources	CRSP, Ken French's website
Construction	We take data on 100 size and book-to-market portfolios from Ken French's website. We take the average market capitalization of each portfolio and multiply it by the number of firms to get its total market capitalization (ME): $ME_t = \text{Avg } ME \text{ of firms in portfolio}_t \times \text{Number of firms in portfolio}_t.$ Note that the data reported by Ken French is the ME at the beginning of the period of the stocks used in the portfolio and, therefore, this needs to be aligned properly to obtain month-end values. We get an estimate of book equity at fiscal year end $Y-1$ (BE_{Y-1}) by multiplying $ME_{June,Y}$ by a second estimate of book-to-market ratios $BM2_{June,Y}$ on Ken's website: $BE_{Y-1} = ME_{June,Y} \times BM2_{June,Y}.$ Note again that the BM_2 reported by Ken French is also at the beginning of the period of the stocks used in the portfolio and, therefore, this needs to be aligned properly to obtain month-end values. The final book-to-market ratio, BM_t, of each portfolio for months t from June of year Y to May of year $Y+1$ is calculated by dividing BE_{Y-1} by ME_t. This elaborate procedure is needed to get close to the book-to-market ratios used by the authors which used data no longer provided by Ken French. We estimate monthly factor from the cross-section of these 100 book-to-market ratios using the Partial Least Squares method. The variable is calculated separately for predicting simple and log returns and separately for predicting returns at monthly, quarterly, and annual frequency. The OOS version of the variable uses data only up to the period for PLS.

Paper 17	Investor attention, psychological anchors, and stock return predictability
Author(s)	Li and Yu (2011)
Variable 1	dtoy : Nearness to the Dow 52-week high
Variable 2	dtoat : Nearness to the Dow historical high
Main Data Sources	CRSP
Construction	We start by obtaining daily price data on the Dow Jones Industrial Average obtained from CRSP. Let p_d denote the level of the Dow at end of day d . Let $p_{52,d}$ and $p_{max,d}$ denote its 52-week high and historical high, respectively, at the end of day d . We compute nearness to the 52-week high as the ratio of the current Dow index and its 52-week high: dtoy = $p_d/p_{52,d}$. Similarly, nearness to the historical high is calculated as the ratio of the current Dow index and its historical high, dtoat = $p_d/p_{max,d}$. The final series takes the month end values of these daily series.

Paper 18	The “Fed Model” and the Predictability of Stock Returns
Author(s)	Maio (2012)
Variable	ygap : Stock-bond yield gap
Main Data Sources	Robert Shiller’s website
Construction	We calculate the difference between stock and bond returns, or yield gap (YG), using the log of the earnings-to-price (E/P) ratio minus the log of 1 plus the 10-year Treasury yield (y_{10}): $YG = \log(E/P) - \log(1 + y_{10})$ Note here that this is in contrast to the author’s code that calculates the yield gap as the log of the earnings-to-price ratio minus 10 times the log of 1 plus the 10-year Treasury yield: $YG' = \log(E/P) - 10 \times \log(1 + y_{10})$

Paper 19	Cross-sectional return dispersion and the equity premium
Author(s)	Maio (2015)
Variable	rdsp : Stock return dispersion
Main Data Sources	Kenneth French's website
Construction	We use monthly returns on 100 portfolios sorted by size and book-to-market ratio, available from Kenneth French's website. We then compute the cross-sectional standard deviation of their returns while ignoring missing values. The final variable is the three-month moving average of this cross-sectional standard deviation.

Paper 20	What is the expected return on the market?
Author(s)	Martin (2017)
Variable	rsvix : scaled risk-neutral vix
Main Data Sources	OptionMetrics
Construction	We retain daily observations on all put and call options data on the S&P 500 with fewer than 550 days and more than 7 days to expiration. We remove duplicated entries and delete options with zero bid. We then calculate the integral: $R_{ft}SVIX_{t,T}^2 = \frac{2}{(T-t)S_t^2} \int_0^\infty \Omega_{t,T}(K) dK,$ where $\Omega_{t,T}(K)$ is the put price ($P_{t,T}(K)$) for $K < S_t$ and the call price ($C_{t,T}(K)$) for $K > S_t$. $C_{t,T}(K)$ is the price (average of bid and ask) of the call option at time t for maturity T and strike K, and similarly for put prices. If $K_1, K_2, \dots, K_N$ are the strikes of observable options then defining $\Delta K_i = (K_{i+1} - K_{i-1})/2$, the integral is calculated as $\sum_{i=1}^N \Omega_{t,T}(K_i) \Delta K_i$. The integral is calculated each day for the expirations available on that date and then linearly interpolated for $T = 30, 60, 90, 180$, and 360 days.

Paper 21	End-of-the-year economic growth and time-varying expected returns
Author(s)	Möller and Rangvid (2014)
Variable 1	gpce : Year-end economic growth based on consumption
Variable 2	gip : Year-end economic growth based on industrial production
Main Data Sources	FRED
Construction	From FRED, we obtain quarterly personal consumption expenditure, which is estimated by taking the sum of “Real personal consumption expenditures per capita: Goods: Nondurable goods (A796RX0Q048SBEA)” and “Real personal consumption expenditures per capita: Services (A797RX0Q048SBEA).” We calculate quarterly percentage changes in these consumption expenditures. gpce is fourth quarter percentage change. gip is calculated by taking the percentage change in year-end “Industrial Production: Total Index (INDPRO)”, also obtained from FRED.

Paper 22	Forecasting the Equity Risk Premium: The Role of Technical Indicators
Author(s)	Neely, Rapach, Tu, and Zhou (2014)
Variable	tchi : Principal components derived from 14 technical indicators.
Main Data Sources	Yahoo Finance
Construction	The 14 technical indicators are based on moving averages (MA), price momentum (MOM), and volume (VOL) for S&P 500 index price and volume data obtained from Yahoo Finance (note: daily volume figures are summed up to obtained the monthly figure). The six binary MA price indicators are based on moving average crossovers which each equal to 1 if the short-term moving average (based on the 1, 2 or 3 month average) crosses above the long-term moving average (based on the 9 or 12 month average), and 0 otherwise. The two binary MOM price indicators are based on whether the current index price exceeds price observed 9 or 12 months ago (they equal 1 if so, and 0 otherwise). The VOL indicators are based on “on-balance” volume for a given period, which is the cumulative sum of the volume for the period if the index price exceeds the previous period’s, and the negative of the same sum if not. The final indicators are derived using the same MA and MOM rules. The final predictor is derived from the first principal component of these 14 series. The OOS version of the variable uses data only up to the period for PCA.

Paper 23	Housing, consumption, and asset pricing
Author(s)	Piazzesi, Schneider, and Tuzel (2007)
Variable	house : Relative share of housing in household consumption basket
Main Data Sources	FRED
Construction	From FRED, we obtain “Personal Consumption Expenditures: Nondurable Goods (PCNDA)”, “Personal Consumption Expenditures: Services (PCESVA)”, “Personal consumption expenditures: Services: Housing and utilities (DHUTRC1A027NBEA)” and “Personal consumption expenditures: Non-durable goods: Clothing and footwear (DCLORC1A027NBEA)”. The housing share is then the log of the following series: $\frac{(Non\ Durables + Services - Clothing\ and\ Footwear) - Housing\ Services}{Non\ Durables + Services - Clothing\ and\ Footwear}$

Paper 24	Average correlation and stock market returns
Author(s)	Pollett and Wilson (2010)
Variable	avgcor : Average correlation of daily stock returns
Main Data Sources	CRSP
Construction	We keep only the 500 largest firms by market capitalization (observed at the end of each month). The daily pairwise correlation of stock returns is then calculated for every pair of stocks in the sample. Each of these is then multiplied by the product of both stock’s weights relative to total sample market capitalization at the end of the same month, then summed to create the measure.

Paper 25	Short interest and aggregate stock returns
Author(s)	Rapach, Ringgenberg, and Zhou (2016)
Variable	shtint : Short interest
Main Data Sources	Compustat, CRSP
Construction	We keep only common stocks traded on the NYSE, AMEX and NASDAQ, with price above \$5. We merge this dataset with Compustat's Supplemental Short Interest File. Shares held short are divided by total shares outstanding for each stock, after adjusting for unit differences, giving us the share of short interest of each stock by month. We take the natural log of the monthly cross-sectional mean of this measure and then linearly detrend it to get the final measure. The OOS version of the variable uses data only up to the period for detrending the variable.

Paper 26	Disagreement and return predictability of stock portfolios
Author(s)	Yu (2011)
Variable	disag : Individual stock analyst forecast dispersions
Main Data Sources	IBES, CRSP
Construction	From IBES, we obtain the month-end standard deviation (σ_t) of long-term growth rate (LTG) for earnings-per-share (EPS). We calculate the cross-sectional value-weighted average of individual-stock disagreement as the final measure.

IA.III Bootstrap Method

Our bootstrap follows [mark1995aer](#) and [kilian1999](#) and imposes the null of no predictability for calculating the critical values. In other words, the data generating process (DGP) is assumed to be:

$$\begin{aligned} y_{t+1} &= \alpha + u_{1t+1} \\ x_{t+1} &= \mu + \rho \cdot x_t + u_{2t+1} \end{aligned} ,$$

where y is the excess return and the predictor variable is x . We estimate the parameters by OLS using the full sample of observations. We correct for the **stambaugh** bias in AR(1) estimates by $\hat{\rho}^* = \hat{\rho} + (1 + 3\hat{\rho})/T$, $\hat{\mu}^* = \bar{x}(1 - \hat{\rho}^*)$, and $\hat{u}_{2t+1}^* = x_{t+1} - \hat{\mu}^* - \hat{\rho}^* \cdot x_t$. We store the residuals for sampling and then generate 10,000 bootstrapped time series by drawing with replacement from the residuals. We generate 10% extra observations, set the initial observation—preceding the sample of data used to estimate the models—to be the sample mean, and then discard the first 10% of generated observations to match the sample time period. This bootstrap procedure not only preserves the autocorrelation structure of the predictor variable but also preserves the cross-correlation structure of the two residuals.

- For x variables that are growth rates (such as [gpce](#) and [gip](#)), the DGP is assumed to be iid instead of an AR(1) process.
- Variables that use macro-economic indicators (such as [pce](#), [ogap](#), and [cay](#)) are bootstrapped as follows. We bootstrap growth rate in (the log of) the macro variable using iid DGP. The bootstrapped time series of the macro variable is then computed applying these growth rates (and using the initial observation as the starting level). The bootstrapped x variable is then computed applying the filter as in the original data. The bootstrapped OOSCT version of the variable uses data only up to the period for running the filter.
- Variables that use PCA (such as [tchi](#)) are bootstrapped as follows. We bootstrap the original technical indicator dummy variables using iid DGP. The bootstrapped x variable is then computed applying PCA to these bootstrapped indicator variables. The bootstrapped OOSCT version of the variable uses data only up to the period for running the PCA.
- Variables that use detrending (such as [shtint](#)) are bootstrapped as follows. We bootstrap growth rate in (log of) the short interest using iid DGP. The bootstrapped

time series of the short interest is then computed applying these growth rates (and using the initial observation as the starting level). The bootstrapped x variable is then computed by linearly detrending the bootstrapped series. The bootstrapped OOSCT version of the variable uses data only up to the period for detrending.

- Variables that use PLS (such as [sntm](#) and [fbm](#)) are bootstrapped as follows. We bootstrap the basis variables using an AR(1) process for DGP.⁶ The bootstrapped variable is then generated by applying PLS to these bootstrapped basis time series. The bootstrapped OOSCT version of the variable uses data only up to the period for running the PLS.

⁶The basis variables for [sntm](#) change each time period. However, for bootstrap purposes, we use the latest vintage of these basis variables.

IA.IV Complete Performance of Investment Strategies

Table IA.1: Untilted \$1-Unscaled Investment Strategy

Vrbl Ppr	Name	Conditional R(V)			#Obs		Unconditional R (U)			Δ (= V – U)		
		Long	Short	L–S	Bull	Bear	L(Eq)	S(TB)	L – S	Mean	SR	
M	BH	vp	4.6	5.5	–0.9 ✗	119	145	8.5	1.5	7.1	–8.0	–0.57
M	BPS	impvar	3.5	9.2	–5.7 ✗	45	147	11.7	1.0	10.6	–16.3	–0.78
M	BTZ	vrp	7.2	2.8	4.4	95	169	8.5	1.5	7.1	–2.6	–0.13
M	CEP	lzrt	6.4	9.3	–2.9 ✗	263	649	11.9	3.8	8.1	–11.0	–0.43
M	CP	ogap	10.6	5.1	5.5	699	213	11.9	3.8	8.1	–2.6	–0.20
M	DJM	wtxas	10.6	5.1	5.5	600	312	11.9	3.8	8.1	–2.6	–0.16
M	HJTZ	sntm	9.5	5.6	4.0	309	117	12.0	3.1	8.9	–4.9	–0.28
M	JT	ndrbl	9.2	8.0	1.2	353	174	13.1	4.2	8.9	–7.7	–0.52
M	JZZ	skvw	5.5	10.2	–4.7 ✗	400	506	11.9	3.8	8.1	–12.7	–0.57
M	KJ	tail	8.7	7.0	1.7	538	368	11.9	3.8	8.1	–6.3	–0.35
M	KP	fbm	8.6	7.0	1.5	469	438	11.8	3.8	8.0	–6.5	–0.30
M	LY	dtoy	7.6	8.1	–0.6 ✗	366	546	11.9	3.8	8.1	–8.7	–0.45
M	LY	dtoat	5.7	9.9	–4.2 ✗	121	791	11.9	3.8	8.1	–12.3	–0.49
M	Maio ⁽¹³⁾	ygap	6.5	9.7	–3.2 ✗	203	382	11.9	4.4	7.6	–10.8	–0.43
M	Maio ⁽¹⁶⁾	rdsp	4.5	11.3	–6.8 ✗	188	716	12.0	3.8	8.2	–15.0	–0.61
M	Mrtn	rsvix	6.3	6.4	0.0	68	124	11.7	1.0	10.6	–10.7	–0.65
M	NRTZ	tchi	9.7	6.6	3.1	345	267	11.9	4.3	7.6	–4.5	–0.19
M	PW	avgcor	9.3	6.4	2.9	412	498	11.9	3.8	8.1	–5.2	–0.27
M	RRZ	shtint	10.9	2.8	8.0	237	111	11.5	2.2	9.2	–1.2	–0.07
M	Y	disag	9.8	1.7	8.1	227	14	10.3	1.2	9.2	–1.0	–0.11
M	BMRR	ntis	10.4	5.7	4.7	569	332	12.2	3.8	8.4	–3.7	–0.22
M	Cmpl	tby	6.2	9.4	–3.2 ✗	257	655	11.9	3.8	8.1	–11.3	–0.46
M	CSa	d/p	6.2	9.5	–3.4 ✗	179	733	11.9	3.8	8.1	–11.5	–0.45
M	CSb	d/y	6.3	9.3	–3.0 ✗	177	734	11.8	3.8	8.0	–11.1	–0.43
M	CSc	e/p	7.5	8.2	–0.7 ✗	298	614	11.9	3.8	8.1	–8.8	–0.38
M	CSD	d/e	4.8	10.9	–6.1 ✗	127	785	11.9	3.8	8.1	–14.2	–0.54
M	CSe	svar	7.6	8.0	–0.4 ✗	379	533	11.9	3.8	8.1	–8.5	–0.47
M	FFa	lty	7.0	8.7	–1.7 ✗	236	676	11.9	3.8	8.1	–9.8	–0.40
M	FFb	ltr	9.7	5.9	3.8	446	466	11.9	3.8	8.1	–4.3	–0.21
M	FFc	tms	8.3	7.4	0.8	403	509	11.9	3.8	8.1	–7.3	–0.33
M	FFd	dfy	7.5	8.2	–0.7 ✗	382	530	11.9	3.8	8.1	–8.8	–0.44
M	FFe	dfr	9.1	6.6	2.4	475	437	11.9	3.8	8.1	–5.7	–0.26
M	FS	infl	10.2	5.5	4.7	413	499	11.9	3.8	8.1	–3.4	–0.16
M	KS	b/m	5.9	9.8	–3.9 ✗	265	647	11.9	3.8	8.1	–12.0	–0.50
Q	AMP	pce	11.3	5.4	5.9	140	53	12.4	4.4	8.0	–2.1	–0.10
Q	BY	govik	4.5	11.9	–7.4 ✗	12	208	11.9	4.5	7.5	–14.9	–0.47
Q	CGP	crdst	8.3	2.0	6.2	48	39	8.8	1.5	7.4	–1.2	–0.04
Q	Crn	i/k	10.4	6.0	4.4	102	118	11.9	4.5	7.5	–3.1	–0.13
Q	LL	cay	9.5	7.0	2.5	127	73	12.2	4.4	7.8	–5.3	–0.24
D	CGMS	skew	12.0	4.9	7.1	34	15	12.3	4.7	7.6	–0.5	–0.02
D	HHT	accrul	8.5	8.1	0.4	15	22	13.5	3.2	10.3	–10.0	–0.40
D	HHT	cfacc	11.7	4.9	6.9	25	12	13.5	3.2	10.3	–3.5	–0.14
D	MR	gpce	11.2	5.8	5.4	32	23	12.3	4.6	7.8	–2.4	–0.12
D	MR	gip	11.7	5.0	6.7	52	24	12.7	3.9	8.8	–2.2	–0.12
D	PST	house	6.8	10.4	–3.6 ✗	22	51	13.2	4.0	9.2	–12.8	–0.41
D	BW	eqis	10.1	6.9	3.2	44	31	13.0	3.9	9.1	–5.9	–0.23
J	CGMS	skew	12.3	4.1	8.2	33	15	11.8	4.7	7.1	1.2	0.06 ✓
J	HHT	accrul	7.3	8.5	–1.2 ✗	15	21	12.7	3.1	9.6	–10.8	–0.42
J	HHT	cfacc	10.0	5.8	4.2	24	12	12.7	3.1	9.6	–5.4	–0.24
J	MR	gpce	12.0	4.6	7.5	32	22	12.0	4.6	7.4	0.1	0.01 ✓
J	MR	gip	10.7	5.8	4.9	52	23	12.6	3.9	8.6	–3.8	–0.20
J	PST	house	10.3	7.0	3.3	50	22	13.2	4.1	9.1	–5.9	–0.27
J	BW	eqis	8.9	8.0	0.9	43	31	12.9	4.0	9.0	–8.0	–0.29

Explanations: Variables and sample periods are defined in Table 1. Investment begins 20 years after the sample has started. ‘D’ means calendar year, ‘J’ is mid-year investing with reporting delay. The conditional timing strategy invests \$1 in the equity premium financed by the T-bill if the predictive coefficient is positive and V is above its median, or the coefficient is negative and V is below its median (both “bullish”); and the opposite otherwise (“bearish”). The unconditional strategy is always bullish and earns the equity premium. All measures are annualized, incl. the Sharpe Ratio (SR), and (none are strong enough to be) starred for statistical significance as in **lo2002SR**. For Jun-Jul, only the (two) positive Δ variables are shown.

Table IA.2: Untilted Z-Scaled Investment Strategy

Vrbl Ppr	Name	Conditional R(V)			#Obs		Unconditional R (U)			$\Delta (= V - U)$	
		Long	Short	L-S	Bull	Bear	L(Eq)	S(TB)	L - S	Mean	SR
M	BH	vp	10.4	2.5	7.9	119	145	11.9	0.9	11.0	-3.2 -0.55
M	BPS	impvar	2.0	5.5	-3.4 ✗	45	147	7.0	0.5	6.6	-10.0 -0.72
M	BTZ	vrp	5.8	4.4	1.4	95	169	9.3	0.9	8.4	-7.0 -0.08
M	CEP	lzrt	4.5	9.0	-4.5 ✗	263	649	10.7	2.8	7.9	-12.4 -0.28
M	CP	ogap	7.3	2.1	5.2	699	213	7.7	1.7	6.0	-0.8 -0.13
M	DJM	wtxas	8.3	3.4	4.9	600	312	8.6	3.1	5.5	-0.6 -0.01
M	HJTZ	sntm	8.6	4.7	3.9	309	117	10.0	3.3	6.7	-2.8 -0.20
M	JT	ndrbl	10.2	5.7	4.5	353	174	12.8	3.2	9.7	-5.2 -0.38
M	JZZ	skvw	2.6	6.2	-3.6 ✗	400	506	6.9	2.0	4.9	-8.5 -0.31
M	KJ	tail	6.1	5.0	1.1	538	368	8.6	2.5	6.1	-5.1 -0.28
M	KP	fbm	2.0	1.8	0.3	469	438	3.1	0.7	2.5	-2.2 -0.32
M	LY	dtoy	4.6	2.7	2.0	366	546	5.7	1.6	4.0	-2.0 -0.40
M	LY	dtoat	3.2	7.1	-4.0 ✗	121	791	8.1	2.2	5.9	-9.8 -0.47
M	Maio ⁽¹³⁾	ygap	7.6	9.5	-1.8 ✗	203	382	11.6	5.5	6.2	-8.0 -0.20
M	Maio ⁽¹⁶⁾	rdsp	1.7	3.5	-1.8 ✗	188	716	4.0	1.2	2.8	-4.6 -0.52
M	Mrtm	rsvix	3.0	2.9	0.1	68	124	5.2	0.7	4.5	-4.4 -0.67
M	NRTZ	tchi	4.1	4.2	-0.1 ✗	345	267	5.0	3.3	1.7	-1.9 -0.04
M	PW	avgor	7.8	3.3	4.4	412	498	8.8	2.3	6.5	-2.1 -0.15
M	RRZ	shtint	16.7	1.5	15.2	237	111	16.1	2.2	13.9	1.3 0.06 ✓
M	Y	disag	15.3	1.0	14.4	227	14	15.2	1.1	14.2	0.2 0.07 ✓
M	BMRR	ntis	5.9	2.6	3.3	569	332	6.4	2.2	4.2	-0.9 -0.16
M	Cmpl	tby	8.9	10.1	-1.1 ✗	257	655	12.4	6.6	5.8	-7.0 -0.15
M	CSa	d/p	6.2	10.8	-4.6 ✗	179	733	12.9	4.1	8.8	-13.4 -0.34
M	CSb	d/y	6.3	10.6	-4.3 ✗	177	734	12.7	4.2	8.6	-12.9 -0.31
M	CSc	e/p	7.5	7.4	0.1	298	614	11.1	3.7	7.4	-7.3 -0.21
M	CSd	d/e	5.2	9.4	-4.3 ✗	127	785	11.2	3.3	7.9	-12.2 -0.43
M	CSe	svar	1.0	1.1	0.0	379	533	1.5	0.7	0.8	-0.8 -0.46
M	FFa	lty	10.7	13.6	-2.9 ✗	236	676	16.0	8.3	7.7	-10.6 -0.18
M	FFb	ltr	12.1	4.3	7.8	446	466	11.7	4.7	7.0	0.8 0.03 ✓
M	FFc	tms	8.2	5.4	2.8	403	509	9.4	4.2	5.2	-2.4 -0.10
M	FFd	dfy	4.1	3.8	0.4	382	530	6.2	1.7	4.5	-4.1 -0.45
M	FFe	dfr	5.4	3.6	1.8	475	437	6.3	2.7	3.7	-1.8 -0.04
M	FS	infl	3.5	2.0	1.5	413	499	3.4	2.1	1.3	0.2 0.01 ✓
M	KS	b/m	5.8	9.0	-3.2 ✗	265	647	10.9	3.8	7.1	-10.2 -0.38
Q	AMP	pce	12.6	4.0	8.6	140	53	12.7	3.9	8.9	-0.3 -0.02
Q	BY	govik	4.2	8.8	-4.6 ✗	12	208	8.8	4.3	4.5	-9.1 -0.27
Q	CGP	crdst	4.5	-1.2	5.7	48	39	2.1	1.3	0.8	5.0 0.11 ✓
Q	Crn	i/k	8.2	5.2	3.0	102	118	9.3	4.1	5.3	-2.3 -0.07
Q	LL	cay	16.6	10.5	6.1	127	73	21.2	5.8	15.4	-9.3 -0.22
D	CGMS	skew	9.5	3.8	5.8	34	15	9.9	3.4	6.5	-0.8 -0.08
D	HHT	accr	3.8	0.3	3.5	15	22	2.4	1.7	0.7	2.8 0.09 ✓
D	HHT	cfacc	7.0	0.6	6.4	25	12	5.8	1.8	4.1	2.3 0.18 ✓
D	MR	gpce	5.9	1.9	4.0	32	23	5.7	2.1	3.6	0.4 0.03 ✓
D	MR	gip	4.3	1.0	3.3	52	24	4.4	1.0	3.4	-0.1 -0.04
D	PST	house	3.0	3.4	-0.5 ✗	22	51	4.9	1.5	3.5	-3.9 -0.35
D	BW	eqis	5.4	2.7	2.7	44	31	5.7	2.4	3.3	-0.6 -0.05
J	CGMS	skew	9.7	3.4	6.3	33	15	9.7	3.4	6.3	-0.1 -0.01
J	HHT	accr	3.1	0.0	3.1	15	21	1.6	1.6	0.1	3.1 0.11 ✓
J	HHT	cfacc	6.5	2.9	3.6	24	12	7.6	1.9	5.7	-2.1 -0.13
J	MR	gpce	6.7	1.1	5.6	32	22	5.7	2.1	3.7	1.9 0.15 ✓
J	MR	gip	3.6	1.2	2.4	52	23	3.9	0.9	3.0	-0.6 -0.11
J	PST	house	3.4	3.3	0.2	50	22	5.2	1.5	3.8	-3.6 -0.29
J	BW	eqis	4.8	4.8	0.0	43	31	7.2	2.4	4.8	-4.8 -0.21

Explanations: See Table IA.1. Here, the investment is not \$1, but \$Z (V realization minus prevailing median, divided by prevailing standard deviation), i.e., the signal strength influences not just the direction but also the amount of the investment. The unconditional strategy is always bullish and invests |\$Z|.

Table IA.3: Long-Equity-tilted \$1-Scaled Investment Strategy

Vrbl Ppr	Name	Conditional R(V)			#Obs		Unconditional R (U)			Δ (= V − U)		
		Long	Short	L-S	Bull	Bear	L(Eq)	S(TB)	L − S	Mean	SR	
M	BH	vp	6.9	3.2	3.7	194	70	8.5	1.5	7.1	−3.4	−0.38
M	BPS	impvar	7.5	5.2	2.4	104	88	11.7	1.0	10.6	−8.2	−0.59
M	BTZ	vrp	5.9	4.1	1.8	155	109	8.5	1.5	7.1	−5.3	−0.30
M	CEP	lzrt	9.1	6.5	2.6	520	392	11.9	3.8	8.1	−5.5	−0.28
M	CP	ogap	11.7	4.0	7.8	868	44	11.9	3.8	8.1	−0.4	−0.04
M	DJM	wtxas	10.5	5.2	5.3	607	305	11.9	3.8	8.1	−2.8	−0.17
M	HJTZ	sntm	11.6	3.5	8.1	408	18	12.0	3.1	8.9	−0.8	−0.09
M	JT	ndrbl	11.6	5.7	6.0	463	64	13.1	4.2	8.9	−3.0	−0.33
M	JZZ	skvw	8.1	7.5	0.6	678	228	11.9	3.8	8.1	−7.5	−0.44
M	KJ	tail	9.9	5.7	4.2	718	188	11.9	3.8	8.1	−3.8	−0.31
M	KP	fbm	10.0	5.6	4.4	785	122	11.8	3.8	8.0	−3.6	−0.32
M	LY	dtoy	10.3	5.4	4.9	643	269	11.9	3.8	8.1	−3.2	−0.28
M	LY	dtoat	8.1	7.6	0.6	475	437	11.9	3.8	8.1	−7.6	−0.45
M	Maio ⁽¹³⁾	ygap	9.5	6.8	2.7	321	264	11.9	4.4	7.6	−4.8	−0.22
M	Maio ⁽¹⁶⁾	rdsp	7.2	8.6	−1.4 ✕	446	458	12.0	3.8	8.2	−9.6	−0.49
M	Mrtm	rsvix	9.3	3.4	5.9	123	69	11.7	1.0	10.6	−4.8	−0.46
M	NRTZ	tchi	12.0	4.2	7.8	481	131	11.9	4.3	7.6	0.3	0.01 ✓
M	PW	avgcor	10.8	4.9	5.9	643	267	11.9	3.8	8.1	−2.2	−0.17
M	RRZ	shtint	11.0	2.7	8.3	286	62	11.5	2.2	9.2	−1.0	−0.09
M	Y	disag	10.7	0.8	10.0	240	1	10.3	1.2	9.2	0.8	0.22 ✓
M	BMRR	ntis	11.8	4.2	7.6	787	114	12.2	3.8	8.4	−0.8	−0.08
M	Cmpl	tby	9.2	6.5	2.7	446	466	11.9	3.8	8.1	−5.4	−0.26
M	CSa	d/p	7.2	8.5	−1.3 ✕	286	626	11.9	3.8	8.1	−9.4	−0.40
M	CSb	d/y	7.2	8.4	−1.3 ✕	286	625	11.8	3.8	8.0	−9.3	−0.39
M	CSc	e/p	9.4	6.2	3.2	497	415	11.9	3.8	8.1	−4.9	−0.25
M	CSD	d/e	7.2	8.5	−1.4 ✕	344	568	11.9	3.8	8.1	−9.5	−0.43
M	CSe	svar	9.5	6.1	3.4	627	285	11.9	3.8	8.1	−4.7	−0.38
M	FFa	lty	9.4	6.2	3.2	480	432	11.9	3.8	8.1	−4.9	−0.24
M	FFb	ltr	11.0	4.6	6.4	594	318	11.9	3.8	8.1	−1.7	−0.10
M	FFc	tms	10.8	4.9	5.9	612	300	11.9	3.8	8.1	−2.2	−0.13
M	FFd	dfy	8.7	7.0	1.8	657	255	11.9	3.8	8.1	−6.3	−0.47
M	FFe	dfr	10.7	4.9	5.8	657	255	11.9	3.8	8.1	−2.3	−0.13
M	FS	infl	11.5	4.2	7.3	708	204	11.9	3.8	8.1	−0.8	−0.06
M	KS	b/m	7.8	7.9	−0.1 ✕	408	504	11.9	3.8	8.1	−8.2	−0.38
Q	AMP	pce	12.7	4.0	8.8	179	14	12.4	4.4	8.0	0.8	0.08 ✓
Q	BY	govik	8.2	8.2	−0.1 ✕	90	130	11.9	4.5	7.5	−7.5	−0.29
Q	CGP	crdst	9.9	0.4	9.5	68	19	8.8	1.5	7.4	2.1	0.10 ✓
Q	Crn	i/k	11.7	4.6	7.1	161	59	11.9	4.5	7.5	−0.4	−0.02
Q	LL	cay	10.6	6.0	4.6	150	50	12.2	4.4	7.8	−3.2	−0.17
D	CGMS	skew	12.1	4.8	7.3	45	4	12.3	4.7	7.6	−0.3	−0.02
D	HHT	accrul	13.2	3.4	9.8	31	6	13.5	3.2	10.3	−0.6	−0.03
D	HHT	cfacc	14.7	1.9	12.8	32	5	13.5	3.2	10.3	2.5	0.18 ✓
D	MR	gpce	12.2	4.7	7.5	46	9	12.3	4.6	7.8	−0.2	−0.02
D	MR	gip	13.0	3.6	9.5	74	2	12.7	3.9	8.8	0.6	0.13 ✓
D	PST	house	11.1	6.1	5.0	53	20	13.2	4.0	9.2	−4.2	−0.24
D	BW	eqis	13.6	3.4	10.3	64	11	13.0	3.9	9.1	1.2	0.10 ✓
J	CGMS	skew	12.2	4.2	8.0	44	4	11.8	4.7	7.1	0.9	0.12 ✓
J	HHT	accrul	12.6	3.2	9.4	31	5	12.7	3.1	9.6	−0.2	−0.01
J	HHT	cfacc	13.2	2.7	10.6	31	5	12.7	3.1	9.6	1.0	0.08 ✓
J	MR	gpce	12.4	4.2	8.2	46	8	12.0	4.6	7.4	0.8	0.06 ✓
J	MR	gip	12.6	4.0	8.6	73	2	12.6	3.9	8.6	0.0	0.00
J	PST	house	12.0	5.3	6.7	68	4	13.2	4.1	9.1	−2.5	−0.20
J	BW	eqis	11.6	5.4	6.2	63	11	12.9	4.0	9.0	−2.8	−0.14

Explanations: See Table IA.1. However, here, the decision cutoff is not the median but the 25th percentile (a more bearish cutoff). Thus, the strategy typically invests more optimistically in equities. (Please note that Y's positive performance is based on only one month, in which it would have been bearish.)

Table IA.4: Long-Equity-tilted Z-Scaled Investment Strategy

Vrbl Ppr	Name	Conditional R(V)			#Obs		Unconditional R (U)			Δ (= V - U)		
		Long	Short	L-S	Bull	Bear	L(Eq)	S(TB)	L - S	Mean	SR	
M	BH	vp	11.7	1.3	10.4	194	70	12.1	1.0	11.1	-0.8	-0.43
M	BPS	impvar	4.4	2.0	2.4	104	88	5.9	0.5	5.4	-3.0	-0.57
M	BTZ	vrp	6.8	4.0	2.8	155	109	10.0	0.8	9.2	-6.5	-0.08
M	CEP	lzrt	7.2	6.4	0.8	520	392	10.9	2.7	8.2	-7.4	-0.21
M	CP	ogap	12.9	3.1	9.8	868	44	13.0	3.0	10.0	-0.2	-0.13
M	DJM	wtxas	8.5	3.1	5.4	607	305	8.6	3.1	5.5	-0.1	0.00
M	HJTZ	sntm	18.1	6.2	11.9	408	18	18.4	5.8	12.6	-0.7	-0.16
M	JT	ndrbl	16.1	5.5	10.5	463	64	17.1	4.6	12.5	-2.0	-0.24
M	JZZ	skvw	5.0	4.2	0.8	678	228	6.6	2.7	3.9	-3.1	-0.16
M	KJ	tail	13.6	4.7	8.9	718	188	13.8	4.4	9.4	-0.5	-0.06
M	KP	fbm	6.1	2.0	4.1	785	122	6.6	1.6	5.0	-0.9	-0.25
M	LY	dtoy	6.4	2.0	4.4	643	269	6.5	1.9	4.7	-0.2	-0.23
M	LY	dtoat	3.7	2.5	1.2	475	437	5.1	1.1	4.0	-2.8	-0.39
M	Maio ⁽¹³⁾	ygap	10.7	8.5	2.3	321	264	13.0	6.2	6.8	-4.5	-0.14
M	Maio ⁽¹⁶⁾	rdsp	1.9	1.6	0.4	446	458	2.7	0.8	1.8	-1.5	-0.40
M	Mrtm	rsvix	4.5	1.0	3.5	123	69	5.0	0.5	4.5	-1.1	-0.54
M	NRTZ	tchi	16.1	5.2	10.9	481	131	15.7	5.5	10.2	0.7	0.04 ✓
M	PW	avgcor	12.0	2.9	9.1	643	267	12.3	2.7	9.6	-0.5	-0.08
M	RRZ	shtint	22.8	1.8	21.1	286	62	21.9	2.7	19.1	2.0	0.13 ✓
M	Y	disag	20.7	1.5	19.2	240	1	20.7	1.5	19.2	0.1	0.22 ✓
M	BMRR	ntis	8.9	3.9	5.0	446	466	8.6	4.2	4.3	0.7	0.02 ✓
M	Cmpl	tby	8.8	2.8	6.1	787	114	8.9	2.8	6.1	-0.1	-0.02
M	CSa	d/p	7.2	6.4	0.8	286	626	10.2	3.4	6.8	-6.0	-0.23
M	CSb	d/y	7.4	6.3	1.1	286	625	10.2	3.4	6.8	-5.7	-0.21
M	CSc	e/p	10.8	5.8	5.0	497	415	12.6	4.1	8.5	-3.5	-0.13
M	CSd	d/e	5.0	5.2	-0.2 ✗	344	568	8.0	2.3	5.7	-5.9	-0.32
M	CSe	svar	1.6	0.8	0.7	627	285	1.7	0.8	0.9	-0.2	-0.32
M	FFa	lty	13.2	7.9	5.3	480	432	15.0	6.1	9.0	-3.7	-0.09
M	FFb	ltr	14.9	3.9	11.0	594	318	13.9	4.9	9.0	2.0	0.09 ✓
M	FFc	tms	12.9	3.9	8.9	612	300	12.5	4.3	8.3	0.7	0.05 ✓
M	FFd	dfy	5.7	2.4	3.3	657	255	6.1	2.0	4.1	-0.8	-0.22
M	FFe	dfr	8.2	3.2	5.0	657	255	8.5	3.0	5.5	-0.5	-0.01
M	FS	infl	7.3	1.2	6.1	708	204	6.5	1.9	4.6	1.5	0.10 ✓
M	KS	b/m	7.6	5.4	2.2	408	504	9.2	3.7	5.5	-3.3	-0.22
Q	AMP	pce	21.6	6.3	15.3	179	14	21.6	6.3	15.3	-0.1	-0.03
Q	BY	govik	2.5	3.1	-0.6 ✗	90	130	3.7	1.9	1.8	-2.4	-0.11
Q	CGP	crdstd	11.6	-0.1	11.7	68	19	10.0	1.5	8.5	3.3	0.11 ✓
Q	Crn	i/k	13.3	3.8	9.5	161	59	13.6	3.5	10.1	-0.6	-0.04
Q	LL	cay	19.6	9.4	10.2	150	50	23.3	5.6	17.7	-7.5	-0.23
D	CGMS	skew	13.0	4.4	8.6	45	4	13.2	4.2	9.0	-0.4	-0.08
D	HHT	accrul	8.8	0.1	8.7	31	6	6.5	2.4	4.1	4.6	0.20 ✓
D	HHT	cfacc	12.2	1.8	10.4	32	5	11.3	2.8	8.5	2.0	0.19 ✓
D	MR	gpce	9.8	2.2	7.7	46	9	9.5	2.6	6.9	0.8	0.12 ✓
D	MR	gip	8.5	1.9	6.6	74	2	8.5	1.9	6.6	0.0	0.16* ✓
D	PST	house	4.4	1.4	3.0	53	20	4.9	0.8	4.1	-1.1	-0.31
D	BW	eqis	11.2	3.1	8.0	64	11	11.0	3.3	7.8	0.3	0.05 ✓
J	CGMS	skew	13.2	4.2	9.0	44	4	13.2	4.2	9.0	0.0	-0.01
J	HHT	accrul	7.8	-0.2	8.0	31	5	5.3	2.2	3.1	4.9	0.22* ✓
J	HHT	cfacc	11.4	3.8	7.6	31	5	12.3	2.9	9.4	-1.8	-0.13
J	MR	gpce	11.1	1.8	9.3	46	8	10.3	2.5	7.8	1.4	0.21* ✓
J	MR	gip	7.6	1.9	5.6	73	2	7.6	1.9	5.7	-0.1	-0.07
J	PST	house	10.8	4.4	6.4	68	4	11.0	4.2	6.8	-0.3	-0.15
J	BW	eqis	9.6	4.2	5.3	63	11	10.5	3.3	7.3	-2.0	-0.16

Explanations: See Tables IA.2 and IA.3.

IA.V Further Results on Martin (QJE 2017)

This appendix describes how **Mrtn** finds and interprets the evidence. We stress that **Mrtn** did *not* cherry pick its presentation: Table II shows that **rsvix** has predictive in-sample power *only* on the 6-month horizon, but not on monthly, bi-monthly, quarterly, and annual horizons. We now deviate from our standard presentation scheme, because we cannot focus on monthly, quarterly, or annual regressions as we could for other papers. (They merely confirm the insignificant performance in **Mrtn** in outperforming the null hypothesis of no predictive ability.)

[Insert Table **IA.5** here: **Appendix: Sensitivity of MA**]

Panel A of Table **IA.5** walks through how our specification differs from Martin's. It matters little whether one uses daily or monthly frequency observations, and/or log or simple equity premium (though results may drop in or out of statistical significance), and/or a sample ending in 2012 or 2020. The real differences are elsewhere.

First, the coefficients are sensitive whether one uses overlapping or non-overlapping observations. With non-overlapping observations, all in-sample coefficients including those on the 6-month horizon turn insignificant in the original sample. In the extended sample, the quarterly but not the semi-annual turn significant.

Second, there is an important difference in how we calculate OOSCT R^2 for all our variables: Our predictions always take variables, run predictive regressions on the prevailing history of the equity premium, and then predict the future equity-premium with the resulting regression. In contrast, under Martin's own theory, Martin's variable is already the correct meaningful expected return. Thus he uses **rsvix** directly as the forecast (equivalent if we set our predictive regression intercept to 0 and the predictive slope to 1). His direct prediction is *always* better than our regression-intermediated prediction. Yet, even his OOSCT performance is *never* statistically significant (on the **DieboldMariano1995** statistic, with Newey-West adjustments for overlaps, where appropriate).

Panel B further investigates the performance on his preferred semi-annual horizon. The **rsvix** variable has no predictive power when used to predict semi-annual equity premia in December and June for first and second halves of calendar years, respectively. It only has good performance when used to predict in March (i.e., at the end of Q1 for equity premia in Q2 and Q3) and September (at the end of Q3 for Q4 and Q1).

The test intent of **MA** is intrinsically different from our own. Our premise is that a variable has to have a coefficient different from 0. In a loose sense, we ask variables to prove that they are better than nothing. The null hypothesis receives 95%, the variable has to beat the 5%. Martin's premise is that **rsvix** has a coefficient of 1. In a loose sense, he asks skeptics to prove that it is not 1. The variable receives the 95%, the alternative gets the 5%.

This makes sense from the perspective that if the theory is correct, it is difficult to reject. However, it also means that its predictions are empirically weak. That is, most of the regression coefficients can similarly not reject (atheoretical) hypotheses that the coefficient on **rsvix** is -1 , 0 , or $+3$, just as they cannot reject that it is 1 . With a good prior in favor of the theory, most of the regression evidence is not contradictory. There is one ironical exception not shown in the table. The theory *can* be rejected when **rsvix** had the best performance against the null hypothesis that it is 0 . Reverting back to simple rates of return, and using Mar/Sep semi-annual prediction, the hypothesis that the coefficient is 1.0 can be rejected with a T-statistic of 2.3 ending in 2012 and 2.6 ending in 2021. Again, with a strong prior, this evidence can be ascribed to sampling.

Figure 15 in the main text plots the performance both IS and OOSCT based on the only specification with good performance, i.e., the Mar/Sep semiannual predictions. The figure shows that **rsvix**' positive OOS performance was based on three months: 2009/03 (predicting to 2009/09), 2011/09 (to 2012/03) and 2020/03 (to 2020/09).⁷ The 2009/03 performance merely undoes the preceding poor 2008/09 prediction. However, the two other observations are principally responsible. Removing either the 2011/09 or 2020/03 prediction both drops the OOS performance into the red and removes the statistical significance of the IS regression. As noted in Section C.2, we leave it to the reader how to think about variables whose performance is based on outliers. There is no obvious answer.

We note that the estimated empirical performance of **rsvix** in the post-publication sample was exceptionally good — in fact, too good. The estimated coefficient was about 10 , much higher (also statistically significantly higher) than the 1.0 suggested by the theory.

Evaluation: With respect to the null hypothesis that the coefficient is zero, we dismiss **rsvix** as useful general predictors of equity premia, primarily based on the sensitivity and specificity in performance. Stock-market volatility fails to predict equity premia on monthly, bimonthly, quarterly, annual, and standard semi-annual frequencies. Even on

⁷The IS performance further benefited from its prediction of the 1999/09 (to 2001/03).

its only good predictive specification (the semi-annual frequency with offset), its good (but not statistically significant) performance was due to two specific outliers.

Table IA.5: Appendix: Sensitivity of MA**Panel A:** Various Prediction Horizons, Log vs Simple Premia, and Sample Ends

F O E			3 month			6 month				12 month				
			Coef	NW-T	OOS R^2		Coef	NW-T	OOS R^2		Coef	NW-T	OOS R^2	
					Reg	Con			Reg	Con			Reg	Con
Reported in Original QJE Paper														
D	Y	S	1.01	0.62	—	1.5	2.10	2.46	—	4.9	1.67	1.32	—	4.7
Sample ending 2012														
D	Y	S	1.06	0.74	−14.8	1.9	2.12	2.19	4.8	5.4	1.71	1.59	−10.1	6.1
M	Y	S	1.24	0.66	−21.2	2.3	2.34	2.19	5.5	6.0	1.71	1.59	−8.4	6.7
Ntv		S	2.62	1.44	−10.9	4.1	1.23	0.96	9.4	4.0	1.68	1.46	−21.2	6.8
Ntv		L	2.29	1.25	−11.6	3.3	1.05	0.87	−11.2	2.6	1.50	1.28	−22.9	5.2
Sample ending 2021														
D	Y	S	1.46	1.12	−7.8	0.2	2.00	2.31	3.6	0.6	1.59	1.95	−6.9	−3.6
M	Y	S	1.48	0.91	−12.4	−0.2	2.09	2.22	3.4	0.2	1.52	1.86	−6.3	−4.5
Ntv		S	2.95	1.91	−1.1	2.2	1.15	0.95	−6.3	−3.7	0.93	0.97	−15.9	−6.1
Ntv		L	2.61	1.69	−2.8	2.9	0.93	0.79	−8.0	−1.3	0.70	0.73	−18.3	−1.2

Panel B: Semi-Annual Predictions: Alignments and Sample Ends

	Dec/Jun				Mar/Sep			
	Coef	NW-T	OOS R^2		Coef	NW-T	OOS R^2	
			Reg	Con			Reg	Con
−2012	1.05	0.87	−11.2	2.6	3.47	2.64	4.5	6.8
−2021	0.93	0.79	−8.0	−1.3	3.15	2.95	5.3	5.7

Explanations: This table shows the effects of various specification choices. The predictor *rsvix* is always measured on the last day of the preceding time-period. We use Newey-West instead of Hansen-Hodrick T-statistics. F means data frequency, either daily (D) or monthly (M). O means overlapping observation. Ntv means use of native frequency non-overlapping natural intervals. E means type of equity premium, either simple (S) or Log (L). The “reg” out-of-sample performance uses a prevailing regression with *rsvix* to predict equity premia. The “con” uses *rsvix* without a linear transformation. (This is possibly only because the natural units of *rsvix* are equity premia.) None of the OOSCT R^2 values are statistically significant on the Diebold-Mariano test (with Newey-West adjustment for overlaps, where appropriate). In Panel B, the predictor is measured either on Dec 31 and Jun 30 on the left or on Mar 31 and Sep 30 on the right (or earlier if these were holidays). Panel B predicts log equity premia and shows Newey-West T-statistics. Again, none are statistically significant.

Interpretation: The 3-month and 12-month results are all insignificant. The only good *rsvix* predictions occurred semi-annually on Mar/Sep ends. They disappear on Dec/Jun ends.

IA.VI Further Tables

In the paper, we report predictions at ‘natural’ frequencies: next month prediction for monthly variables, next quarter prediction for quarterly variables, and next year prediction for annual variables. In this internet appendix, we report predictions for all frequencies, wherever possible, for all variables.

Notes on Tables IA.6-IA.9

Variables are defined in the paper. Prediction frequencies in Tables IA.6 to IA.9 are monthly, quarterly, annual (Jan to Dec), and annual (Jul to Jun), respectively. Table IA.6 considers only monthly variables and Table IA.7 considers only monthly and quarterly variables. All regressions are for non-overlapping observations. Therefore, for example, we forecast next quarter returns using month-end data in March, June, September, and December for monthly variables. Predictions in Table IA.9 are done using December-end data for the 12-months starting in July of next year. The independent variable is always the Z-normalized (which matters for the coefficient magnitude but not the t -statistic) and without further controls. The sample dates as well as the OOS starting dates are given in each panel of each table. The IS sample period always begins when data is available, but no earlier than 1926. The OOS sample period usually begins 20 years after the IS period. The IS statistics report the R^2 (in percent; not adjusted), the OLS beta together with its t -statistic in parenthesis (computed using Newey-West correction with one lag). The OOS statistics are the OOS- R^2 (in percent) and the MSE-F statistic of **ClarkMcCracken2001**. The significance is denoted of * (90%), ** (95%), and *** (99%). The last column reports the utility difference for investment strategy based on the variable and based on unconditional forecast for a quadratic risk-averse investor with risk aversion parameter 5 and investment limits of 0 and 1.5 times equity, just as in **campbell2008predicting**. We calculate the the statistical significance of the utility difference following the procedure in footnote 16 of **demiguel** and report the resulting t -statistic in the parenthesis besides the annualized utility difference.

Notes on Tables IA.10-IA.13

Variables are defined in the paper. Prediction frequencies in Tables IA.10 to IA.13 are monthly, quarterly, annual (Jan to Dec), and annual (Jul to Jun), respectively. Table IA.10 considers only monthly variables and Table IA.11 considers only monthly

and quarterly variables. The conditional timing strategy invests \$1 or \$Z in the equity premium financed by the T-bill if the predictive coefficient is positive and the variable is above its *median*, or if the predictive coefficient is negative and the variable is below its median (both “bullish”); and the opposite otherwise (“bearish”). The unconditional strategy is always bullish and earns the equity premium. The table reports the long-short returns for the model and the unconditional. The last two column report the differences in the two long-short strategies. All measures are annualized, including the Sharpe Ratio (SR), and (none are strong enough to be) starred for statistical significance as in **lo2002SR**.

Notes on Tables **IA.14-IA.17**

Variables are defined in the paper. Prediction frequencies in Tables **IA.14** to **IA.17** are monthly, quarterly, annual (Jan to Dec), and annual (Jul to Jun), respectively. Table **IA.14** considers only monthly variables and Table **IA.15** considers only monthly and quarterly variables. The conditional timing strategy invests \$1 or \$Z in the equity premium financed by the T-bill if the predictive coefficient is positive and the variable is above its *25th percentile*, or if the predictive coefficient is negative and the variable is below its 25th percentile (both “bullish”); and the opposite otherwise (“bearish”). The unconditional strategy is always bullish and earns the equity premium. The table reports the long-short returns for the model and the unconditional. The last two column report the differences in the two long-short strategies. All measures are annualized, including the Sharpe Ratio (SR), and starred for statistical significance as in **lo2002SR**.

Table IA.6: Predicting Next Month Returns

Variable s		Sample		OOS Beg	R^2	IS Slope	OOS		OOS-CT		Util Difference		
		Beg	End				R^2	MSE-F	R^2	MSE-F	$U_{\text{cond}} - U_{\text{uncond}}$	$U_{\text{cond}} - U_{100\%}$	
Panel A: Predicting log returns													
BH	vp	199001	202112	200001	1.47	0.51 (1.44)	-0.42	-1.11	-0.23	-0.60	-2.50 (-1.75)	-4.67 (-2.98)	
BPS	impvar	199601	202112	200601	0.09	0.13 (0.35)	-3.32	-6.13	-3.19	-5.90	-2.19 (-1.38)	-5.41 (-2.25)	
BTZ	vrp	199001	202112	200001	0.06	0.10 (0.17)	-8.61	-20.85	0.21	0.54	0.85 (0.39)	-1.32 (-0.61)	
CEP	lzrt	192601	202112	194601	0.01	0.04 (0.20)	-0.46	-4.15	0.02	0.19	1.18 (1.64)	0.72 (0.66)	
CP	ogap	192601	202112	194601	0.17	-0.22 (-0.66)	-0.20	-1.79	-0.20	-1.79	-0.26 (-0.90)	-0.71 (-1.00)	
DJM	wtexas	192601	202112	194601	0.26	-0.27 (-1.41)	-0.17	-1.59	0.14	1.25*	0.26 (0.37)	-0.19 (-0.19)	
HJTZ	sntm	196507	202012	198507	1.04	0.45 (2.66)	-1.24	-5.20	-0.94	-3.97	-1.05 (-0.70)	-1.79 (-1.26)	
JT	ndrbl	195802	202112	197802	0.41	-0.27 (-1.40)	-0.77	-4.01	-0.47	-2.46	-1.06 (-1.06)	-1.86 (-1.47)	
JZZ	skvw	192607	202112	194607	0.00	-0.02 (-0.07)	-0.57	-5.10	-0.47	-4.27	-0.53 (-1.61)	-0.97 (-1.15)	
KJ	tail	192607	202112	194607	0.01	0.04 (0.26)	-0.35	-3.13	-0.35	-3.13	-0.43 (-1.41)	-0.87 (-1.08)	
KP	fbm	192606	202112	194606	3.16	0.96 (3.00)	-1.76	-15.66	-1.50	-13.42	-0.77 (-1.05)	-1.19 (-1.35)	
LY	dtoy	192601	202112	194601	0.10	0.17 (0.41)	-0.55	-4.97	-0.36	-3.30	0.39 (0.78)	-0.06 (-0.07)	
LY	dtoat	192601	202112	194601	0.02	-0.07 (-0.27)	-0.08	-0.73	-0.08	-0.73	-0.13 (-0.68)	-0.58 (-0.70)	
M1	ygap	195304	202112	197304	0.06	0.11 (0.54)	-1.27	-7.34	-0.77	-4.48	-0.19 (-0.24)	-0.25 (-0.18)	
M2	rdsp	192609	202112	194609	0.30	0.29 (0.78)	-1.34	-11.91	-0.99	-8.88	-1.35 (-2.06)	-1.87 (-1.67)	
MA	rsvix	199601	202112	200601	0.12	0.15 (0.33)	-4.78	-8.71	-3.27	-6.05	-2.19 (-1.52)	-5.41 (-2.20)	
NRTZ	tchi	195101	202112	197101	0.51	0.30 (1.69)	0.26	1.62**	0.48	2.93**	1.46 (1.64)	1.33 (0.91)	
PW	avgcor	192603	202112	194603	0.13	0.19 (0.82)	-0.18	-1.67	-0.18	-1.60	-0.29 (-0.91)	-0.77 (-1.00)	
RRZ	shtint	197301	202112	199301	0.67	-0.37 (-1.79)	1.41	4.97***	1.26	4.42***	3.13 (2.07)	1.87 (1.19)	
Y	disag	198112	202112	200112	0.02	0.06 (0.33)	-1.02	-2.42	-0.20	-0.49	2.00 (1.26)	-1.08 (-0.48)	
CSa	d/p	192601	202112	194601	0.10	0.17 (0.71)	-0.20	-1.86	-0.06	-0.50	0.47 (0.49)	0.02 (0.02)	
CSb	d/y	192602	202112	194602	0.17	0.22 (0.90)	-0.43	-3.93	-0.06	-0.55	0.69 (0.59)	0.26 (0.19)	
CSc	e/p	192601	202112	194601	0.26	0.27 (1.59)	-1.33	-12.00	-0.64	-5.76	0.23 (0.19)	-0.22 (-0.17)	
CSd	d/e	192601	202112	194601	0.04	-0.11 (-0.38)	-1.16	-10.45	-0.93	-8.44	-0.51 (-0.77)	-0.96 (-1.23)	
CSe	svar	192601	202112	194601	0.02	-0.08 (-0.19)	-0.31	-2.82	-0.01	-0.12	0.32 (1.28)	-0.13 (-0.16)	
KS	b/m	192601	202112	194601	0.28	0.29 (0.88)	-1.45	-12.98	-1.06	-9.57	-0.72 (-0.60)	-1.17 (-0.90)	
BMRR	ntis	192612	202112	194612	0.44	-0.36 (-1.44)	-0.47	-4.24	-0.47	-4.18	-0.08 (-0.11)	-0.72 (-0.99)	
Ca	tby	192601	202112	194601	0.29	-0.29 (-1.82)	0.20	1.82**	0.37	3.35**	0.65 (1.03)	0.20 (0.19)	
FFa	lty	192601	202112	194601	0.19	-0.24 (-1.57)	-0.57	-5.18	0.25	2.26**	0.53 (0.61)	0.08 (0.06)	
FFb	ltr	192601	202112	194601	0.17	0.22 (1.42)	-0.99	-8.89	-0.82	-7.44	-1.02 (-1.50)	-1.47 (-1.61)	
FFc	tms	192601	202112	194601	0.11	0.18 (1.10)	0.02	0.16	0.02	0.20	0.04 (0.08)	-0.41 (-0.51)	
FFd	dfy	192601	202112	194601	0.01	0.06 (0.12)	-0.12	-1.05	-0.12	-1.05	-0.31 (-1.91)	-0.76 (-0.96)	
FFe	dfr	192601	202112	194601	0.19	0.23 (0.91)	-0.33	-3.03	-0.30	-2.75	0.41 (0.78)	-0.04 (-0.05)	
FS	infl	192601	202112	194601	0.15	-0.21 (-0.95)	0.07	0.63*	0.10	0.92*	0.13 (0.35)	-0.32 (-0.36)	

Variable s	Sample		OOS Beg	IS		OOS		OOS-CT		Util Difference		
	Beg	End		R ²	Slope	R ²	MSE-F	R ²	MSE-F	U _{cond} − U _{uncond}	U _{cond} − U _{100%}	
Panel B: Predicting simple returns												
BH	vp	199001	202112	200001	1.93	0.58 (1.63)	0.02	0.05	0.09	0.23	−2.40 (−1.68)	−5.17 (−3.52)
BPS	impvar	199601	202112	200601	0.21	0.20 (0.52)	−3.37	−6.23	−3.34	−6.18	−1.45 (−1.00)	−5.24 (−2.49)
BTZ	vrp	199001	202112	200001	0.05	0.09 (0.16)	−9.56	−22.95	0.45	1.19*	1.05 (0.47)	−1.72 (−0.84)
CEP	lzrt	192601	202112	194601	0.04	0.11 (0.47)	−0.76	−6.91	0.04	0.38*	1.39 (1.41)	0.50 (0.49)
CP	ogap	192601	202112	194601	0.42	−0.35 (−0.98)	−0.48	−4.33	−0.48	−4.33	0.15 (0.39)	−0.74 (−1.16)
DJM	wtexas	192601	202112	194601	0.26	−0.28 (−1.49)	−0.05	−0.44	0.16	1.50**	0.44 (0.64)	−0.46 (−0.52)
HJTZ	sntm	196507	202012	198507	1.07	0.45 (2.69)	−1.23	−5.14	−0.97	−4.09	−0.48 (−0.35)	−1.91 (−1.43)
JT	ndrbl	195802	202112	197802	0.47	−0.29 (−1.49)	−0.72	−3.75	−0.43	−2.24	−1.00 (−1.01)	−2.29 (−1.89)
JZZ	skvw	192607	202112	194607	0.00	0.01 (0.03)	−0.75	−6.74	−0.69	−6.17	−0.41 (−1.07)	−1.30 (−1.79)
KJ	tail	192607	202112	194607	0.02	0.08 (0.56)	−0.27	−2.44	−0.27	−2.44	−0.21 (−1.31)	−1.10 (−1.68)
KP	fbm	192606	202112	194606	2.33	0.83 (3.06)	−1.90	−16.89	−1.83	−16.26	−0.63 (−1.19)	−1.54 (−2.01)
LY	dtoy	192601	202112	194601	0.00	−0.01 (−0.03)	−0.22	−2.00	−0.22	−2.00	0.02 (0.11)	−0.87 (−1.28)
LY	dtoat	192601	202112	194601	0.14	−0.20 (−0.72)	0.09	0.85*	0.09	0.85*	−0.13 (−0.44)	−1.02 (−1.47)
M2	ygap	195304	202112	197304	0.06	0.10 (0.52)	−1.33	−7.66	−0.81	−4.71	−0.21 (−0.25)	−0.81 (−0.67)
M1	rdsp	192609	202112	194609	0.68	0.45 (1.07)	−1.47	−13.09	−1.26	−11.22	−1.69 (−1.89)	−2.57 (−2.51)
MA	rsvix	199601	202112	200601	0.30	0.24 (0.51)	−4.73	−8.62	−3.73	−6.88	−2.17 (−1.57)	−5.95 (−2.59)
NRTZ	tchi	195101	202112	197101	0.42	0.27 (1.53)	0.16	0.96*	0.34	2.06**	1.56 (1.69)	0.92 (0.66)
PW	avgcor	192603	202112	194603	0.31	0.30 (1.26)	0.43	3.91**	0.43	3.91**	0.18 (0.50)	−0.74 (−1.17)
RRZ	shtint	197301	202112	199301	0.66	−0.36 (−1.82)	1.37	4.81***	1.30	4.57***	3.36 (2.31)	1.61 (1.12)
Y	disag	198112	202112	200112	0.02	0.06 (0.33)	−0.98	−2.34	−0.47	−1.12	1.46 (0.92)	−1.87 (−0.90)
CSa	d/p	192601	202112	194601	0.24	0.26 (1.05)	−0.34	−3.07	0.02	0.16	0.44 (0.35)	−0.46 (−0.38)
CSb	d/y	192602	202112	194602	0.33	0.31 (1.20)	−0.82	−7.38	−0.12	−1.12	0.39 (0.27)	−0.52 (−0.38)
CSc	e/p	192601	202112	194601	0.28	0.28 (1.63)	−1.29	−11.58	−0.69	−6.25	0.57 (0.46)	−0.33 (−0.28)
CSd	d/e	192601	202112	194601	0.00	0.01 (0.05)	−0.58	−5.22	−0.55	−5.01	−0.16 (−0.49)	−1.06 (−1.47)
CSe	svar	192601	202112	194601	0.02	0.08 (0.18)	−0.58	−5.26	−0.57	−5.20	−0.39 (−1.46)	−1.29 (−1.92)
KS	b/m	192601	202112	194601	0.53	0.40 (1.12)	−2.25	−20.06	−1.73	−15.51	−0.94 (−0.68)	−1.83 (−1.44)
BMRR	ntis	192612	202112	194612	0.48	−0.38 (−1.50)	−0.75	−6.67	−0.75	−6.67	−0.11 (−0.16)	−1.03 (−1.52)
Ca	tby	192601	202112	194601	0.37	−0.33 (−2.05)	0.36	3.26**	0.54	4.98***	1.24 (1.53)	0.34 (0.33)
FFa	lty	192601	202112	194601	0.24	−0.27 (−1.77)	−0.28	−2.59	0.47	4.30***	0.99 (1.05)	0.10 (0.09)
FFb	ltr	192601	202112	194601	0.21	0.25 (1.56)	−0.57	−5.16	−0.55	−4.94	−0.34 (−0.54)	−1.24 (−1.54)
FFc	tms	192601	202112	194601	0.15	0.21 (1.27)	0.10	0.89*	0.12	1.08*	0.85 (1.36)	−0.05 (−0.07)
FFd	dfy	192601	202112	194601	0.24	0.27 (0.53)	−0.18	−1.62	−0.16	−1.48	−0.37 (−0.64)	−1.27 (−1.72)
FFe	dfr	192601	202112	194601	0.11	0.18 (0.70)	−0.29	−2.67	−0.22	−1.98	0.46 (0.93)	−0.44 (−0.59)
FS	infl	192601	202112	194601	0.26	−0.28 (−1.24)	0.37	3.38**	0.39	3.60**	0.82 (1.94)	−0.07 (−0.09)

Table IA.7: Predicting Next Quarter Returns

Variable s		Sample		OOS Beg	IS		OOS		OOS-CT		Util Difference		
		Beg	End		R ²	Slope	R ²	MSE-F	R ²	MSE-F	U _{cond} − U _{uncond}	U _{cond} − U _{100%}	
Panel A: Predicting log returns													
AMP	pce	19534	20214	19734	4.61	−1.73 (−3.70)	−0.66	−1.25	−0.38	−0.73	−0.51 (−0.34)	−0.14 (−0.10)	
BH	vp	19901	20214	20001	8.18	2.28 (2.74)	1.81	1.60**	2.75	2.46**	−0.21 (−0.09)	−1.86 (−0.80)	
BPS	impvar	19961	20214	20061	2.10	1.23 (1.00)	−6.42	−3.80	−5.77	−3.44	−4.61 (−1.97)	−7.72 (−2.86)	
BTZ	vrp	19901	20214	20001	0.18	−0.34 (−0.30)	−130.10	−49.19	7.88	7.44***	5.41 (2.00)	3.76 (1.24)	
BY	govik	19471	20214	19671	0.59	0.60 (1.52)	−1.40	−3.02	0.07	0.14	1.86 (1.55)	1.42 (0.79)	
CEP	lzrt	19261	20214	19461	0.05	0.23 (0.34)	−4.17	−12.13	−1.09	−3.27	0.63 (0.59)	0.25 (0.18)	
CGP	crdstd	19902	20214	20002	4.56	−1.72 (−1.67)	2.67	2.36**	4.54	4.09***	6.16 (3.09)	4.58 (1.59)	
CP	ogap	19261	20214	19461	0.63	−0.83 (−0.91)	−1.52	−4.54	−1.52	−4.54	−0.17 (−0.42)	−0.56 (−0.71)	
DJM	wtxas	19261	20214	19461	0.20	0.47 (0.73)	−2.78	−8.19	−1.46	−4.35	−0.07 (−0.08)	−0.45 (−0.41)	
HJTZ	sntm	19653	20204	19853	3.37	1.52 (2.87)	−6.09	−8.10	−5.17	−6.93	−2.69 (−1.33)	−2.78 (−1.61)	
JT	ndrbl	19581	20214	19781	2.34	−1.24 (−2.24)	0.33	0.57	0.59	1.04*	−0.48 (−0.42)	−0.82 (−0.61)	
JZZ	skvw	19263	20214	19463	0.18	0.44 (0.39)	−5.20	−14.87	−2.03	−6.00	−0.32 (−0.54)	−0.58 (−0.57)	
KJ	tail	19263	20214	19463	0.03	0.19 (0.37)	−0.73	−2.19	−0.73	−2.19	−0.70 (−2.06)	−0.96 (−1.07)	
KP	fbm	19262	20214	19462	7.49	2.86 (4.18)	−3.57	−10.41	−3.15	−9.23	−1.35 (−1.53)	−1.66 (−1.64)	
LY	dtoy	19261	20214	19461	0.03	0.17 (0.14)	−1.21	−3.63	−1.10	−3.31	0.01 (0.03)	−0.37 (−0.39)	
LY	dtoat	19261	20214	19461	0.11	−0.34 (−0.43)	−0.13	−0.38	−0.13	−0.38	−0.05 (−0.21)	−0.43 (−0.46)	
M2	ygap	19532	20214	19732	0.14	0.30 (0.44)	−4.23	−7.87	−2.53	−4.78	−0.81 (−0.69)	−0.42 (−0.26)	
M1	rdsp	19263	20214	19463	0.00	0.06 (0.06)	−0.51	−1.52	−0.50	−1.49	−0.18 (−0.87)	−0.43 (−0.44)	
MA	rsvix	19961	20214	20061	5.58	2.00 (1.69)	−2.79	−1.71	−1.36	−0.85	−3.75 (−1.30)	−6.86 (−2.44)	
NRTZ	tchi	19511	20214	19711	0.35	0.47 (0.77)	−0.48	−0.97	−0.13	−0.26	1.03 (1.34)	1.27 (0.82)	
PW	avgcor	19261	20214	19461	0.87	0.97 (1.35)	1.04	3.17**	1.04	3.17**	0.00 (0.01)	−0.38 (−0.43)	
RRZ	shtint	19731	20214	19931	1.46	−1.04 (−1.57)	3.18	3.77***	3.17	3.76***	2.71 (1.90)	2.16 (1.19)	
Y	disag	19814	20214	20014	0.06	−0.20 (−0.32)	−3.97	−3.05	−0.96	−0.76	2.47 (1.04)	−0.66 (−0.22)	
CSa	d/p	19261	20214	19461	0.49	0.73 (1.01)	−1.19	−3.58	−0.07	−0.21	1.31 (1.01)	0.93 (0.59)	
CSb	d/y	19261	20214	19461	0.61	0.82 (1.16)	−2.46	−7.29	−0.12	−0.36	1.43 (1.02)	1.05 (0.64)	
CSc	e/p	19261	20214	19461	0.82	0.95 (1.51)	−6.15	−17.55	−3.12	−9.17	0.43 (0.30)	0.05 (0.03)	
CSd	d/e	19261	20214	19461	0.03	−0.18 (−0.21)	−3.27	−9.59	−2.73	−8.04	−0.95 (−1.30)	−1.34 (−1.47)	
CSe	svar	19261	20214	19461	0.84	0.96 (0.78)	0.31	0.95*	0.31	0.95*	−0.17 (−0.37)	−0.55 (−0.58)	
KS	b/m	19261	20214	19461	1.29	1.19 (1.30)	−6.15	−17.55	−3.74	−10.92	−0.94 (−0.62)	−1.33 (−0.83)	
BMRR	ntis	19264	20214	19464	1.83	−1.41 (−2.06)	−2.57	−7.52	−2.54	−7.43	−1.03 (−1.09)	−1.55 (−1.73)	
Ca	tby	19261	20214	19461	0.59	−0.80 (−1.58)	0.09	0.27	0.88	2.71**	0.78 (1.11)	0.40 (0.33)	
FFa	lty	19261	20214	19461	0.40	−0.66 (−1.40)	−1.81	−5.39	0.62	1.88**	0.84 (0.85)	0.46 (0.34)	
FFb	ltr	19261	20214	19461	1.19	1.14 (2.38)	1.00	3.07**	0.87	2.66**	0.70 (0.74)	0.32 (0.28)	
FFc	tms	19261	20214	19461	0.19	0.46 (0.86)	−0.32	−0.97	−0.26	−0.80	−0.26 (−0.44)	−0.64 (−0.73)	
FFd	dfy	19261	20214	19461	0.15	0.41 (0.32)	−0.52	−1.56	−0.47	−1.41	−0.49 (−1.38)	−0.88 (−0.97)	
FFe	dfr	19261	20214	19461	0.42	−0.68 (−0.85)	−7.84	−22.04	−6.22	−17.76	−1.67 (−1.98)	−2.05 (−1.95)	
FS	infl	19261	20214	19461	0.07	−0.28 (−0.37)	−0.30	−0.90	−0.30	−0.90	−0.31 (−1.46)	−0.69 (−0.79)	
Co	i/k	19471	20214	19671	4.41	−1.65 (−3.73)	2.73	6.16***	3.47	7.88***	2.37 (1.62)	1.93 (1.23)	
LL	cay	19521	20214	19721	0.01	−0.09 (−0.18)	−11.91	−21.18	−6.11	−11.46	−1.96 (−1.06)	−1.55 (−0.81)	

Variable s		Sample		OOS Beg	IS		OOS		OOS-CT		Util Difference		
		Beg	End		R ²	Slope	R ²	MSE-F	R ²	MSE-F	U _{cond} − U _{uncond}	U _{cond} − U _{100%}	
Panel B: Predicting simple returns													
AMP	pce	19534	20214	19734	4.59	−1.72 (−3.72)	−0.86	−1.64	−0.66	−1.27	−0.21 (−0.14)	−0.73 (−0.54)	
BH	vp	19901	20214	20001	9.96	2.51 (2.96)	2.76	2.47**	3.56	3.21**	0.48 (0.21)	−2.06 (−0.96)	
BPS	impvar	19961	20214	20061	3.09	1.48 (1.21)	−5.65	−3.37	−5.07	−3.04	−3.81 (−1.76)	−7.60 (−2.96)	
BTZ	vrp	19901	20214	20001	0.30	−0.43 (−0.36)	−146.17	−51.66	7.96	7.52***	5.18 (1.78)	2.65 (0.94)	
BY	govik	19471	20214	19671	0.53	0.57 (1.41)	−1.52	−3.27	−0.14	−0.30	1.63 (1.30)	0.43 (0.26)	
CEP	lzrt	19261	20214	19461	0.26	0.59 (0.69)	−8.66	−24.16	−1.74	−5.17	0.60 (0.41)	−0.67 (−0.50)	
CGP	crdstd	19902	20214	20002	3.85	−1.57 (−1.55)	1.99	1.75**	4.05	3.63**	5.82 (2.71)	3.30 (1.20)	
CP	ogap	19261	20214	19461	1.83	−1.50 (−1.30)	−6.08	−17.38	−6.08	−17.38	0.03 (0.06)	−1.24 (−1.55)	
DJM	wtexas	19261	20214	19461	0.12	0.38 (0.58)	−2.65	−7.81	−1.77	−5.27	0.21 (0.28)	−1.06 (−1.10)	
HJTZ	sntm	19653	20204	19853	3.45	1.53 (2.88)	−5.99	−7.96	−5.19	−6.95	−1.95 (−1.03)	−2.99 (−1.77)	
JT	ndrbl	19581	20214	19781	2.79	−1.35 (−2.42)	0.60	1.05*	0.77	1.37*	−0.81 (−0.68)	−1.84 (−1.40)	
JZZ	skvw	19263	20214	19463	0.60	0.86 (0.58)	−9.20	−25.36	−4.03	−11.65	−0.86 (−1.00)	−2.11 (−2.28)	
KJ	tail	19263	20214	19463	0.13	0.41 (0.80)	−0.30	−0.90	−0.30	−0.90	−0.25 (−1.06)	−1.50 (−2.10)	
KP	fbm	19262	20214	19462	6.54	2.85 (2.92)	−2.44	−7.21	−2.44	−7.21	−0.85 (−1.50)	−2.11 (−2.49)	
LY	dtoy	19261	20214	19461	0.45	−0.74 (−0.51)	0.76	2.31**	0.76	2.31**	−0.16 (−0.49)	−1.43 (−1.88)	
LY	dtoat	19261	20214	19461	0.72	−0.94 (−1.01)	0.86	2.64**	0.86	2.64**	0.26 (0.49)	−1.01 (−1.28)	
M2	ygap	19532	20214	19732	0.12	0.27 (0.40)	−4.39	−8.15	−2.72	−5.13	−0.56 (−0.48)	−1.06 (−0.74)	
M1	rdsp	19263	20214	19463	0.55	0.83 (0.62)	−1.01	−3.01	−1.01	−3.00	−0.25 (−0.34)	−1.50 (−1.58)	
MA	rsvix	19961	20214	20061	7.21	2.26 (1.91)	−1.15	−0.72	−0.04	−0.03	−2.14 (−0.80)	−5.92 (−2.31)	
NRTZ	tchi	19511	20214	19711	0.19	0.35 (0.58)	−0.65	−1.30	−0.57	−1.15	0.84 (1.26)	0.29 (0.22)	
PW	avgcor	19261	20214	19461	1.71	1.46 (1.85)	3.84	12.09***	3.84	12.09***	1.19 (2.00)	−0.07 (−0.10)	
RRZ	shtint	19731	20214	19931	1.46	−1.04 (−1.61)	3.05	3.61**	3.05	3.61**	2.72 (2.14)	1.35 (0.89)	
Y	disag	19814	20214	20014	0.06	−0.19 (−0.31)	−3.91	−3.01	−2.10	−1.65	1.89 (0.83)	−1.78 (−0.65)	
CSa	d/p	19261	20214	19461	1.21	1.23 (1.41)	−4.14	−12.04	−0.49	−1.49	0.95 (0.55)	−0.32 (−0.20)	
CSb	d/y	19261	20214	19461	1.34	1.29 (1.51)	−6.83	−19.38	−1.08	−3.24	0.98 (0.54)	−0.29 (−0.18)	
CSc	e/p	19261	20214	19461	0.90	1.06 (1.56)	−7.72	−21.72	−5.03	−14.52	0.96 (0.63)	−0.31 (−0.23)	
CSd	d/e	19261	20214	19461	0.12	0.39 (0.42)	−0.67	−2.02	−0.67	−2.02	−0.02 (−0.08)	−1.29 (−1.57)	
CSe	svar	19261	20214	19461	3.04	1.94 (1.27)	0.18	0.54*	0.18	0.54*	0.52 (0.61)	−0.75 (−0.88)	
KS	b/m	19261	20214	19461	2.81	1.87 (1.52)	−13.74	−36.59	−8.70	−24.25	−1.07 (−0.61)	−2.34 (−1.47)	
BMRR	ntis	19264	20214	19464	2.03	−1.59 (−2.05)	−4.52	−12.99	−4.53	−13.00	−0.53 (−0.61)	−1.68 (−1.93)	
Ca	tby	19261	20214	19461	0.77	−0.98 (−1.85)	0.73	2.24**	1.55	4.78***	1.46 (1.50)	0.20 (0.17)	
FFa	lty	19261	20214	19461	0.51	−0.80 (−1.64)	−0.72	−2.17	1.55	4.78***	1.22 (1.20)	−0.05 (−0.04)	
FFb	ltr	19261	20214	19461	1.11	1.17 (2.45)	0.92	2.83**	0.78	2.39**	1.13 (1.25)	−0.14 (−0.14)	
FFc	tms	19261	20214	19461	0.27	0.58 (1.06)	0.00	−0.01	0.10	0.30	0.83 (1.13)	−0.44 (−0.53)	
FFd	dfy	19261	20214	19461	1.69	1.45 (0.90)	−1.64	−4.88	−1.18	−3.54	−0.60 (−0.58)	−1.87 (−2.10)	
FFe	dfr	19261	20214	19461	0.46	−0.75 (−0.98)	−8.16	−22.87	−7.01	−19.84	−0.81 (−1.21)	−2.08 (−2.28)	
FS	infl	19261	20214	19461	0.42	−0.72 (−0.87)	0.70	2.15**	0.71	2.18**	0.51 (1.22)	−0.76 (−0.90)	
Co	i/k	19471	20214	19671	4.48	−1.66 (−3.70)	2.65	5.97***	3.39	7.68***	2.16 (1.40)	0.96 (0.64)	
LL	cay	19521	20214	19721	0.01	−0.10 (−0.19)	−12.02	−21.35	−6.64	−12.39	−1.95 (−0.99)	−2.39 (−1.28)	

Table IA.8: Predicting Next Year (Jan to Dec) Returns

Variable s		Sample		OOS Beg	R^2	<u>IS</u>	R^2	<u>OOS</u>	R^2	<u>OOS-CT</u>	<u>Util Difference</u>		
		Beg	End			Slope		MSE-F		MSE-F	$U_{\text{cond}} - U_{\text{uncond}}$	$U_{\text{cond}} - U_{100\%}$	
Panel A: Predicting log returns													
AMP	pce	1953	2021	1973	15.71	-6.62 (-3.37)	-7.09	-3.18	-5.25	-2.39	-0.39 (-0.20)	-2.38 (-1.30)	
BH	vp	1990	2021	2000	0.41	1.07 (0.48)	-11.01	-2.08	-9.10	-1.75	-0.63 (-0.69)	-6.73 (-2.36)	
BPS	impvar	1996	2021	2006	0.23	0.86 (0.31)	-10.19	-1.39	-2.94	-0.43	-3.22 (-1.01)	-10.49 (-3.28)	
BTZ	vrp	1990	2021	2000	0.54	-1.25 (-0.45)	-6.41	-1.27	-6.39	-1.26	-0.84 (-0.92)	-6.94 (-2.34)	
BY	govik	1947	2021	1967	2.29	2.49 (1.48)	-5.17	-2.65	-0.08	-0.04	2.11 (1.40)	-0.12 (-0.06)	
CEP	lzrt	1926	2021	1946	7.59	7.36 (2.55)	-0.46	-0.34	2.69	2.08**	2.75 (1.59)	0.93 (0.52)	
CGMS	skew	1951	2019	1971	0.05	0.38 (0.16)	-3.62	-1.68	-2.29	-1.08	0.20 (0.26)	-1.67 (-0.99)	
CGP	crdstd	1990	2021	2000	1.32	-1.95 (-0.55)	-9.41	-1.81	1.71	0.36	1.70 (1.24)	-4.40 (-1.43)	
CP	ogap	1926	2021	1946	2.72	-3.20 (-1.35)	-9.27	-6.36	-9.27	-6.36	0.26 (0.74)	-1.55 (-1.53)	
DJM	wtexas	1926	2021	1946	0.88	-1.85 (-0.92)	-5.76	-4.08	-5.34	-3.80	0.27 (0.26)	-1.54 (-1.11)	
HHT	accrul	1965	2021	1985	8.25	4.76 (2.29)	10.18	4.08***	4.40	1.66**	-1.52 (-1.54)	-4.30 (-1.94)	
HHT	cfacc	1965	2021	1985	11.19	-5.53 (-3.09)	6.32	2.43**	16.35	7.04***	4.32 (4.30)	1.54 (0.88)	
HJTZ	sntm	1965	2020	1985	5.40	3.84 (1.55)	-5.06	-1.69	-3.10	-1.05	-2.44 (-1.43)	-5.03 (-2.45)	
JT	ndrbl	1958	2021	1978	11.32	-5.39 (-2.10)	-2.91	-1.22	1.01	0.44	3.23 (1.51)	0.45 (0.20)	
JZZ	skvw	1926	2021	1946	1.54	2.43 (0.99)	-0.33	-0.25	0.35	0.27	2.03 (2.08)	0.22 (0.17)	
KJ	tail	1926	2021	1946	0.72	1.65 (0.79)	-0.39	-0.29	-0.32	-0.24	-0.20 (-0.73)	-2.02 (-1.81)	
KP	fbm	1926	2021	1946	17.63	8.14 (3.27)	-7.06	-4.95	-6.98	-4.89	0.73 (0.56)	-1.09 (-0.80)	
LY	dtoy	1926	2021	1946	1.05	1.99 (0.60)	-6.64	-4.67	-5.93	-4.20	-0.85 (-1.08)	-2.66 (-1.99)	
LY	dtoat	1926	2021	1946	0.66	-1.58 (-0.70)	-1.58	-1.16	-1.58	-1.16	0.59 (1.32)	-1.22 (-1.09)	
M2	ygap	1953	2021	1973	1.18	1.82 (0.82)	-16.73	-6.88	-12.40	-5.30	0.23 (0.12)	-1.76 (-0.93)	
M1	rdsp	1926	2021	1946	1.11	-2.04 (-0.64)	-1.13	-0.84	-1.13	-0.84	-0.33 (-0.55)	-2.14 (-1.84)	
MA	rsvix	1996	2021	2006	0.67	1.46 (0.73)	-18.32	-2.32	-1.82	-0.27	1.51 (1.46)	-5.77 (-2.23)	
MR	gpce	1947	2021	1967	12.48	-5.78 (-3.61)	5.86	3.36**	5.43	3.10**	-0.02 (-0.01)	-2.25 (-1.37)	
MR	gip	1926	2021	1946	0.02	-0.28 (-0.09)	-1.95	-1.43	-2.05	-1.50	0.03 (0.10)	-1.78 (-1.62)	
NRTZ	tchi	1951	2021	1971	0.03	-0.27 (-0.15)	-5.89	-2.78	-5.89	-2.78	-1.62 (-2.13)	-3.60 (-2.46)	
PST	house	1929	2021	1949	0.92	1.86 (0.96)	0.46	0.33	0.46	0.33	0.37 (0.84)	-1.55 (-1.22)	
PW	avgcor	1926	2021	1946	0.31	1.09 (0.46)	-7.77	-5.41	-7.49	-5.23	-1.02 (-1.30)	-2.83 (-2.14)	
RRZ	shtint	1973	2021	1993	6.19	-4.72 (-1.68)	2.62	0.75*	2.62	0.75*	4.15 (2.01)	-0.36 (-0.18)	
Y	disag	1981	2021	2001	0.01	0.19 (0.09)	-17.96	-3.05	2.83	0.58	8.52 (1.84)	3.06 (0.72)	
CSa	d/p	1926	2021	1946	1.30	2.26 (0.94)	-1.27	-0.94	0.31	0.23	2.32 (1.57)	0.51 (0.31)	
CSb	d/y	1926	2021	1946	1.61	2.51 (1.09)	-2.42	-1.77	1.28	0.97*	2.56 (1.48)	0.75 (0.41)	
CSc	e/p	1926	2021	1946	1.69	2.54 (1.42)	-1.60	-1.18	2.53	1.95**	1.76 (1.13)	-0.06 (-0.04)	
CSD	d/e	1926	2021	1946	0.00	-0.01 (0.00)	-3.65	-2.64	-3.65	-2.64	-1.40 (-1.98)	-3.22 (-2.84)	
CSe	svar	1926	2021	1946	0.82	-1.75 (-0.65)	-5.47	-3.89	-0.55	-0.41	-0.47 (-0.65)	-2.29 (-1.76)	
KS	b/m	1926	2021	1946	2.55	3.13 (1.69)	-8.80	-6.07	-6.15	-4.34	1.17 (0.66)	-0.64 (-0.38)	
BMRR	ntis	1926	2021	1946	6.27	-4.86 (-1.60)	-7.80	-5.43	-7.81	-5.43	-0.31 (-0.34)	-2.13 (-1.93)	
BW	eqis	1927	2021	1947	8.17	-5.54 (-2.73)	0.18	0.14	1.49	1.12*	1.13 (1.24)	-0.79 (-0.59)	
Ca	tby	1926	2021	1946	2.25	-2.92 (-1.85)	-3.89	-2.81	0.66	0.50*	0.97 (0.96)	-0.84 (-0.54)	
FFa	lty	1926	2021	1946	1.02	-1.98 (-1.23)	-6.34	-4.47	1.80	1.38**	0.88 (0.86)	-0.94 (-0.60)	
FFb	ltr	1926	2021	1946	5.53	4.57 (2.71)	-9.06	-6.23	0.52	0.39*	2.09 (1.41)	0.28 (0.17)	
FFc	tms	1926	2021	1946	1.78	2.59 (1.74)	-2.36	-1.73	-2.19	-1.61	0.69 (0.87)	-1.13 (-0.86)	
FFd	dfy	1926	2021	1946	0.03	0.31 (0.15)	-0.98	-0.73	-0.98	-0.73	-0.45 (-1.80)	-2.27 (-2.07)	
FFe	dfr	1926	2021	1946	0.15	-0.76 (-0.43)	-3.95	-2.85	-0.91	-0.68	-0.34 (-0.54)	-2.15 (-1.74)	
FS	infl	1926	2021	1946	0.07	0.52 (0.14)	-12.02	-8.05	-11.09	-7.49	-1.28 (-2.14)	-3.09 (-2.76)	
Co	i/k	1947	2021	1967	12.31	-5.76 (-2.98)	7.06	4.10***	10.94	6.64***	4.56 (2.68)	2.33 (1.32)	
LL	cay	1952	2021	1972	0.19	0.79 (0.29)	-30.15	-11.35	-14.44	-6.18	2.52 (0.98)	0.59 (0.24)	

Variable s		Sample		OOS Beg	R^2	IS Slope	OOS		OOS-CT		Util Difference	
		Beg	End				R^2	MSE-F	R^2	MSE-F	$U_{\text{cond}} - U_{\text{uncond}}$	$U_{\text{cond}} - U_{100\%}$
Panel B: Predicting simple returns												
AMP	pce	1953	2021	1973	15.65	-7.00 (-3.32)	-9.77	-4.27	-8.19	-3.63	-1.05 (-0.45)	-3.03 (-1.43)
BH	vp	1990	2021	2000	0.92	1.63 (0.71)	-10.94	-2.07	-10.94	-2.07	-0.55 (-0.53)	-5.70 (-2.04)
BPS	impvar	1996	2021	2006	0.78	1.59 (0.54)	-11.75	-1.58	-6.65	-0.93	-2.18 (-0.63)	-10.61 (-2.65)
BTZ	vrp	1990	2021	2000	0.29	-0.94 (-0.31)	-8.02	-1.56	-8.02	-1.56	-0.92 (-0.74)	-6.07 (-2.04)
BY	govik	1947	2021	1967	2.52	2.77 (1.47)	-5.60	-2.87	-0.41	-0.22	0.98 (0.58)	-1.27 (-0.63)
CEP	lzrt	1926	2021	1946	8.32	7.94 (2.69)	1.46	1.11*	2.64	2.03**	2.54 (1.17)	-0.14 (-0.08)
CGMS	skew	1951	2019	1971	0.07	0.47 (0.18)	-3.87	-1.79	-2.98	-1.39	0.20 (0.21)	-1.69 (-1.03)
CGP	crdstd	1990	2021	2000	0.83	-1.57 (-0.42)	-10.11	-1.93	-0.76	-0.16	0.33 (0.20)	-4.82 (-1.51)
CP	ogap	1926	2021	1946	3.04	-3.49 (-1.35)	-10.91	-7.38	-10.91	-7.38	0.33 (0.86)	-2.35 (-1.93)
DJM	wtxas	1926	2021	1946	0.80	-1.82 (-0.90)	-5.98	-4.23	-6.27	-4.43	0.32 (0.29)	-2.35 (-1.56)
HHT	accrul	1965	2021	1985	9.69	5.32 (2.66)	11.85	4.84***	6.33	2.43**	0.81 (0.76)	-3.18 (-1.42)
HHT	cfacc	1965	2021	1985	12.14	-5.95 (-3.17)	6.83	2.64**	16.88	7.31***	4.19 (3.61)	0.20 (0.13)
HJTZ	sntm	1965	2020	1985	5.63	4.04 (1.64)	-4.11	-1.38	-2.88	-0.98	-0.77 (-0.44)	-4.67 (-2.19)
JT	ndrbl	1958	2021	1978	10.97	-5.49 (-2.16)	-3.32	-1.38	0.06	0.02	2.15 (0.95)	-0.98 (-0.41)
JZZ	skvw	1926	2021	1946	2.01	2.85 (1.18)	0.26	0.19	0.57	0.43*	1.15 (1.22)	-1.53 (-1.15)
KJ	tail	1926	2021	1946	1.15	2.14 (0.99)	0.07	0.05	0.07	0.05	0.00 (-0.02)	-2.68 (-2.15)
KP	fbm	1926	2021	1946	16.36	8.08 (3.93)	-9.05	-6.22	-9.17	-6.30	-0.33 (-0.30)	-3.01 (-2.04)
LY	dtoy	1926	2021	1946	0.28	1.06 (0.34)	-5.91	-4.18	-5.91	-4.18	-0.07 (-0.12)	-2.75 (-2.09)
LY	dtoat	1926	2021	1946	1.46	-2.43 (-1.06)	-0.38	-0.29	-0.38	-0.29	0.58 (0.96)	-2.10 (-1.88)
M2	ygap	1953	2021	1973	1.41	2.11 (0.87)	-21.33	-8.44	-17.14	-7.02	-1.08 (-0.53)	-3.07 (-1.58)
M1	rdsp	1926	2021	1946	0.29	-1.07 (-0.34)	-2.07	-1.52	-2.07	-1.52	-0.03 (-0.12)	-2.70 (-2.16)
MA	rsvix	1996	2021	2006	1.20	1.95 (0.97)	-15.95	-2.06	-5.22	-0.74	-0.68 (-0.60)	-9.12 (-2.77)
MR	gpce	1947	2021	1967	14.69	-6.65 (-3.65)	7.93	4.65***	7.50	4.38***	0.62 (0.46)	-1.63 (-0.98)
MR	gip	1926	2021	1946	0.13	-0.71 (-0.24)	-1.54	-1.14	-1.54	-1.14	0.62 (1.62)	-2.05 (-1.88)
NRTZ	tchi	1951	2021	1971	0.21	-0.81 (-0.41)	-5.89	-2.78	-5.89	-2.78	-1.39 (-1.92)	-3.27 (-2.24)
PST	house	1929	2021	1949	0.74	1.70 (0.82)	0.56	0.41*	0.56	0.41*	0.27 (0.73)	-2.29 (-1.85)
PW	avgcor	1926	2021	1946	0.79	1.77 (0.77)	-5.90	-4.18	-5.90	-4.18	-0.14 (-0.17)	-2.81 (-2.10)
RRZ	shtint	1973	2021	1993	5.36	-4.52 (-1.69)	1.73	0.49	1.73	0.49	0.43 (0.23)	-3.27 (-1.34)
Y	disag	1981	2021	2001	0.01	0.14 (0.06)	-21.17	-3.49	-1.55	-0.31	4.49 (1.14)	0.18 (0.05)
CSa	d/p	1926	2021	1946	2.09	2.95 (1.25)	-0.03	-0.02	1.35	1.03*	1.94 (1.07)	-0.73 (-0.46)
CSb	d/y	1926	2021	1946	2.42	3.17 (1.38)	-1.13	-0.84	2.47	1.90**	2.45 (1.17)	-0.23 (-0.13)
CSc	e/p	1926	2021	1946	1.88	2.76 (1.46)	-0.76	-0.56	2.13	1.63**	0.39 (0.24)	-2.29 (-1.53)
CSd	d/e	1926	2021	1946	0.13	0.74 (0.30)	-2.76	-2.02	-2.76	-2.02	-0.70 (-1.71)	-3.38 (-2.56)
CSe	svar	1926	2021	1946	0.36	-1.20 (-0.46)	-5.22	-3.72	-2.49	-1.82	-0.20 (-0.29)	-2.87 (-2.08)
KS	b/m	1926	2021	1946	3.03	3.51 (1.75)	-7.30	-5.10	-5.99	-4.24	0.30 (0.15)	-2.37 (-1.45)
BMRR	ntis	1926	2021	1946	4.75	-4.35 (-1.59)	-5.49	-3.90	-5.49	-3.90	-0.27 (-0.34)	-2.95 (-2.36)
BW	eqis	1927	2021	1947	8.13	-5.69 (-2.93)	0.35	0.26	1.43	1.07*	1.26 (1.25)	-1.31 (-0.95)
Ca	tby	1926	2021	1946	2.86	-3.40 (-2.06)	-2.96	-2.15	1.63	1.25*	1.84 (1.29)	-0.84 (-0.52)
FFa	lty	1926	2021	1946	1.49	-2.46 (-1.46)	-5.04	-3.60	2.80	2.16**	1.30 (0.96)	-1.37 (-0.82)
FFb	ltr	1926	2021	1946	5.98	4.90 (2.76)	-9.42	-6.46	-1.36	-1.00	1.61 (0.96)	-1.06 (-0.63)
FFc	tms	1926	2021	1946	1.81	2.69 (1.67)	-2.81	-2.05	-2.64	-1.93	1.47 (1.29)	-1.21 (-0.91)
FFd	dfy	1926	2021	1946	0.20	0.91 (0.41)	-0.77	-0.57	-0.77	-0.57	-0.08 (-0.54)	-2.76 (-2.24)
FFe	dfr	1926	2021	1946	0.27	-1.05 (-0.55)	-4.38	-3.15	-1.91	-1.40	-0.67 (-0.61)	-3.34 (-2.28)
FS	infl	1926	2021	1946	0.06	-0.50 (-0.15)	-7.14	-5.00	-7.14	-5.00	-0.44 (-1.02)	-3.12 (-2.43)
Co	i/k	1947	2021	1967	11.25	-5.84 (-2.89)	5.74	3.29**	9.55	5.70***	3.20 (1.80)	0.95 (0.53)
LL	cay	1952	2021	1972	0.17	0.78 (0.27)	-30.11	-11.34	-17.06	-7.14	0.33 (0.12)	-1.51 (-0.61)

Table IA.9: Predicting Next Year (Jul to Jun) Returns

Variable s		Sample		OOS Beg	IS		OOS		OOS-CT		Util Difference		
		Beg	End		R ²	Slope	R ²	MSE-F	R ²	MSE-F	U _{cond} - U _{uncond}	U _{cond} - U _{100%}	
Panel A: Predicting log returns													
AMP	pce	1954	2021	1974	11.04	-5.19 (-2.83)	0.36	0.17	0.75	0.35	-0.24 (-0.18)	-0.65 (-0.48)	
BH	vp	1991	2021	2001	1.34	-1.67 (-0.70)	-10.80	-1.95	0.98	0.20	1.20 (1.11)	-1.53 (-0.64)	
BPS	impvar	1997	2021	2007	3.23	-2.75 (-1.03)	-0.45	-0.06	1.99	0.28	0.23 (0.42)	-0.99 (-0.27)	
BTZ	vrp	1991	2021	2001	1.54	-1.84 (-0.83)	-3.07	-0.60	-2.32	-0.45	-0.69 (-0.84)	-3.43 (-1.57)	
BY	govik	1948	2021	1968	1.82	2.11 (1.22)	-8.13	-3.98	0.81	0.43	2.53 (1.57)	0.80 (0.40)	
CEP	lzrt	1927	2021	1947	6.08	7.97 (3.20)	-2.42	-1.75	4.36	3.37**	1.78 (1.25)	1.02 (0.61)	
CGMS	skew	1952	2019	1972	1.37	1.78 (0.81)	-0.43	-0.20	-0.43	-0.20	0.37 (1.25)	-0.06 (-0.04)	
CGP	crdstd	1991	2021	2001	5.98	-3.66 (-1.51)	-4.95	-0.94	7.63	1.65**	3.28 (2.39)	0.54 (0.22)	
CP	ogap	1927	2021	1947	1.21	-2.56 (-0.95)	-12.35	-8.14	-12.35	-8.14	-0.12 (-0.29)	-0.87 (-0.93)	
DJM	wtexas	1927	2021	1947	0.03	-0.43 (-0.26)	-5.73	-4.01	-3.99	-2.84	-0.14 (-0.19)	-0.90 (-0.84)	
HHT	accrul	1966	2021	1986	10.76	5.12 (2.61)	12.39	4.95***	6.31	2.36**	-0.48 (-0.55)	-2.04 (-1.06)	
HHT	cfacc	1966	2021	1986	4.51	-3.29 (-1.44)	-6.45	-2.12	12.08	4.81***	1.62 (1.77)	0.07 (0.04)	
HJTZ	sntm	1966	2020	1986	4.73	3.32 (1.32)	-19.37	-5.52	-14.55	-4.32	-3.91 (-2.71)	-5.33 (-2.89)	
JT	ndrbl	1959	2021	1979	7.61	-4.20 (-2.11)	6.79	3.06**	6.74	3.03**	1.98 (1.54)	1.28 (0.61)	
JZZ	skvw	1927	2021	1947	1.47	2.83 (0.98)	-19.51	-12.08	-14.66	-9.46	0.99 (0.82)	0.24 (0.15)	
KJ	tail	1927	2021	1947	0.33	1.33 (0.59)	-2.35	-1.70	-2.20	-1.59	-0.87 (-2.29)	-1.62 (-1.44)	
KP	fbm	1927	2021	1947	17.15	9.70 (1.99)	-9.30	-6.30	-5.58	-3.91	-0.35 (-0.24)	-1.10 (-0.76)	
LY	dtoy	1927	2021	1947	0.00	-0.05 (-0.01)	-1.34	-0.98	-1.34	-0.98	-0.36 (-1.42)	-1.12 (-1.07)	
LY	dtoat	1927	2021	1947	0.87	-2.17 (-0.82)	-4.18	-2.97	-3.95	-2.81	0.58 (0.66)	-0.18 (-0.13)	
M2	ygap	1954	2021	1974	1.27	1.80 (1.00)	-12.94	-5.39	-9.37	-4.03	0.94 (0.92)	0.54 (0.33)	
M1	rdsp	1927	2021	1947	0.96	-2.27 (-0.62)	-3.66	-2.61	-3.66	-2.61	-1.21 (-1.65)	-1.97 (-1.78)	
MA	rsvix	1997	2021	2007	0.18	-0.64 (-0.27)	-23.34	-2.65	-5.80	-0.77	-1.28 (-0.81)	-2.51 (-0.66)	
MR	gpce	1948	2021	1968	7.50	-4.25 (-1.94)	8.33	4.81***	8.33	4.81***	0.99 (2.15)	-0.74 (-0.62)	
MR	gip	1927	2021	1947	0.05	-0.52 (-0.17)	-0.47	-0.35	-0.47	-0.35	0.08 (0.47)	-0.67 (-0.67)	
NRTZ	tchi	1952	2021	1972	2.45	-2.42 (-1.28)	-3.72	-1.76	-3.72	-1.76	-0.54 (-0.44)	-1.04 (-0.70)	
PST	house	1930	2021	1950	0.36	1.38 (0.65)	0.10	0.07	0.10	0.07	0.39 (1.15)	-0.72 (-0.63)	
PW	avgcor	1927	2021	1947	0.05	0.52 (0.19)	-4.90	-3.45	-4.90	-3.45	-1.37 (-2.57)	-2.12 (-1.83)	
RRZ	shtint	1974	2021	1994	4.24	-3.72 (-1.20)	-0.97	-0.26	-0.97	-0.26	-1.14 (-0.73)	-2.97 (-1.57)	
Y	disag	1982	2021	2002	2.58	-3.17 (-0.94)	-18.02	-2.90	9.08	1.90**	5.16 (1.62)	1.54 (0.45)	
CSa	d/p	1927	2021	1947	2.04	3.43 (1.24)	-11.84	-7.84	-0.40	-0.30	0.33 (0.20)	-0.43 (-0.23)	
CSb	d/y	1927	2021	1947	1.93	3.33 (1.30)	-13.20	-8.63	-0.01	-0.01	0.35 (0.20)	-0.41 (-0.22)	
CSc	e/p	1927	2021	1947	2.11	3.48 (2.00)	-4.74	-3.35	3.17	2.42**	0.68 (0.41)	-0.07 (-0.04)	
CSd	d/e	1927	2021	1947	0.06	0.59 (0.19)	-4.14	-2.94	-2.31	-1.67	-0.05 (-0.17)	-0.81 (-0.68)	
CSe	svar	1927	2021	1947	0.52	1.67 (0.37)	-4.36	-3.09	-3.95	-2.81	0.26 (0.43)	-0.49 (-0.39)	
KS	b/m	1927	2021	1947	3.05	4.12 (1.61)	-21.80	-13.25	-12.06	-7.96	-0.46 (-0.26)	-1.21 (-0.69)	
BMRR	ntis	1927	2021	1947	10.88	-7.66 (-1.64)	-27.90	-16.14	-27.11	-15.78	-0.72 (-0.58)	-1.47 (-1.29)	
BW	eqis	1928	2021	1948	4.05	-4.68 (-1.76)	-13.89	-8.90	-6.99	-4.77	-0.37 (-0.41)	-1.26 (-0.89)	
Ca	tby	1927	2021	1947	0.67	-1.93 (-1.14)	-4.02	-2.86	-1.70	-1.23	0.16 (0.29)	-0.59 (-0.45)	
FFa	lty	1927	2021	1947	0.13	-0.86 (-0.45)	-10.05	-6.76	-2.18	-1.58	0.31 (0.47)	-0.45 (-0.31)	
FFb	ltr	1927	2021	1947	0.20	-1.04 (-0.46)	-11.42	-7.59	-10.95	-7.31	-1.49 (-1.78)	-2.25 (-1.83)	
FFc	tms	1927	2021	1947	1.22	2.56 (1.56)	-0.78	-0.57	-0.78	-0.57	0.15 (0.35)	-0.61 (-0.60)	
FFd	dfy	1927	2021	1947	1.33	2.68 (0.76)	-10.52	-7.05	-5.93	-4.14	-0.32 (-0.35)	-1.07 (-0.84)	
FFe	dfr	1927	2021	1947	0.04	0.44 (0.26)	-3.77	-2.69	-1.66	-1.21	-0.38 (-0.71)	-1.13 (-0.97)	
FS	infl	1927	2021	1947	1.67	3.00 (0.62)	-37.70	-20.26	-32.99	-18.36	-2.30 (-2.44)	-3.05 (-2.57)	
Co	i/k	1948	2021	1968	7.18	-4.20 (-2.25)	4.95	2.76**	4.98	2.78**	2.71 (2.46)	0.98 (0.65)	
LL	cay	1953	2021	1973	0.53	1.49 (0.54)	-16.46	-6.78	-14.90	-6.23	0.69 (0.42)	0.14 (0.07)	

Variable s		Sample		OOS Beg	IS		OOS		OOS-CT		Util Difference	
		Beg	End		R ²	Slope	R ²	MSE-F	R ²	MSE-F	U _{cond} − U _{uncond}	U _{cond} − U _{100%}
Panel B: Predicting simple returns												
AMP	pce	1954	2021	1974	10.34	−5.52 (−2.79)	−0.01	−0.01	0.04	0.02	−0.06 (−0.04)	−0.86 (−0.58)
BH	vp	1991	2021	2001	1.51	−1.87 (−0.74)	−10.60	−1.92	0.05	0.01	1.06 (0.92)	−2.66 (−1.09)
BPS	impvar	1997	2021	2007	2.75	−2.64 (−0.96)	−1.02	−0.14	0.39	0.05	−0.20 (−0.34)	−2.21 (−0.63)
BTZ	vrp	1991	2021	2001	1.50	−1.91 (−0.82)	−3.47	−0.67	−3.09	−0.60	0.17 (0.19)	−3.56 (−1.57)
BY	govik	1948	2021	1968	1.85	2.34 (1.19)	−8.70	−4.24	−0.17	−0.09	2.30 (1.19)	0.39 (0.18)
CEP	lzrt	1927	2021	1947	7.94	10.17 (2.96)	1.61	1.21*	4.82	3.75**	2.02 (0.89)	−0.64 (−0.36)
CGMS	skew	1952	2019	1972	1.65	2.14 (0.85)	−0.37	−0.17	−0.37	−0.17	0.11 (0.33)	−0.75 (−0.52)
CGP	crdstd	1991	2021	2001	6.10	−3.88 (−1.55)	−5.77	−1.09	6.35	1.36*	3.23 (2.32)	−0.50 (−0.21)
CP	ogap	1927	2021	1947	2.38	−4.00 (−1.05)	−26.05	−15.29	−26.05	−15.29	−0.23 (−0.41)	−2.89 (−2.28)
DJM	wtexas	1927	2021	1947	0.18	−1.12 (−0.63)	−14.75	−9.51	−7.79	−5.35	−0.17 (−0.16)	−2.84 (−2.24)
HHT	accrul	1966	2021	1986	9.38	5.21 (2.58)	12.03	4.79***	7.64	2.90**	0.20 (0.20)	−2.13 (−1.04)
HHT	cfacc	1966	2021	1986	4.13	−3.43 (−1.53)	−5.25	−1.75	10.53	4.12***	1.50 (1.43)	−0.82 (−0.52)
HJTZ	sntm	1966	2020	1986	5.32	3.81 (1.40)	−22.24	−6.19	−18.52	−5.31	−4.36 (−3.16)	−6.72 (−3.14)
JT	ndrbl	1959	2021	1979	7.45	−4.53 (−2.12)	5.86	2.61**	5.73	2.55**	1.95 (1.16)	0.98 (0.44)
JZZ	skvw	1927	2021	1947	4.62	5.61 (1.33)	−39.77	−21.05	−30.12	−17.13	1.81 (0.84)	−0.86 (−0.48)
KJ	tail	1927	2021	1947	1.00	2.60 (1.21)	−0.25	−0.18	−0.25	−0.18	0.22 (0.32)	−2.44 (−2.14)
KP	fbm	1927	2021	1947	24.06	12.72 (2.62)	−23.01	−13.84	−19.90	−12.28	−1.56 (−1.06)	−4.22 (−2.64)
LY	dtoy	1927	2021	1947	4.00	−5.20 (−0.85)	2.02	1.53**	2.02	1.53**	0.79 (0.71)	−1.88 (−1.46)
LY	dtoat	1927	2021	1947	3.54	−4.91 (−1.38)	0.60	0.44*	0.60	0.44*	2.49 (1.55)	−0.18 (−0.15)
M2	ygap	1954	2021	1974	1.65	2.25 (1.10)	−13.81	−5.70	−10.84	−4.60	0.81 (0.66)	0.01 (0.01)
M1	rdsp	1927	2021	1947	0.12	0.88 (0.20)	−2.49	−1.80	−2.49	−1.80	0.14 (0.24)	−2.52 (−2.00)
MA	rsvix	1997	2021	2007	0.20	−0.71 (−0.29)	−21.14	−2.44	−6.02	−0.79	−0.82 (−0.55)	−2.83 (−0.80)
MR	gpce	1948	2021	1968	8.04	−4.84 (−1.99)	9.07	5.29***	9.07	5.29***	1.44 (2.22)	−0.48 (−0.42)
MR	gip	1927	2021	1947	1.59	−3.27 (−0.81)	−0.02	−0.01	−0.02	−0.01	0.09 (0.25)	−2.57 (−2.24)
NRTZ	tchi	1952	2021	1972	2.81	−2.84 (−1.29)	−3.87	−1.82	−3.87	−1.82	−0.35 (−0.23)	−1.15 (−0.76)
PST	house	1930	2021	1950	0.00	−0.06 (−0.02)	−5.73	−3.85	−5.73	−3.85	−0.43 (−0.71)	−2.83 (−2.17)
PW	avgcor	1927	2021	1947	0.95	2.52 (0.70)	0.33	0.25	0.33	0.25	1.66 (2.15)	−1.00 (−0.95)
RRZ	shtint	1974	2021	1994	4.00	−3.98 (−1.23)	−2.58	−0.68	−2.58	−0.68	−0.47 (−0.29)	−3.19 (−1.53)
Y	disag	1982	2021	2002	2.56	−3.48 (−0.97)	−14.29	−2.38	4.76	0.95*	2.85 (1.06)	−0.26 (−0.08)
CSa	d/p	1927	2021	1947	5.75	6.44 (1.79)	−26.91	−15.69	−0.34	−0.25	−0.28 (−0.10)	−2.95 (−1.36)
CSb	d/y	1927	2021	1947	4.93	5.95 (1.82)	−23.63	−14.15	−0.77	−0.57	0.13 (0.05)	−2.53 (−1.17)
CSc	e/p	1927	2021	1947	2.65	4.36 (2.19)	−2.38	−1.72	1.99	1.51**	0.51 (0.29)	−2.15 (−1.37)
CSd	d/e	1927	2021	1947	2.16	3.87 (0.94)	−15.16	−9.74	−6.27	−4.37	1.78 (0.96)	−0.89 (−0.52)
CSe	svar	1927	2021	1947	6.08	6.39 (0.95)	−19.90	−12.28	−18.98	−11.80	1.70 (0.88)	−0.96 (−0.68)
KS	b/m	1927	2021	1947	5.36	6.11 (1.60)	−24.57	−14.60	−16.97	−10.74	−0.93 (−0.40)	−3.59 (−1.82)
BMRR	ntis	1927	2021	1947	5.94	−6.33 (−1.58)	−24.57	−14.59	−24.57	−14.59	−1.24 (−1.00)	−3.90 (−2.67)
BW	eqis	1928	2021	1948	2.40	−4.03 (−1.58)	−14.02	−8.98	−11.84	−7.73	0.44 (0.32)	−1.87 (−1.15)
Ca	tby	1927	2021	1947	0.83	−2.39 (−1.34)	−5.17	−3.64	−4.48	−3.17	0.77 (0.60)	−1.89 (−1.18)
FFa	lty	1927	2021	1947	0.28	−1.40 (−0.68)	−22.17	−13.43	−16.87	−10.68	0.40 (0.25)	−2.26 (−1.28)
FFb	ltr	1927	2021	1947	1.11	−2.74 (−0.97)	−34.36	−18.92	−28.20	−16.28	−1.39 (−0.71)	−4.05 (−2.42)
FFc	tms	1927	2021	1947	0.95	2.53 (1.38)	−0.85	−0.62	−0.85	−0.62	0.33 (0.48)	−2.34 (−1.98)
FFd	dfy	1927	2021	1947	7.30	7.02 (1.22)	−17.59	−11.07	−11.21	−7.46	−0.60 (−0.28)	−3.26 (−2.14)
FFe	dfr	1927	2021	1947	0.00	−0.06 (−0.03)	−7.01	−4.85	−4.63	−3.27	−0.19 (−0.17)	−2.85 (−2.02)
FS	infl	1927	2021	1947	0.00	−0.08 (−0.02)	−10.85	−7.24	−10.85	−7.24	−0.40 (−1.18)	−3.06 (−2.46)
Co	i/k	1948	2021	1968	5.67	−4.10 (−2.04)	3.05	1.67**	3.05	1.67**	2.05 (1.59)	0.13 (0.08)
LL	cay	1953	2021	1973	0.42	1.45 (0.48)	−12.34	−5.27	−12.00	−5.14	0.15 (0.08)	−0.73 (−0.36)

Table IA.10: Untilted Investment Strategies on Monthly Returns

Variable		Sample		OOS		Difference		
		Beg	End	Beg	Model	Uncond	Return	SR
Panel A: Investing \$1								
BH	vp	199001	202112	200001	-0.91	7.05	-7.97	-0.57
BPS	impvar	199601	202112	200601	-5.67	10.62	-16.29	-0.78
BTZ	vrp	199001	202112	200001	4.41	7.05	-2.64	-0.13
CEP	lzrt	192601	202112	194601	-2.85	8.10	-10.96	-0.43
CP	ogap	192601	202112	194601	5.52	8.10	-2.58	-0.20
DJM	wtexas	192601	202112	194601	5.47	8.10	-2.63	-0.16
HJTZ	sntm	196507	202012	198507	3.96	8.88	-4.92	-0.28
JT	ndrbl	195802	202112	197802	1.20	8.94	-7.74	-0.52
JZZ	skvw	192607	202112	194607	-4.65	8.05	-12.70	-0.57
KJ	tail	192607	202112	194607	1.74	8.05	-6.32	-0.35
KP	fbm	192606	202112	194606	1.53	8.00	-6.46	-0.30
LY	dtoy	192601	202112	194601	-0.55	8.10	-8.66	-0.45
LY	dtoat	192601	202112	194601	-4.20	8.10	-12.30	-0.49
M2	ygap	195304	202112	197304	-3.22	7.55	-10.77	-0.43
M1	rdsp	192609	202112	194609	-6.82	8.20	-15.02	-0.61
MA	rsvix	199601	202112	200601	-0.03	10.62	-10.66	-0.65
NRTZ	tchi	195101	202112	197101	3.08	7.56	-4.48	-0.19
PW	avgcor	192603	202112	194603	2.91	8.12	-5.21	-0.27
RRZ	shtint	197301	202112	199301	8.02	9.24	-1.22	-0.07
Y	disag	198112	202112	200112	8.11	9.15	-1.04	-0.11
CSa	d/p	192601	202112	194601	-3.35	8.10	-11.46	-0.45
CSb	d/y	192602	202112	194602	-3.04	8.02	-11.06	-0.43
CSc	e/p	192601	202112	194601	-0.71	8.10	-8.82	-0.38
CSD	d/e	192601	202112	194601	-6.09	8.10	-14.19	-0.54
CSe	svar	192601	202112	194601	-0.37	8.10	-8.48	-0.47
KS	b/m	192601	202112	194601	-3.92	8.10	-12.03	-0.50
BMRR	ntis	192612	202112	194612	4.70	8.38	-3.68	-0.22
Ca	tby	192601	202112	194601	-3.19	8.10	-11.30	-0.46
FFa	lty	192601	202112	194601	-1.74	8.10	-9.84	-0.40
FFb	ltr	192601	202112	194601	3.78	8.10	-4.32	-0.21
FFc	tms	192601	202112	194601	0.84	8.10	-7.26	-0.33
FFd	dfy	192601	202112	194601	-0.70	8.10	-8.80	-0.44
FFe	dfr	192601	202112	194601	2.44	8.10	-5.66	-0.26
FS	infl	192601	202112	194601	4.72	8.10	-3.38	-0.16

Variable		Sample		OOS			Difference	
		Beg	End	Beg	Model	Uncond	Return	SR
Panel B: Investing \$z								
BH	vp	199001	202112	200001	7.89	11.04	-3.15	-0.55
BPS	impvar	199601	202112	200601	-3.43	6.56	-9.99	-0.72
BTZ	vrp	199001	202112	200001	1.43	8.44	-7.01	-0.08
CEP	lzrt	192601	202112	194601	-4.49	7.90	-12.39	-0.28
CP	ogap	192601	202112	194601	5.18	5.96	-0.78	-0.13
DJM	wtexas	192601	202112	194601	4.87	5.47	-0.60	-0.01
HJTZ	sntm	196507	202012	198507	3.94	6.71	-2.78	-0.20
JT	ndrbl	195802	202112	197802	4.47	9.66	-5.18	-0.38
JZZ	skvw	192607	202112	194607	-3.56	4.92	-8.48	-0.31
KJ	tail	192607	202112	194607	1.05	6.14	-5.09	-0.28
KP	fbm	192606	202112	194606	0.25	2.47	-2.22	-0.32
LY	dtoy	192601	202112	194601	1.98	4.02	-2.04	-0.40
LY	dtoat	192601	202112	194601	-3.95	5.89	-9.84	-0.47
M2	ygap	195304	202112	197304	-1.83	6.17	-8.00	-0.20
M1	rdsp	192609	202112	194609	-1.75	2.82	-4.57	-0.52
MA	rsvix	199601	202112	200601	0.08	4.51	-4.43	-0.67
NRTZ	tchi	195101	202112	197101	-0.14	1.74	-1.88	-0.04
PW	avgcor	192603	202112	194603	4.43	6.50	-2.07	-0.15
RRZ	shtint	197301	202112	199301	15.21	13.90	1.31	0.06
Y	disag	198112	202112	200112	14.35	14.16	0.20	0.07
CSa	d/p	192601	202112	194601	-4.58	8.78	-13.36	-0.34
CSb	d/y	192602	202112	194602	-4.33	8.55	-12.89	-0.31
CSc	e/p	192601	202112	194601	0.07	7.41	-7.33	-0.21
CSd	d/e	192601	202112	194601	-4.29	7.91	-12.20	-0.43
CSe	svar	192601	202112	194601	-0.03	0.78	-0.81	-0.46
KS	b/m	192601	202112	194601	-3.17	7.06	-10.23	-0.38
BMRR	ntis	192612	202112	194612	3.29	4.18	-0.89	-0.16
Ca	tby	192601	202112	194601	-1.14	5.82	-6.96	-0.15
FFa	lty	192601	202112	194601	-2.93	7.69	-10.62	-0.18
FFb	ltr	192601	202112	194601	7.77	7.00	0.77	0.03
FFc	tms	192601	202112	194601	2.78	5.19	-2.41	-0.10
FFd	dfy	192601	202112	194601	0.38	4.47	-4.09	-0.45
FFe	dfr	192601	202112	194601	1.82	3.65	-1.83	-0.04
FS	infl	192601	202112	194601	1.46	1.31	0.16	0.01

Table IA.11: Untilted Investment Strategies on Quarterly Returns

Variable		Sample		OOS			Difference	
		Beg	End	Beg	Model	Uncond	Return	SR
Panel A: Investing \$1								
AMP	pce	19534	20214	19734	5.91	8.00	-2.09	-0.10
BH	vp	19901	20214	20001	3.97	7.36	-3.39	-0.17
BPS	impvar	19961	20214	20061	-2.31	10.95	-13.26	-0.58
BTZ	vrp	19901	20214	20001	-5.37	7.36	-12.73	-0.61
BY	govik	19471	20214	19671	-7.42	7.45	-14.87	-0.47
CEP	lzrt	19261	20214	19461	-3.31	8.36	-11.67	-0.44
CGP	crdstd	19902	20214	20002	6.21	7.39	-1.18	-0.04
CP	ogap	19261	20214	19461	5.90	8.36	-2.46	-0.17
DJM	wtxas	19261	20214	19461	4.75	8.36	-3.61	-0.20
HJTZ	sntm	19653	20204	19853	4.76	9.16	-4.41	-0.22
JT	ndrbl	19581	20214	19781	1.62	8.99	-7.37	-0.46
JZZ	skvw	19263	20214	19463	-2.60	8.31	-10.92	-0.43
KJ	tail	19263	20214	19463	0.00	8.31	-8.32	-0.41
KP	fbm	19262	20214	19462	2.76	8.32	-5.56	-0.25
LY	dtoy	19261	20214	19461	0.06	8.36	-8.30	-0.38
LY	dtoat	19261	20214	19461	-1.93	8.36	-10.28	-0.37
M2	ygap	19532	20214	19732	-3.22	7.83	-11.05	-0.41
M1	rdsp	19263	20214	19463	-4.90	8.31	-13.22	-0.49
MA	rsvix	19961	20214	20061	4.77	10.95	-6.18	-0.30
NRTZ	tchi	19511	20214	19711	1.99	7.83	-5.84	-0.22
PW	avgcor	19261	20214	19461	3.30	8.36	-5.06	-0.24
RRZ	shtint	19731	20214	19931	9.21	9.55	-0.34	-0.02
Y	disag	19814	20214	20014	-8.15	9.89	-18.04	-0.60
CSa	d/p	19261	20214	19461	-3.07	8.36	-11.43	-0.40
CSb	d/y	19261	20214	19461	-3.06	8.36	-11.42	-0.39
CSc	e/p	19261	20214	19461	-1.31	8.36	-9.67	-0.37
CSd	d/e	19261	20214	19461	-6.52	8.36	-14.87	-0.52
CSe	svar	19261	20214	19461	1.38	8.36	-6.98	-0.34
KS	b/m	19261	20214	19461	-4.19	8.36	-12.55	-0.47
BMRR	ntis	19264	20214	19464	4.42	8.59	-4.18	-0.25
Ca	tby	19261	20214	19461	-3.63	8.36	-11.99	-0.46
FFa	lty	19261	20214	19461	-1.85	8.36	-10.20	-0.38
FFb	ltr	19261	20214	19461	2.16	8.36	-6.20	-0.30
FFc	tms	19261	20214	19461	1.30	8.36	-7.06	-0.30
FFd	dfy	19261	20214	19461	-0.69	8.36	-9.05	-0.41
FFe	dfr	19261	20214	19461	-0.84	8.36	-9.19	-0.42
FS	infl	19261	20214	19461	3.91	8.36	-4.45	-0.20
Co	i/k	19471	20214	19671	4.35	7.45	-3.10	-0.13
LL	cay	19521	20214	19721	2.52	7.79	-5.27	-0.24

Variable	Sample		OOS			Difference		
		Beg	End	Beg	Model	Uncond	Return	SR
Panel B: Investing \$z								
AMP	pce	19534	20214	19734	8.59	8.87	-0.28	-0.02
BH	vp	19901	20214	20001	11.63	14.13	-2.50	-0.39
BPS	impvar	19961	20214	20061	2.26	9.56	-7.30	-0.46
BTZ	vrp	19901	20214	20001	-0.11	17.17	-17.27	-0.58
BY	govik	19471	20214	19671	-4.62	4.47	-9.09	-0.27
CEP	lzrt	19261	20214	19461	-5.85	8.82	-14.67	-0.29
CGP	crdstd	19902	20214	20002	5.74	0.78	4.96	0.11
CP	ogap	19261	20214	19461	5.30	5.95	-0.65	-0.10
DJM	wtxas	19261	20214	19461	3.23	9.31	-6.07	-0.14
HJTZ	sntm	19653	20204	19853	3.43	6.29	-2.86	-0.17
JT	ndrbl	19581	20214	19781	7.87	11.33	-3.46	-0.28
JZZ	skvw	19263	20214	19463	-3.33	6.20	-9.53	-0.41
KJ	tail	19263	20214	19463	0.80	6.00	-5.20	-0.25
KP	f6m	19262	20214	19462	0.50	2.42	-1.93	-0.28
LY	dtoy	19261	20214	19461	3.22	5.01	-1.79	-0.35
LY	dtoat	19261	20214	19461	-3.71	6.26	-9.97	-0.42
M2	ygap	19532	20214	19732	-1.96	7.24	-9.21	-0.20
M1	rdsp	19263	20214	19463	-1.94	2.68	-4.62	-0.42
MA	rsvix	19961	20214	20061	9.02	12.12	-3.10	-0.37
NRTZ	tchi	19511	20214	19711	-2.46	3.56	-6.02	-0.12
PW	avgcor	19261	20214	19461	6.27	7.75	-1.48	-0.10
RRZ	shtint	19731	20214	19931	15.55	13.37	2.18	0.10
Y	disag	19814	20214	20014	-12.32	12.14	-24.46	-0.44
CSa	d/p	19261	20214	19461	-4.31	8.40	-12.71	-0.30
CSb	d/y	19261	20214	19461	-4.36	8.59	-12.95	-0.29
CSc	e/p	19261	20214	19461	-0.17	7.83	-8.00	-0.20
CSd	d/e	19261	20214	19461	-3.67	7.94	-11.61	-0.39
CSe	svar	19261	20214	19461	4.20	4.91	-0.71	-0.31
KS	b/m	19261	20214	19461	-2.91	7.32	-10.23	-0.35
BMRR	ntis	19264	20214	19464	3.61	4.40	-0.79	-0.14
Ca	tby	19261	20214	19461	-1.78	6.51	-8.30	-0.17
FFa	lty	19261	20214	19461	-3.55	8.47	-12.03	-0.19
FFb	ltr	19261	20214	19461	7.13	8.48	-1.35	-0.05
FFc	tms	19261	20214	19461	2.67	4.92	-2.25	-0.08
FFd	dfy	19261	20214	19461	0.49	4.53	-4.04	-0.44
FFe	dfr	19261	20214	19461	0.72	4.62	-3.89	-0.16
FS	infl	19261	20214	19461	0.40	2.93	-2.53	-0.14
Co	i/k	19471	20214	19671	2.99	5.26	-2.27	-0.07
LL	cay	19521	20214	19721	6.12	15.44	-9.32	-0.22

Table IA.12: Untilted Investment Strategies on Annual (Jan to Dec) Returns

Variable		Sample		OOS			Difference	
		Beg	End	Beg	Model	Uncond	Return	SR
Panel A: Investing \$1								
AMP	pce	1953	2021	1973	4.05	8.04	-3.99	-0.21
BH	vp	1990	2021	2000	-5.59	7.73	-13.31	-0.63
BPS	impvar	1996	2021	2006	-0.96	11.49	-12.45	-0.40
BTZ	vrp	1990	2021	2000	3.23	7.73	-4.49	-0.20
BY	govik	1947	2021	1967	-8.14	7.77	-15.91	-0.49
CEP	lzrt	1926	2021	1946	0.04	8.84	-8.80	-0.30
CGMS	skew	1951	2019	1971	7.10	7.59	-0.49	-0.02
CGP	crdst	1990	2021	2000	2.49	7.73	-5.24	-0.16
CP	ogap	1926	2021	1946	6.83	8.84	-2.02	-0.10
DJM	wtxas	1926	2021	1946	3.05	8.84	-5.79	-0.22
HHT	accrul	1965	2021	1985	0.37	10.32	-9.95	-0.40
HHT	cfacc	1965	2021	1985	6.85	10.32	-3.47	-0.14
HJTZ	sntm	1965	2020	1985	2.99	9.81	-6.82	-0.38
JT	ndrbl	1958	2021	1978	3.15	9.38	-6.23	-0.27
JZZ	skvw	1926	2021	1946	-3.45	8.84	-12.29	-0.39
KJ	tail	1926	2021	1946	5.16	8.84	-3.68	-0.13
KP	fbm	1926	2021	1946	-2.00	8.84	-10.84	-0.37
LY	dtoy	1926	2021	1946	-1.40	8.84	-10.24	-0.34
LY	dtoat	1926	2021	1946	-4.28	8.84	-13.13	-0.39
M2	ygap	1953	2021	1973	-2.67	8.04	-10.71	-0.36
M1	rdsp	1926	2021	1946	6.01	8.84	-2.84	-0.17
MA	rsvix	1996	2021	2006	-1.61	11.49	-13.10	-0.62
MR	gpce	1947	2021	1967	5.41	7.77	-2.36	-0.12
MR	gip	1926	2021	1946	6.69	8.84	-2.16	-0.12
NRTZ	tchi	1951	2021	1971	-1.31	8.22	-9.53	-0.41
PST	house	1929	2021	1949	-3.61	9.22	-12.82	-0.41
PW	avgcor	1926	2021	1946	0.02	8.84	-8.83	-0.34
RRZ	shtint	1973	2021	1993	7.04	10.14	-3.10	-0.15
Y	disag	1981	2021	2001	6.98	8.78	-1.80	-0.22
CSa	d/p	1926	2021	1946	-2.78	8.84	-11.62	-0.37
CSb	d/y	1926	2021	1946	-2.27	8.84	-11.11	-0.35
CSc	e/p	1926	2021	1946	-0.47	8.84	-9.31	-0.34
CSd	d/e	1926	2021	1946	-6.44	8.84	-15.29	-0.46
CSe	svar	1926	2021	1946	1.67	8.84	-7.17	-0.27
KS	b/m	1926	2021	1946	-3.25	8.84	-12.09	-0.41
BMRR	ntis	1926	2021	1946	4.57	8.84	-4.27	-0.21
BW	eqis	1927	2021	1947	3.17	9.09	-5.92	-0.23
Ca	tby	1926	2021	1946	-3.84	8.84	-12.68	-0.41
FFa	lty	1926	2021	1946	-2.88	8.84	-11.72	-0.38
FFb	ltr	1926	2021	1946	5.27	8.84	-3.58	-0.14
FFc	tms	1926	2021	1946	-0.03	8.84	-8.87	-0.29
FFd	dfy	1926	2021	1946	-0.80	8.84	-9.64	-0.39
FFe	dfr	1926	2021	1946	2.17	8.84	-6.67	-0.26
FS	infl	1926	2021	1946	5.81	8.84	-3.04	-0.15
Co	i/k	1947	2021	1967	3.93	7.77	-3.84	-0.14
LL	cay	1952	2021	1972	-1.10	8.19	-9.28	-0.30

Variable		Sample		OOS		Difference		
		Beg	End	Beg	Model	Uncond	Return	SR
Panel B: Investing \$z								
AMP	pce	1953	2021	1973	8.17	9.62	-1.44	-0.11
BH	vp	1990	2021	2000	0.53	6.94	-6.41	-0.57
BPS	impvar	1996	2021	2006	-1.44	9.84	-11.29	-0.70
BTZ	vrp	1990	2021	2000	5.11	8.69	-3.58	-0.16
BY	govik	1947	2021	1967	-4.45	4.39	-8.83	-0.26
CEP	lzrt	1926	2021	1946	0.01	5.21	-5.20	-0.12
CGMS	skew	1951	2019	1971	5.75	6.54	-0.79	-0.08
CGP	crdst	1990	2021	2000	0.04	4.13	-4.09	-0.10
CP	ogap	1926	2021	1946	5.06	5.66	-0.59	-0.08
DJM	wtexas	1926	2021	1946	1.36	6.07	-4.71	-0.09
HHT	accrul	1965	2021	1985	3.48	0.74	2.75	0.09
HHT	cfacc	1965	2021	1985	6.35	4.07	2.29	0.18
HJTZ	sntm	1965	2020	1985	6.31	9.23	-2.91	-0.27
JT	ndrbl	1958	2021	1978	8.39	9.86	-1.47	-0.06
JZZ	skvw	1926	2021	1946	-1.20	5.28	-6.49	-0.18
KJ	tail	1926	2021	1946	0.73	6.56	-5.83	-0.26
KP	fbm	1926	2021	1946	0.56	4.28	-3.73	-0.24
LY	dtoy	1926	2021	1946	-2.81	4.48	-7.30	-0.26
LY	dtoat	1926	2021	1946	-4.43	6.33	-10.76	-0.34
M2	ygap	1953	2021	1973	-1.14	7.48	-8.62	-0.21
M1	rdsp	1926	2021	1946	4.89	4.06	0.83	0.07
MA	rsvix	1996	2021	2006	1.14	9.97	-8.83	-0.65
MR	gpce	1947	2021	1967	3.99	3.64	0.35	0.03
MR	gip	1926	2021	1946	3.30	3.43	-0.13	-0.04
NRTZ	tchi	1951	2021	1971	4.46	5.87	-1.41	-0.38
PST	house	1929	2021	1949	-0.45	3.49	-3.93	-0.35
PW	avgcor	1926	2021	1946	3.53	6.93	-3.41	-0.17
RRZ	shtint	1973	2021	1993	17.52	14.36	3.16	0.10
Y	disag	1981	2021	2001	12.53	13.86	-1.33	-0.22
CSa	d/p	1926	2021	1946	-4.24	8.17	-12.41	-0.27
CSb	d/y	1926	2021	1946	-4.39	8.52	-12.91	-0.27
CSc	e/p	1926	2021	1946	0.82	8.46	-7.64	-0.20
CSd	d/e	1926	2021	1946	-4.71	7.95	-12.66	-0.37
CSe	svar	1926	2021	1946	-1.96	3.49	-5.46	-0.15
KS	b/m	1926	2021	1946	-2.87	7.36	-10.23	-0.32
BMRR	ntis	1926	2021	1946	3.63	4.41	-0.78	-0.17
BW	eqis	1927	2021	1947	2.71	3.32	-0.61	-0.05
Ca	tby	1926	2021	1946	-3.72	8.55	-12.28	-0.24
FFa	lty	1926	2021	1946	-4.42	8.83	-13.26	-0.23
FFb	ltr	1926	2021	1946	7.73	9.93	-2.20	-0.08
FFc	tms	1926	2021	1946	0.95	6.84	-5.90	-0.17
FFd	dfy	1926	2021	1946	0.93	4.77	-3.84	-0.34
FFe	dfr	1926	2021	1946	-1.42	10.00	-11.42	-0.27
FS	infl	1926	2021	1946	1.53	4.03	-2.50	-0.12
Co	i/k	1947	2021	1967	1.94	4.40	-2.46	-0.07
LL	cay	1952	2021	1972	-3.11	12.83	-15.94	-0.25

Table IA.13: Untilted Investment Strategies on Annual (Jul to Jun) Returns

Variable		Sample		OOS			Difference	
		Beg	End	Beg	Model	Uncond	Return	SR
Panel A: Investing \$1								
AMP	pce	1954	2021	1974	5.34	8.03	-2.69	-0.15
BH	vp	1991	2021	2001	5.71	7.27	-1.56	-0.06
BPS	impvar	1997	2021	2007	8.35	11.07	-2.73	-0.35
BTZ	vrp	1991	2021	2001	2.63	7.27	-4.64	-0.23
BY	govik	1948	2021	1968	-7.46	7.37	-14.83	-0.44
CEP	lzrt	1927	2021	1947	0.61	8.62	-8.01	-0.29
CGMS	skew	1952	2019	1972	8.23	7.07	1.16	0.06
CGP	crdstd	1991	2021	2001	0.85	7.27	-6.42	-0.21
CP	ogap	1927	2021	1947	7.52	8.62	-1.10	-0.06
DJM	wtexas	1927	2021	1947	2.57	8.62	-6.05	-0.28
HHT	accrul	1966	2021	1986	-1.19	9.59	-10.79	-0.42
HHT	cfacc	1966	2021	1986	4.22	9.59	-5.37	-0.24
HJTZ	sntm	1966	2020	1986	-3.59	8.70	-12.29	-0.66
JT	ndrbl	1959	2021	1979	4.83	9.31	-4.49	-0.20
JZZ	skvw	1927	2021	1947	-5.49	8.62	-14.11	-0.44
KJ	tail	1927	2021	1947	4.01	8.62	-4.61	-0.17
KP	fbm	1927	2021	1947	-0.39	8.62	-9.01	-0.38
LY	dtoy	1927	2021	1947	1.15	8.62	-7.47	-0.30
LY	dtoat	1927	2021	1947	-5.68	8.62	-14.30	-0.42
M2	ygap	1954	2021	1974	-1.80	8.03	-9.82	-0.36
M1	rdsp	1927	2021	1947	-5.49	8.62	-14.11	-0.44
MA	rsvix	1997	2021	2007	5.89	11.07	-5.18	-0.21
MR	gpce	1948	2021	1968	7.49	7.37	0.12	0.01
MR	gip	1927	2021	1947	4.86	8.62	-3.76	-0.20
NRTZ	tchi	1952	2021	1972	0.24	7.74	-7.50	-0.29
PST	house	1930	2021	1950	3.27	9.13	-5.86	-0.27
PW	avgcor	1927	2021	1947	-0.67	8.62	-9.29	-0.31
RRZ	shtint	1974	2021	1994	8.38	9.61	-1.23	-0.07
Y	disag	1982	2021	2002	-4.54	8.64	-13.19	-0.46
CSa	d/p	1927	2021	1947	-2.83	8.62	-11.45	-0.40
CSb	d/y	1927	2021	1947	-2.16	8.62	-10.78	-0.37
CSc	e/p	1927	2021	1947	0.29	8.62	-8.33	-0.31
CSd	d/e	1927	2021	1947	-6.14	8.62	-14.76	-0.43
CSe	svar	1927	2021	1947	-2.48	8.62	-11.10	-0.39
KS	b/m	1927	2021	1947	-1.51	8.62	-10.13	-0.37
BMRR	ntis	1927	2021	1947	4.13	8.62	-4.49	-0.20
BW	eqis	1928	2021	1948	0.94	8.95	-8.01	-0.29
Ca	tby	1927	2021	1947	-2.47	8.62	-11.09	-0.35
FFa	lty	1927	2021	1947	-2.99	8.62	-11.61	-0.37
FFb	ltr	1927	2021	1947	1.22	8.62	-7.40	-0.33
FFc	tms	1927	2021	1947	0.29	8.62	-8.33	-0.28
FFd	dfy	1927	2021	1947	-1.22	8.62	-9.84	-0.39
FFe	dfr	1927	2021	1947	-0.25	8.62	-8.87	-0.31
FS	infl	1927	2021	1947	5.74	8.62	-2.88	-0.13
Co	i/k	1948	2021	1968	1.34	7.37	-6.03	-0.20
LL	cay	1953	2021	1973	-2.99	7.76	-10.75	-0.36

Variable		Sample		OOS			Difference	
		Beg	End	Beg	Model	Uncond	Return	SR
Panel B: Investing \$z								
AMP	pce	1954	2021	1974	8.20	8.98	-0.78	-0.08
BH	vp	1991	2021	2001	2.37	4.21	-1.83	-0.08
BPS	impvar	1997	2021	2007	4.85	8.95	-4.10	-0.28
BTZ	vrp	1991	2021	2001	4.03	5.58	-1.55	-0.11
BY	govik	1948	2021	1968	-5.20	5.16	-10.36	-0.29
CEP	lzrt	1927	2021	1947	0.36	4.55	-4.19	-0.11
CGMS	skew	1952	2019	1972	6.27	6.34	-0.07	-0.01
CGP	crdst	1991	2021	2001	1.38	2.39	-1.01	-0.04
CP	ogap	1927	2021	1947	4.42	4.46	-0.04	-0.01
DJM	wtexas	1927	2021	1947	-0.65	6.65	-7.30	-0.24
HHT	accrul	1966	2021	1986	3.11	0.05	3.06	0.11
HHT	cfacc	1966	2021	1986	3.58	5.68	-2.10	-0.13
HJTZ	sntm	1966	2020	1986	-0.37	5.14	-5.51	-0.47
JT	ndrbl	1959	2021	1979	10.58	11.72	-1.14	-0.06
JZZ	skvw	1927	2021	1947	-3.78	5.71	-9.49	-0.29
KJ	tail	1927	2021	1947	0.69	6.58	-5.89	-0.21
KP	fbm	1927	2021	1947	0.14	3.00	-2.86	-0.22
LY	dtoy	1927	2021	1947	2.60	4.51	-1.91	-0.35
LY	dtoat	1927	2021	1947	-4.57	5.91	-10.47	-0.35
M2	ygap	1954	2021	1974	-0.24	6.75	-6.99	-0.27
M1	rdsp	1927	2021	1947	-4.82	3.75	-8.57	-0.35
MA	rsvix	1997	2021	2007	1.11	7.85	-6.74	-0.33
MR	gpce	1948	2021	1968	5.59	3.66	1.94	0.15
MR	gip	1927	2021	1947	2.39	2.98	-0.59	-0.11
NRTZ	tchi	1952	2021	1972	6.01	7.21	-1.20	-0.28
PST	house	1930	2021	1950	0.18	3.77	-3.58	-0.29
PW	avgcor	1927	2021	1947	1.48	7.66	-6.17	-0.24
RRZ	shtint	1974	2021	1994	15.11	11.70	3.42	0.15
Y	disag	1982	2021	2002	-7.54	10.57	-18.11	-0.38
CSa	d/p	1927	2021	1947	-3.94	7.81	-11.75	-0.27
CSb	d/y	1927	2021	1947	-4.08	8.05	-12.13	-0.27
CSc	e/p	1927	2021	1947	1.60	8.08	-6.48	-0.23
CSd	d/e	1927	2021	1947	-5.28	7.34	-12.62	-0.36
CSe	svar	1927	2021	1947	0.83	2.35	-1.52	-0.31
KS	b/m	1927	2021	1947	-2.92	7.01	-9.93	-0.33
BMRR	ntis	1927	2021	1947	2.71	3.93	-1.22	-0.18
BW	eqis	1928	2021	1948	0.03	4.82	-4.79	-0.21
Ca	tby	1927	2021	1947	-5.33	9.95	-15.28	-0.26
FFa	lty	1927	2021	1947	-6.47	10.30	-16.77	-0.22
FFb	ltr	1927	2021	1947	0.84	9.68	-8.84	-0.25
FFc	tms	1927	2021	1947	1.34	8.25	-6.91	-0.19
FFd	dfy	1927	2021	1947	-0.04	4.07	-4.12	-0.33
FFe	dfr	1927	2021	1947	-3.29	7.06	-10.35	-0.28
FS	infl	1927	2021	1947	1.18	3.43	-2.24	-0.11
Co	i/k	1948	2021	1968	0.90	4.37	-3.48	-0.10
LL	cay	1953	2021	1973	-1.73	7.23	-8.96	-0.19

Table IA.14: Equity-tilted Investment Strategies on Monthly Returns

Variable		Sample		OOS			Difference	
		Beg	End	Beg	Model	Uncond	Return	SR
Panel A: Investing \$1								
BH	<i>vp</i>	199001	202112	200001	3.66	7.05	-3.39	-0.38
BPS	<i>impvar</i>	199601	202112	200601	2.39	10.62	-8.24	-0.59
BTZ	<i>vrp</i>	199001	202112	200001	1.78	7.05	-5.27	-0.30
CEP	<i>lzrt</i>	192601	202112	194601	2.62	8.10	-5.49	-0.28
CP	<i>ogap</i>	192601	202112	194601	7.76	8.10	-0.35	-0.04
DJM	<i>wtexas</i>	192601	202112	194601	5.34	8.10	-2.77	-0.17
HJTZ	<i>sntm</i>	196507	202012	198507	8.11	8.88	-0.78	-0.09
JT	<i>ndrbl</i>	195802	202112	197802	5.95	8.94	-2.98	-0.33
JZZ	<i>skvw</i>	192607	202112	194607	0.58	8.05	-7.47	-0.44
KJ	<i>tail</i>	192607	202112	194607	4.23	8.05	-3.82	-0.31
KP	<i>fbm</i>	192606	202112	194606	4.44	8.00	-3.56	-0.32
LY	<i>dtoy</i>	192601	202112	194601	4.90	8.10	-3.20	-0.28
LY	<i>dtoat</i>	192601	202112	194601	0.55	8.10	-7.55	-0.45
M2	<i>ygap</i>	195304	202112	197304	2.72	7.55	-4.83	-0.22
M1	<i>rdsp</i>	192609	202112	194609	-1.37	8.20	-9.57	-0.49
MA	<i>rsvix</i>	199601	202112	200601	5.86	10.62	-4.76	-0.46
NRTZ	<i>tchi</i>	195101	202112	197101	7.81	7.56	0.25	0.01
PW	<i>avgcor</i>	192603	202112	194603	5.92	8.12	-2.20	-0.17
RRZ	<i>shtint</i>	197301	202112	199301	8.29	9.24	-0.96	-0.09
Y	<i>disag</i>	198112	202112	200112	9.98	9.15	0.83	0.22
CSa	<i>d/p</i>	192601	202112	194601	-1.30	8.10	-9.41	-0.40
CSb	<i>d/y</i>	192602	202112	194602	-1.26	8.02	-9.29	-0.39
CSc	<i>e/p</i>	192601	202112	194601	3.22	8.10	-4.88	-0.25
CSd	<i>d/e</i>	192601	202112	194601	-1.37	8.10	-9.47	-0.43
CSe	<i>svar</i>	192601	202112	194601	3.41	8.10	-4.70	-0.38
KS	<i>b/m</i>	192601	202112	194601	-0.08	8.10	-8.18	-0.38
BMRR	<i>ntis</i>	192612	202112	194612	7.59	8.38	-0.78	-0.08
Ca	<i>tby</i>	192601	202112	194601	2.68	8.10	-5.42	-0.26
FFa	<i>lty</i>	192601	202112	194601	3.19	8.10	-4.91	-0.24
FFb	<i>ltr</i>	192601	202112	194601	6.42	8.10	-1.69	-0.10
FFc	<i>tms</i>	192601	202112	194601	5.89	8.10	-2.21	-0.13
FFd	<i>dfy</i>	192601	202112	194601	1.77	8.10	-6.34	-0.47
FFe	<i>dfr</i>	192601	202112	194601	5.78	8.10	-2.33	-0.13
FS	<i>infl</i>	192601	202112	194601	7.26	8.10	-0.84	-0.06

Variable	Sample		OOS			Difference	
	Beg	End	Beg	Model	Uncond	Return	SR
Panel B: Investing \$z							
BH	vp	199001	202112	200001	10.37	11.13	-0.76 -0.43
BPS	impvar	199601	202112	200601	2.44	5.43	-2.99 -0.57
BTZ	vrp	199001	202112	200001	2.75	9.23	-6.48 -0.08
CEP	lzrt	192601	202112	194601	0.82	8.23	-7.41 -0.21
CP	ogap	192601	202112	194601	9.78	9.97	-0.19 -0.13
DJM	wtexas	192601	202112	194601	5.38	5.47	-0.08 0.00
HJTZ	sntm	196507	202012	198507	11.93	12.62	-0.69 -0.16
JT	ndrbl	195802	202112	197802	10.54	12.52	-1.98 -0.24
JZZ	skvw	192607	202112	194607	0.84	3.91	-3.07 -0.16
KJ	tail	192607	202112	194607	8.90	9.41	-0.50 -0.06
KP	fbm	192606	202112	194606	4.09	5.03	-0.94 -0.25
LY	dtoy	192601	202112	194601	4.44	4.67	-0.23 -0.23
LY	dtoat	192601	202112	194601	1.17	4.01	-2.84 -0.39
M2	ygap	195304	202112	197304	2.29	6.79	-4.51 -0.14
M1	rdsp	192609	202112	194609	0.37	1.83	-1.46 -0.40
MA	rsvix	199601	202112	200601	3.47	4.52	-1.05 -0.54
NRTZ	tchi	195101	202112	197101	10.89	10.23	0.66 0.04
PW	avgcor	192603	202112	194603	9.13	9.60	-0.47 -0.08
RRZ	shtint	197301	202112	199301	21.08	19.14	1.95 0.13
Y	disag	198112	202112	200112	19.21	19.15	0.07 0.22
CSa	d/p	192601	202112	194601	0.83	6.82	-6.00 -0.23
CSb	d/y	192602	202112	194602	1.12	6.78	-5.66 -0.21
CSc	e/p	192601	202112	194601	5.03	8.51	-3.48 -0.13
CSd	d/e	192601	202112	194601	-0.19	5.69	-5.88 -0.32
CSe	svar	192601	202112	194601	0.73	0.90	-0.17 -0.32
KS	b/m	192601	202112	194601	2.20	5.49	-3.28 -0.22
BMRR	ntis	192612	202112	194612	6.05	6.11	-0.06 -0.02
Ca	tby	192601	202112	194601	5.04	4.32	0.71 0.02
FFa	lty	192601	202112	194601	5.33	8.98	-3.65 -0.09
FFb	ltr	192601	202112	194601	10.96	8.97	1.99 0.09
FFc	tms	192601	202112	194601	8.93	8.28	0.66 0.05
FFd	dfy	192601	202112	194601	3.30	4.06	-0.76 -0.22
FFe	dfr	192601	202112	194601	4.98	5.46	-0.48 -0.01
FS	infl	192601	202112	194601	6.08	4.59	1.49 0.10

Table IA.15: Equity-tilted Investment Strategies on Quarterly Returns

Variable		Sample		OOS			Difference	
		Beg	End	Beg	Model	Uncond	Return	SR
Panel A: Investing \$1								
AMP	pce	19534	20214	19734	8.75	8.00	0.75	0.08
BH	vp	19901	20214	20001	2.97	7.36	-4.39	-0.44
BPS	impvar	19961	20214	20061	3.64	10.95	-7.31	-0.49
BTZ	vrp	19901	20214	20001	-3.16	7.36	-10.52	-0.70
BY	govik	19471	20214	19671	-0.06	7.45	-7.52	-0.29
CEP	lzrt	19261	20214	19461	2.28	8.36	-6.08	-0.29
CGP	crdst	19902	20214	20002	9.49	7.39	2.11	0.10
CP	ogap	19261	20214	19461	7.97	8.36	-0.39	-0.05
DJM	wtxas	19261	20214	19461	4.75	8.36	-3.61	-0.20
HJTZ	sntm	19653	20204	19853	8.24	9.16	-0.92	-0.10
JT	ndrbl	19581	20214	19781	6.97	8.99	-2.02	-0.22
JZZ	skvw	19263	20214	19463	1.09	8.31	-7.23	-0.41
KJ	tail	19263	20214	19463	4.62	8.31	-3.70	-0.28
KP	fbm	19262	20214	19462	5.14	8.32	-3.18	-0.29
LY	dtoy	19261	20214	19461	5.37	8.36	-2.99	-0.22
LY	dtoat	19261	20214	19461	0.59	8.36	-7.77	-0.40
M2	ygap	19532	20214	19732	2.44	7.83	-5.39	-0.23
M1	rdsp	19263	20214	19463	-1.52	8.31	-9.83	-0.44
MA	rsvix	19961	20214	20061	6.57	10.95	-4.38	-0.28
NRTZ	tchi	19511	20214	19711	6.32	7.83	-1.51	-0.07
PW	avgcor	19261	20214	19461	6.44	8.36	-1.92	-0.14
RRZ	shtint	19731	20214	19931	8.92	9.55	-0.63	-0.05
Y	disag	19814	20214	20014	1.18	9.89	-8.71	-0.31
CSa	d/p	19261	20214	19461	-1.38	8.36	-9.74	-0.38
CSb	d/y	19261	20214	19461	-1.30	8.36	-9.66	-0.38
CSc	e/p	19261	20214	19461	4.05	8.36	-4.31	-0.20
CSd	d/e	19261	20214	19461	0.23	8.36	-8.13	-0.35
CSe	svar	19261	20214	19461	5.12	8.36	-3.23	-0.23
KS	b/m	19261	20214	19461	0.14	8.36	-8.22	-0.36
BMRR	ntis	19264	20214	19464	8.08	8.59	-0.51	-0.06
Ca	tby	19261	20214	19461	3.76	8.36	-4.60	-0.20
FFa	lty	19261	20214	19461	2.73	8.36	-5.63	-0.25
FFb	ltr	19261	20214	19461	5.16	8.36	-3.20	-0.17
FFc	tms	19261	20214	19461	5.67	8.36	-2.69	-0.14
FFd	dfy	19261	20214	19461	1.92	8.36	-6.44	-0.47
FFe	dfr	19261	20214	19461	4.98	8.36	-3.37	-0.21
FS	infl	19261	20214	19461	7.81	8.36	-0.55	-0.04
Co	i/k	19471	20214	19671	7.08	7.45	-0.38	-0.02
LL	cay	19521	20214	19721	4.60	7.79	-3.19	-0.17

Variable		Sample		OOS			Difference	
		Beg	End	Beg	Model	Uncond	Return	SR
Panel B: Investing \$z								
AMP	pce	19534	20214	19734	15.28	15.34	-0.07	-0.03
BH	vp	19901	20214	20001	14.02	15.03	-1.01	-0.57
BPS	impvar	19961	20214	20061	8.21	10.93	-2.71	-0.38
BTZ	vrp	19901	20214	20001	3.75	13.24	-9.48	-0.45
BY	govik	19471	20214	19671	-0.59	1.78	-2.38	-0.11
CEP	lzrt	19261	20214	19461	-0.01	8.71	-8.72	-0.22
CGP	crdstd	19902	20214	20002	11.72	8.47	3.26	0.11
CP	ogap	19261	20214	19461	10.02	10.10	-0.08	-0.05
DJM	wtexas	19261	20214	19461	3.24	9.31	-6.07	-0.14
HJTZ	sntm	19653	20204	19853	12.01	12.52	-0.51	-0.09
JT	ndrbl	19581	20214	19781	14.09	14.48	-0.39	-0.07
JZZ	skvw	19263	20214	19463	0.44	5.81	-5.36	-0.33
KJ	tail	19263	20214	19463	9.24	9.55	-0.31	-0.03
KP	fbm	19262	20214	19462	5.36	6.01	-0.65	-0.23
LY	dtoy	19261	20214	19461	5.23	5.55	-0.33	-0.23
LY	dtoat	19261	20214	19461	1.65	4.55	-2.90	-0.37
M2	ygap	19532	20214	19732	2.20	7.99	-5.79	-0.15
M1	rdsp	19263	20214	19463	0.35	1.88	-1.53	-0.31
MA	rsvix	19961	20214	20061	12.66	13.66	-1.00	-0.41
NRTZ	tchi	19511	20214	19711	9.23	10.05	-0.82	-0.05
PW	avgcor	19261	20214	19461	11.13	11.02	0.11	0.02
RRZ	shtint	19731	20214	19931	21.51	18.51	3.00	0.19
Y	disag	19814	20214	20014	-3.18	9.55	-12.73	-0.36
CSa	d/p	19261	20214	19461	1.28	6.74	-5.45	-0.19
CSb	d/y	19261	20214	19461	1.33	6.96	-5.64	-0.19
CSc	e/p	19261	20214	19461	4.93	9.05	-4.12	-0.13
CSd	d/e	19261	20214	19461	0.52	6.01	-5.49	-0.29
CSe	svar	19261	20214	19461	5.02	5.16	-0.14	-0.20
KS	b/m	19261	20214	19461	2.60	5.92	-3.32	-0.20
BMRR	ntis	19264	20214	19464	6.45	6.55	-0.10	-0.04
Ca	tby	19261	20214	19461	4.75	5.11	-0.36	-0.01
FFa	lty	19261	20214	19461	4.71	9.46	-4.75	-0.10
FFb	ltr	19261	20214	19461	10.92	10.33	0.60	0.03
FFc	tms	19261	20214	19461	8.98	7.98	1.00	0.07
FFd	dfy	19261	20214	19461	3.49	4.21	-0.72	-0.22
FFe	dfr	19261	20214	19461	4.02	5.54	-1.51	-0.08
FS	infl	19261	20214	19461	5.88	6.63	-0.76	-0.08
Co	i/k	19471	20214	19671	9.47	10.05	-0.58	-0.04
LL	cay	19521	20214	19721	10.21	17.67	-7.46	-0.23

Table IA.16: Equity-tilted Investment Strategies on Annual (Jan to Dec) Returns

Variable		Sample		OOS			Difference	
		Beg	End	Beg	Model	Uncond	Return	SR
Panel A: Investing \$1								
AMP	pce	1953	2021	1973	7.90	8.04	-0.14	-0.01
BH	vp	1990	2021	2000	7.01	7.73	-0.71	-0.13
BPS	impvar	1996	2021	2006	3.46	11.49	-8.03	-0.40
BTZ	vrp	1990	2021	2000	6.06	7.73	-1.67	-0.08
BY	govik	1947	2021	1967	0.92	7.77	-6.85	-0.25
CEP	lzrt	1926	2021	1946	5.74	8.84	-3.11	-0.13
CGMS	skew	1951	2019	1971	7.31	7.59	-0.29	-0.02
CGP	crdstd	1990	2021	2000	6.30	7.73	-1.43	-0.07
CP	ogap	1926	2021	1946	8.55	8.84	-0.29	-0.03
DJM	wtxas	1926	2021	1946	3.09	8.84	-5.75	-0.22
HHT	accrul	1965	2021	1985	9.75	10.32	-0.56	-0.03
HHT	cfacc	1965	2021	1985	12.81	10.32	2.49	0.18
HJTZ	sntm	1965	2020	1985	8.98	9.81	-0.84	-0.17
JT	ndrbl	1958	2021	1978	7.78	9.38	-1.60	-0.09
JZZ	skvw	1926	2021	1946	5.92	8.84	-2.92	-0.13
KJ	tail	1926	2021	1946	4.52	8.84	-4.32	-0.24
KP	fbm	1926	2021	1946	7.23	8.84	-1.61	-0.11
LY	dtoy	1926	2021	1946	5.22	8.84	-3.62	-0.18
LY	dtoat	1926	2021	1946	0.83	8.84	-8.01	-0.32
M2	ygap	1953	2021	1973	2.42	8.04	-5.62	-0.22
M1	rdsp	1926	2021	1946	9.64	8.84	0.79*	0.16*
MA	rsvix	1996	2021	2006	6.74	11.49	-4.75	-0.37
MR	gpce	1947	2021	1967	7.53	7.77	-0.24	-0.02
MR	gip	1926	2021	1946	9.45	8.84	0.61	0.13
NRTZ	tchi	1951	2021	1971	7.37	8.22	-0.86	-0.12
PST	house	1929	2021	1949	4.98	9.22	-4.23	-0.24
PW	avgcor	1926	2021	1946	5.38	8.84	-3.47	-0.16
RRZ	shtint	1973	2021	1993	11.46	10.14	1.32	0.09
Y	disag	1981	2021	2001	8.78	8.78	0.00	
CSa	d/p	1926	2021	1946	0.69	8.84	-8.15	-0.30
CSb	d/y	1926	2021	1946	0.11	8.84	-8.73	-0.31
CSc	e/p	1926	2021	1946	4.68	8.84	-4.16	-0.18
CSd	d/e	1926	2021	1946	-2.11	8.84	-10.96	-0.34
CSe	svar	1926	2021	1946	6.22	8.84	-2.62	-0.15
KS	b/m	1926	2021	1946	0.68	8.84	-8.16	-0.31
BMRR	ntis	1926	2021	1946	7.68	8.84	-1.17	-0.09
BW	eqis	1927	2021	1947	10.27	9.09	1.18	0.10
Ca	tby	1926	2021	1946	4.06	8.84	-4.78	-0.20
FFa	lty	1926	2021	1946	2.61	8.84	-6.23	-0.25
FFb	ltr	1926	2021	1946	7.08	8.84	-1.76	-0.10
FFc	tms	1926	2021	1946	5.48	8.84	-3.36	-0.16
FFd	dfy	1926	2021	1946	1.99	8.84	-6.85	-0.32
FFe	dfr	1926	2021	1946	3.79	8.84	-5.05	-0.25
FS	infl	1926	2021	1946	7.73	8.84	-1.12	-0.08
Co	i/k	1947	2021	1967	6.60	7.77	-1.17	-0.06
LL	cay	1952	2021	1972	1.61	8.19	-6.58	-0.24

Variable		Sample		OOS			Difference	
		Beg	End	Beg	Model	Uncond	Return	SR
Panel B: Investing \$z								
AMP	pce	1953	2021	1973	14.29	14.66	-0.37	-0.11
BH	vp	1990	2021	2000	6.12	6.17	-0.05	-0.08
BPS	impvar	1996	2021	2006	3.76	7.09	-3.33	-0.37
BTZ	vrp	1990	2021	2000	9.92	12.18	-2.26	-0.27
BY	govik	1947	2021	1967	-0.49	1.72	-2.21	-0.10
CEP	lzrt	1926	2021	1946	7.05	6.92	0.14	0.00
CGMS	skew	1951	2019	1971	8.60	9.03	-0.43	-0.08
CGP	crdstd	1990	2021	2000	5.91	10.57	-4.66	-0.21
CP	ogap	1926	2021	1946	10.14	10.20	-0.07	-0.03
DJM	wtxas	1926	2021	1946	3.71	6.47	-2.76	-0.06
HHT	accrul	1965	2021	1985	8.68	4.11	4.58	0.20
HHT	cfacc	1965	2021	1985	10.42	8.47	1.96	0.19
HJTZ	sntm	1965	2020	1985	13.01	13.73	-0.72	-0.17
JT	ndrbl	1958	2021	1978	14.69	13.80	0.89	0.06
JZZ	skvw	1926	2021	1946	5.14	5.89	-0.75	-0.03
KJ	tail	1926	2021	1946	9.79	10.64	-0.85	-0.13
KP	fbm	1926	2021	1946	6.83	7.69	-0.86	-0.11
LY	dtoy	1926	2021	1946	3.30	6.68	-3.38	-0.21
LY	dtoat	1926	2021	1946	1.31	4.11	-2.80	-0.23
M2	ygap	1953	2021	1973	3.35	8.31	-4.96	-0.15
M1	rdsp	1926	2021	1946	10.67	9.63	1.04*	0.16*
MA	rsvix	1996	2021	2006	8.08	8.46	-0.38	-0.20
MR	gpce	1947	2021	1967	7.66	6.90	0.76	0.12
MR	gip	1926	2021	1946	6.64	6.60	0.04*	0.16*
NRTZ	tchi	1951	2021	1971	5.81	5.81	0.00	-0.12
PST	house	1929	2021	1949	2.98	4.07	-1.09	-0.31
PW	avgcor	1926	2021	1946	8.31	8.58	-0.27	-0.03
RRZ	shtint	1973	2021	1993	22.58	19.13	3.45	0.16
Y	disag	1981	2021	2001	19.22	19.22	0.00	
CSa	d/p	1926	2021	1946	1.56	7.00	-5.44	-0.18
CSb	d/y	1926	2021	1946	1.77	7.20	-5.44	-0.17
CSc	e/p	1926	2021	1946	6.20	10.15	-3.95	-0.14
CSd	d/e	1926	2021	1946	-0.29	5.91	-6.21	-0.29
CSe	svar	1926	2021	1946	2.58	6.66	-4.08	-0.13
KS	b/m	1926	2021	1946	2.64	6.06	-3.42	-0.18
BMRR	ntis	1926	2021	1946	6.22	6.25	-0.03	-0.04
BW	eqis	1927	2021	1947	8.04	7.76	0.28	0.05
Ca	tby	1926	2021	1946	3.51	6.12	-2.61	-0.08
FFa	lty	1926	2021	1946	4.20	9.99	-5.80	-0.14
FFb	ltr	1926	2021	1946	14.11	15.85	-1.74	-0.11
FFc	tms	1926	2021	1946	7.71	8.02	-0.31	-0.02
FFd	dfy	1926	2021	1946	4.00	4.61	-0.61	-0.23
FFe	dfr	1926	2021	1946	4.08	11.93	-7.84	-0.24
FS	infl	1926	2021	1946	8.08	9.01	-0.93	-0.10
Co	i/k	1947	2021	1967	8.63	9.93	-1.30	-0.08
LL	cay	1952	2021	1972	1.18	12.45	-11.26	-0.24

Table IA.17: Equity-tilted Investment Strategies on Annual (Jul to Jun) Returns

Variable		Sample		OOS			Difference	
		Beg	End	Beg	Model	Uncond	Return	SR
Panel A: Investing \$1								
AMP	pce	1954	2021	1974	8.83	8.03	0.80	0.09
BH	vp	1991	2021	2001	8.04	7.27	0.77	0.05
BPS	impvar	1997	2021	2007	9.22	11.07	-1.85	-0.26
BTZ	vrp	1991	2021	2001	7.50	7.27	0.23	0.02
BY	govik	1948	2021	1968	-0.11	7.37	-7.48	-0.25
CEP	lzrt	1927	2021	1947	5.18	8.62	-3.44	-0.17
CGMS	skew	1952	2019	1972	7.97	7.07	0.90	0.12
CGP	crdst	1991	2021	2001	7.91	7.27	0.64	0.06
CP	ogap	1927	2021	1947	9.13	8.62	0.51	0.07
DJM	wtxas	1927	2021	1947	2.86	8.62	-5.76	-0.27
HHT	accrul	1966	2021	1986	9.43	9.59	-0.17	-0.01
HHT	cfacc	1966	2021	1986	10.56	9.59	0.96	0.08
HJTZ	sntm	1966	2020	1986	6.31	8.70	-2.39	-0.23
JT	ndrbl	1959	2021	1979	7.46	9.31	-1.85	-0.14
JZZ	skvw	1927	2021	1947	1.74	8.62	-6.88	-0.30
KJ	tail	1927	2021	1947	4.45	8.62	-4.17	-0.20
KP	fbm	1927	2021	1947	6.47	8.62	-2.15	-0.13
LY	dtoy	1927	2021	1947	4.19	8.62	-4.43	-0.26
LY	dtoat	1927	2021	1947	0.42	8.62	-8.20	-0.32
M2	ygap	1954	2021	1974	1.01	8.03	-7.02	-0.28
M1	rdsp	1927	2021	1947	-0.06	8.62	-8.68	-0.29
MA	rsvix	1997	2021	2007	6.49	11.07	-4.59	-0.37
MR	gpce	1948	2021	1968	8.17	7.37	0.80	0.06
MR	gip	1927	2021	1947	8.60	8.62	-0.02	0.00
NRTZ	tchi	1952	2021	1972	6.16	7.74	-1.58	-0.20
PST	house	1930	2021	1950	6.67	9.13	-2.45	-0.20
PW	avgcor	1927	2021	1947	3.49	8.62	-5.13	-0.21
RRZ	shtint	1974	2021	1994	11.64	9.61	2.03	0.16
Y	disag	1982	2021	2002	1.64	8.64	-7.00	-0.27
CSa	d/p	1927	2021	1947	0.92	8.62	-7.70	-0.28
CSb	d/y	1927	2021	1947	0.45	8.62	-8.18	-0.29
CSc	e/p	1927	2021	1947	4.26	8.62	-4.36	-0.18
CSd	d/e	1927	2021	1947	-1.39	8.62	-10.02	-0.33
CSe	svar	1927	2021	1947	2.55	8.62	-6.07	-0.27
KS	b/m	1927	2021	1947	0.25	8.62	-8.37	-0.35
BMRR	ntis	1927	2021	1947	7.48	8.62	-1.14	-0.07
BW	eqis	1928	2021	1948	6.19	8.95	-2.76	-0.14
Ca	tby	1927	2021	1947	2.65	8.62	-5.97	-0.24
FFa	lty	1927	2021	1947	2.73	8.62	-5.89	-0.24
FFb	ltr	1927	2021	1947	3.06	8.62	-5.56	-0.26
FFc	tms	1927	2021	1947	4.17	8.62	-4.45	-0.19
FFd	dfy	1927	2021	1947	1.64	8.62	-6.98	-0.33
FFe	dfr	1927	2021	1947	3.35	8.62	-5.28	-0.23
FS	infl	1927	2021	1947	7.16	8.62	-1.46	-0.11
Co	i/k	1948	2021	1968	4.73	7.37	-2.64	-0.11
LL	cay	1953	2021	1973	2.75	7.76	-5.00	-0.21

Variable		Sample		OOS			Difference	
		Beg	End	Beg	Model	Uncond	Return	SR
Panel B: Investing \$z								
AMP	pce	1954	2021	1974	14.37	14.45	-0.08	-0.04
BH	vp	1991	2021	2001	5.86	9.94	-4.08	-0.25
BPS	impvar	1997	2021	2007	14.80	17.75	-2.94	-0.26
BTZ	vrp	1991	2021	2001	8.64	9.38	-0.75	-0.16
BY	govik	1948	2021	1968	-1.13	2.05	-3.18	-0.16
CEP	lzrt	1927	2021	1947	7.26	6.53	0.72	0.03
CGMS	skew	1952	2019	1972	9.00	9.02	-0.01	-0.01
CGP	crdstd	1991	2021	2001	6.85	8.23	-1.39	-0.12
CP	ogap	1927	2021	1947	9.40	9.29	0.11	0.06
DJM	wtexas	1927	2021	1947	1.54	7.15	-5.62	-0.23
HHT	accrul	1966	2021	1986	7.99	3.12	4.87*	0.22*
HHT	cfacc	1966	2021	1986	7.55	9.37	-1.83	-0.13
HJTZ	sntm	1966	2020	1986	4.38	5.09	-0.71	-0.18
JT	ndrbl	1959	2021	1979	16.63	16.08	0.55	0.06
JZZ	skvw	1927	2021	1947	2.42	4.73	-2.30	-0.14
KJ	tail	1927	2021	1947	9.48	10.58	-1.10	-0.10
KP	fbm	1927	2021	1947	3.94	4.31	-0.37	-0.07
LY	dtoy	1927	2021	1947	4.67	4.84	-0.18	-0.19
LY	dtoat	1927	2021	1947	1.02	3.92	-2.90	-0.31
M2	ygap	1954	2021	1974	4.56	7.49	-2.94	-0.16
M1	rdsp	1927	2021	1947	-0.58	2.14	-2.72	-0.29
MA	rsvix	1997	2021	2007	6.38	10.78	-4.40	-0.27
MR	gpce	1948	2021	1968	9.25	7.83	1.42*	0.21*
MR	gip	1927	2021	1947	5.64	5.70	-0.06	-0.07
NRTZ	tchi	1952	2021	1972	7.32	7.32	0.00	-0.20
PST	house	1930	2021	1950	6.44	6.77	-0.34	-0.15
PW	avgcor	1927	2021	1947	6.15	8.21	-2.06	-0.16
RRZ	shtint	1974	2021	1994	19.94	16.71	3.23	0.21
Y	disag	1982	2021	2002	-0.49	8.76	-9.25	-0.37
CSa	d/p	1927	2021	1947	1.68	6.82	-5.13	-0.18
CSb	d/y	1927	2021	1947	1.90	7.10	-5.20	-0.17
CSc	e/p	1927	2021	1947	6.89	10.26	-3.37	-0.20
CSd	d/e	1927	2021	1947	-1.06	4.93	-6.00	-0.27
CSe	svar	1927	2021	1947	2.12	2.43	-0.31	-0.25
KS	b/m	1927	2021	1947	2.49	6.41	-3.92	-0.22
BMRR	ntis	1927	2021	1947	5.24	5.57	-0.32	-0.12
BW	eqis	1928	2021	1948	5.32	7.27	-1.95	-0.16
Ca	tby	1927	2021	1947	1.65	7.80	-6.16	-0.15
FFa	lty	1927	2021	1947	1.96	11.23	-9.27	-0.17
FFb	ltr	1927	2021	1947	4.34	10.09	-5.75	-0.21
FFc	tms	1927	2021	1947	7.81	9.21	-1.40	-0.06
FFd	dfy	1927	2021	1947	2.92	3.57	-0.65	-0.20
FFe	dfr	1927	2021	1947	2.08	8.31	-6.23	-0.24
FS	infl	1927	2021	1947	7.61	8.23	-0.62	-0.07
Co	i/k	1948	2021	1968	7.64	7.53	0.12	0.01
LL	cay	1953	2021	1973	2.32	7.47	-5.15	-0.16

IA.VII Log and Non-Log Returns

Our key tables in the main paper were for log returns, while our key figures were for simple (non-log) returns.

While Internet Appendix [IA.VI](#) shows further results for both log and simple equity premia, this section provides a visual summary for predicting simple equity premia, much in the same way as that in Table 6. Table [IA.18](#) shows the performance of variables if we do not predict log equity premia but simple equity premia. (The omitted columns from Table 6 are unaffected by definition.) The inference is almost always the same, with many of the marked changes (yellow where it improves, orange where it deteriorates) being merely small drops in and out of statistical significance. The overall picture remains the same, with monthly variables coming out a tiny bit better than log returns, and longer-frequency variables coming out a tiny bit worse.

Table IA.18: Visual Table for Log and Non-Log Returns**Panel A:** Monthly-Frequency Variables

		Log Returns (as in Paper)				Simple Returns			
		IS Performance		Other		IS Performance		Other	
BH	vp	X	..	X	X	X	..	.	.
BPS	impvar	X	..	X	X	X	..	X	X
BTZ	vrp	X	.X	.	X	X	.X	.	.
CEP	lzrt	X	.X	.	.	X	.X	.	.
CP	ogap	†X	..	†X	†X	†X	..	†X	†X
DJM	wtexas	X	..	✓	.	X	..	✓	.
HJTZ	sntm	X	..	X	.	X	..	X	.
JT	ndrbl	X	..	X	X	X	..	X	X
JZZ	skvw	X	.X	X	X	X	.X	X	X
KJ	tail	†X	.X	†X	†X	†X	.X	†X	†X
KP	fbm	X	..	X	X	X	..	X	X
LY	dtoy	X	XX	X	X	X	XX	.	X
LY	dtoat	X	..	X	X	X	..	X	.
Maio ₍₁₃₎	ygap	X	..	X	X	X	..	X	X
Maio ₍₁₆₎	rdsp	X	.X	X	X	X	.X	X	X
Mrtm	rsvix	X	.X	X	X	X	.X	X	X
NRTZ	tchi	X	..	✓	✓	X	..	✓	✓
PW	avgcov	†X	..	†X	†X	†X	..	†✓	†✓
RRZ	shtint	X	..	✓	✓	X	..	✓	✓
Y	disag	X	.X	X	X	X	.X	X	X
pre-2008									
BMRR	ntis	X	n/a	X	X	X	n/a	X	X
Cmpl	tby	✓	n/a	✓	✓	✓	n/a	✓	✓
CSa	d/p	X	n/a	X	X	X	n/a	.	X
CSb	d/y	X	n/a	X	X	X	n/a	X	X
CSc	e/p	X	n/a	X	X	X	n/a	X	X
CSd	d/e	X	n/a	X	X	X	n/a	X	X
CSe	svar	X	n/a	X	X	X	n/a	X	X
FFa	lty	X	n/a	✓	✓	X	n/a	✓	✓
FFb	ltr	X	n/a	X	X	X	n/a	X	X
FFc	tms	X	n/a	.	X	X	n/a	.	✓
FFd	dfy	X	n/a	X	X	X	n/a	X	X
FFe	dfr	X	n/a	X	X	X	n/a	X	X
FS	infl	X	n/a	✓	.	X	n/a	✓	✓
KS	b/m	X	n/a	X	X	X	n/a	X	X

(Table IA.18 continues.)

Panel B: Quarterly-Frequency Variables

		Log Returns (as in Paper)				Simple Returns			
		IS Performance		Other		IS Performance		Other	
AMP	pce	✓	..	✗	✓	✓	..	✗	✓
BY	govik	✗	✗✗	.	.	✗	✗✗	✗	.
CGP	crdstd	✗	..	✓	✓	✗	..	✓	✓
Crn	i/k	✓	n/a .	✓	✓	✓	n/a .	✓	✓
LL	cay	✗	n/a ✗	✗	✗	✗	n/a ✗	✗	✗

Panel C: Annual-Frequency Variables, Jan-Dec

		Log Returns (as in Paper)				Simple Returns			
		IS Performance		Other		IS Performance		Other	
CGMS	skew	✗	✗✗	✗	✗	✗	✗✗	✗	✗
HHT	accrul	✓	..	✓	✓	✓	..	✓	✓
HHT	cfacc	✓	..	✓	✓	✓	..	✓	✓
MR	gpce	✓	..	✓	✓	✓	..	✓	✓
MR	gip	✗	. ✗	✗	✗	✗	. ✗	✗	✗
PST	house	✗	✗✗	✓	.	✗	✗✗	✓	.
BW	eqis	✓	n/a .	✓	✓	✓	n/a .	✓	✓

Panel D: Annual-Frequency Variables, Jul-Jun

		Log Returns (as in Paper)				Simple Returns			
		IS Performance		Other		IS Performance		Other	
CGMS	skew	✗	✗.	✗	✗	✗	✗.	✗	✗
HHT	accrul	✓	..	✓	✓	✓	..	✓	✓
HHT	cfacc	✗	..	✓	✓	✗	..	✓	✓
MR	gpce	✓	..	✓	✓	✓	..	✓	✓
MR	gip	✗	..	✗	✗	✗	..	✗	✗
PST	house	✗	✗✗	✗	✗	✗	✗.	✗	✗
BW	eqis	✗	n/a ✗	✗	✓	✗	n/a ✗	✗	✗